

States of Matter : Gases and Liquids



➡ Case Based MCQs

Case I : Read the passage given below and answer the following questions from 31 to 35.

Intermolecular forces are the forces of attraction and repulsion that exist between molecules of a compound. These cause the compound to exist in a certain state of matter – solid, liquid or gas and affect the melting and boiling points of compounds as well as the solubilities of one substance in another. Attractive intermolecular forces are also called van der Waals' forces. These are weak forces.

31. Dipole-dipole forces act between the molecules possessing permanent dipole. Ends of dipoles possess 'partial charges'. The partial charge is

- (a) more than unit electronic charge
- (b) equal to unit electronic charge
- (c) less than unit electronic charge
- (d) double the unit electronic charge.

32. The nature of inter-particle forces in benzene is

- (a) dipole-dipole interaction
- (b) dispersion force
- (c) ion-dipole interaction
- (d) H-bonding.

33. The interaction energy between two temporary dipoles is proportional to (where r is the distance between the two particles)

- (a) $1/r^4$
- (b) $1/r^2$
- (c) $1/r^5$
- (d) $1/r^6$

34. Attractive intermolecular forces known as van der Waals forces do not include which of the following types of interactions?

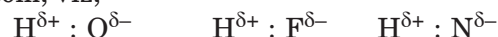
- (a) London forces
- (b) Dipole-dipole forces
- (c) Ion-dipole forces
- (d) Dipole-induced dipole forces

35. In which of the following molecules, the van der Waals forces are likely to be the most important in determining the m.pt. and b.pt?

- (a) CO
- (b) H₂S
- (c) Br₂
- (d) HCl

Case II : Read the passage given below and answer the following questions from 36 to 40.

If a hydrogen atom is bonded to a highly electronegative element such as fluorine, oxygen, nitrogen, then the shared pair of electrons lies more towards the electronegative element. This leads to a polarity in the bond in such a way that a slight positive charge gets developed on H-atom, viz,



Such a bond between the hydrogen atom of one molecule and the more electronegative atom of the same or another molecule is called hydrogen bond.

36. Which of the following compounds can form hydrogen bond?

- (a) CH_4 (b) H_2O
(c) NaCl (d) CHCl_3

37. The boiling point is not affected due to hydrogen bonding in

- (a) water (b) ammonia
(c) methyl alcohol (d) hydrogen chloride.

38. Unusual high b.p. of water is result of

- (a) intermolecular hydrogen bonding
(b) intramolecular hydrogen bonding
(c) both intra and intermolecular hydrogen bonding
(d) high specific heat.

39. Boiling point of hydrogen fluoride is highest amongst HF , HCl , HBr and HI . Which type of intermolecular forces are present in hydrogen fluoride?

- (a) $\text{H}-\text{F}$ has highest van der Waals forces and dipole moment.
(b) $\text{H}-\text{F}$ has highest London forces.
(c) $\text{H}-\text{F}$ has highest dipole moment hence has dipole-dipole, London forces and hydrogen bonding.
(d) $\text{H}-\text{F}$ has strong intermolecular interactions like dipole-induced dipole.

40. Which of the following statements is not true?

- (a) Intermolecular hydrogen bonds are formed between two different molecules of compounds.
(b) Intramolecular hydrogen bonds are formed between two different molecules of the same compound.
(c) Intramolecular hydrogen bonds are formed within the same molecule.
(d) Hydrogen bonds have strong influence on the physical properties of a compound.

Case III : Read the passage given below and answer the following questions from 41 to 45.

An ideal gas is a gas to which the laws of Boyle and Charles are strictly applicable under all conditions of temperatures and pressures.

From Boyle's law we get, $V \propto \frac{1}{P}$
(at constant n and T)

From Charles' law we get, $V \propto T$
(at constant n and P)

From Avogadro's law we get, $V \propto n$ (at constant T and P)

Combining the above three equations we get

$$V \propto \frac{nT}{P} \text{ or, } V = R \frac{nT}{P} \text{ [where } R = \text{ideal gas constant]}$$

$$\text{or } PV = nRT$$

Ideal gas equation is a relation between four variables and it describes the state of any gas. For this reason, it is also called equation of state.

41. At 25°C and 730 mm pressure, 380 mL of dry oxygen was collected. If the temperature is constant, what volume will the oxygen occupy at 760 mm pressure?

- (a) 365 mL
(b) 449 mL
(c) 569 mL
(d) 621 mL

42. 7.0 g of a gas at 300 K and 1 atm occupies a volume of 4.1 litre. What is the molecular mass of the gas?

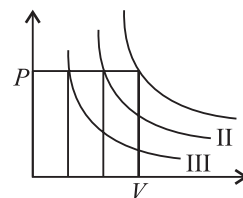
- (a) 42 (b) 38.24
(c) 14.5 (d) 46.5

43. If P is the pressure and ρ is the density of a gas, then P and ρ are related as

- (a) $P \propto \rho$ (b) $P \propto \rho^2$
(c) $P \propto 1/\rho$ (d) $P \propto 1/\rho^2$

44. I, II, III are three isotherms respectively at T_1 , T_2 and T_3 . Temperature will be in order

- (a) $T_1 = T_2 = T_3$
(b) $T_1 < T_2 < T_3$
(c) $T_1 > T_2 > T_3$
(d) $T_1 > T_2 = T_3$



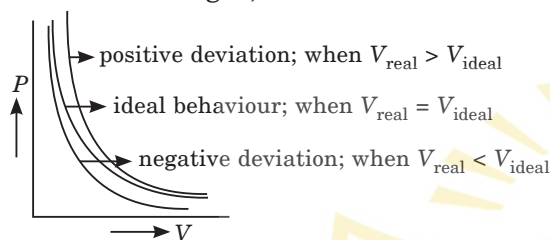
45. If volume of 2 moles of an ideal gas at 540 K is 44.8 litre, then its pressure will be

- (a) 1 atm (b) 3 atm
(c) 2 atm (d) 4 atm

Case IV: Read the passage given below and answer the following questions from 46 to 48.

Real gases do not obey ideal gas equation under all conditions. They nearly obey ideal gas equation at higher temperatures and very low pressures. However, they show deviations from ideality at low temperatures and high pressures.

The isotherms obtained by plotting pressure, P against volume, V for real gases do not coincide with that of ideal gas, as shown :



V_{real} = Volume of the real gas at given pressure.

V_{ideal} = Volume of the gas calculated by ideal gas equation at given pressure.

The deviation from ideal gas behaviour can also be expressed by compressibility factor, Z .

46. The gas equation $PV = nZRT$ becomes ideal gas equation when

- (a) $Z = 0$
- (b) $Z = 0.5$
- (c) $Z = 1$
- (d) $Z = 2$

47. The units of van der Waals' constants a and b respectively are

- (a) $\text{L atm}^2 \text{mol}^{-1}$ and mol L^{-1}
- (b) L atm mol^2 and mol L
- (c) $\text{L}^2 \text{atm mol}^{-2}$ and $\text{mol}^{-1} \text{L}$
- (d) $\text{L}^{-2} \text{atm}^{-1} \text{mol}^{-1}$ and L mol^{-2}

48. The correction factor ' b ' to the ideal gas equation corresponds to

- (a) density of the gas molecules
- (b) excluded volume or covolume
- (c) electric field present between the gas molecules
- (d) forces of attraction between the gas molecules.

➡ Assertion & Reasoning Based MCQs

For question numbers 49-55, a statement of assertion followed by a statement of reason is given. Choose the correct answer out of the following choices.

- (a) Assertion and reason both are correct statements and reason is correct explanation for assertion.
- (b) Assertion and reason both are correct statements but reason is not correct explanation for assertion.
- (c) Assertion is correct statement but reason is wrong statement.
- (d) Assertion is wrong statement but reason is correct statement.

49. **Assertion :** Three states of matter are the result of balance between intermolecular forces and thermal energy of the molecules.

Reason : Intermolecular forces tend to keep the molecules together but thermal energy of molecules tends to keep them apart.

50. **Assertion :** At constant temperature, PV vs V plot for real gases is not a straight line.

Reason : At high pressure all gases have $Z > 1$ but at intermediate pressure most gases have $Z < 1$.

51. **Assertion :** The plot of volume (V) versus pressure (P) at constant temperature is a hyperbola in the first quadrant.

Reason : $V \propto 1/P$ at constant temperature.

52. **Assertion :** Compressibility factor (Z) for non ideal gases is always greater than 1.

Reason : The gases which have $Z > 1$ are difficult to compress.

53. **Assertion :** van der Waals equation is applicable only to non-ideal gases.

Reason : Ideal gases obey the equation $PV = nRT$.

54. **Assertion :** Vapour pressure of NH_3 is higher than $\text{C}_2\text{H}_5\text{OH}$.

Reason : H-bonding is observed in both the molecules.

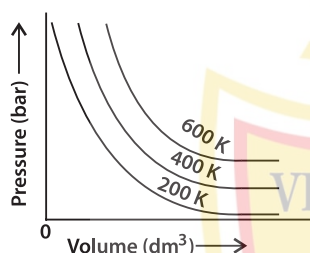
55. **Assertion :** The graph between P vs $1/V$ is a straight line.

Reason : At constant temperature, $P \propto 1/V$.

SUBJECTIVE TYPE QUESTIONS

Very Short Answer Type Questions (VSA)

1. Calculate the volume occupied by 4.0 mole of an ideal gas under NTP condition.
2. How is the partial pressure of a gas in a mixture related to the total pressure of the gaseous mixture?
3. The variation of pressure with volume of the gas at different temperatures can be graphically represented as shown in figure.
4. Under which of the following two conditions applied together, a gas deviates most from the ideal behaviour?
5. Physical properties of ice, water and steam are very different. What is the chemical composition of water in all the three states.
6. Name two intermolecular forces that exist between HF molecules in liquid state.



On the basis of this graph given, how will the volume of a gas change if its pressure is increased at constant temperature?

7. Define the partial pressure of gas.
8. Define compressibility factor.
9. Using the equation of state $PV = nRT$; show that at a given temperature, density of a gas is proportional to gas pressure P .
10. Why is boiling point of hydrogen fluoride higher than that of hydrogen chloride?

Short Answer Type Questions (SA-I)

11. 40 mL of O_2 was collected at $100^\circ C$ and 1 bar pressure. Calculate its volume (in mL) at 273 K and 1.013 bar.
12. 0.068 dm^3 of a sample of nitrogen is collected over water at $20^\circ C$ and 0.92 bar. What will be the volume of dry nitrogen at STP (in mL) (Aqueous tension of water at $20^\circ C = 0.023 \text{ bar}$)?
13. Calculate the volume (in m^3) occupied by 2 moles of an ideal gas at $25 \times 10^5 \text{ Nm}^{-2}$ pressure and 300 K temperature.
14. At $0^\circ C$, the density of a certain oxide of a gas at 2 bar is same as that of dinitrogen at 5 bar. What is the molecular mass of the oxide?
15. How much time (in years) would it take to distribute one Avogadro number of wheat grains, if 10^{10} grains are distributed each second?
16. Calculate the volume (in litres) occupied by 8.8 g of CO_2 at $31.1^\circ C$ and 1 bar pressure.
17. Which type of intermolecular forces exist among the following molecules?
(i) He atoms and HCl molecules
(ii) HF molecules
(iii) N_2 molecules
(iv) HCl molecules
18. Explain the physical significance of van der Waals' parameters.
19. Calculate the total pressure (in bar) in a mixture of 8 g of dioxygen and 4 g of dihydrogen confined in a vessel of 1 dm^3 at $27^\circ C$.
 $R = 0.083 \text{ bar dm}^3 \text{ K}^{-1} \text{ mol}^{-1}$
20. 1 mole of sulphur dioxide occupies a volume of 350 mL at $27^\circ C$ and $5 \times 10^6 \text{ Pa}$ pressure. Calculate the compressibility factor of the gas. Is it less or more compressible than an ideal gas?

Short Answer Type Questions (SA-II)

21. Describe London forces or dispersion forces with example.

22. The drain cleaner, Drainex contains small bits of aluminium which react with caustic soda to produce dihydrogen. What volume of dihydrogen (in mL) at 20°C and one bar will be released when 0.15 g of aluminium reacts?

23. On the basis of intermolecular forces and thermal energy explain why substances exist in three different states of matter?

24. Two moles of ammonia gas are enclosed in a vessel of 5 litre capacity at 27°C . Calculate the pressure exerted by the gas, assuming that

(i) the gas behaves like an ideal gas (using ideal gas equation)

(ii) the gas behaves like a real gas (using van der Waals equation)

Given that for ammonia, $a = 4.17 \text{ atm litre}^2 \text{ mol}^{-2}$ and $b = 0.037 \text{ litre mol}^{-1}$.

25. 2.9 g of a gas at 95°C occupied the same volume as 0.184 g of dihydrogen at 17°C at the same pressure. What is the molar mass of the gas?

26. On the basis of their interaction energy and thermal energy explain why

(i) a solid has high rigidity?

(ii) In gas, molecules are sufficiently apart from one another?

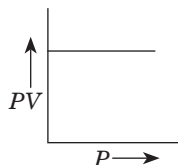
(iii) liquid has no definite shape?

27. Explain the following :

(i) Boyle's law

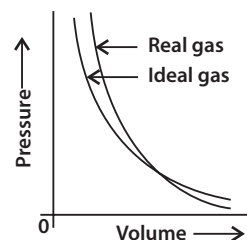
(ii) Avogadro's law

28. (a) Which gas law is shown by the following graph?



(b) At 25°C and 760 mm Hg pressure, a gas occupies 600 mL volume. What will be its pressure (in mmHg) at a height where temperature is 10°C and volume of the gas is 640 mL?

29. Pressure versus volume graph for a real gas and an ideal gas are shown in the figure.



Answer the following questions on the basis of this graph.

(i) Interpret the behaviour of real gas with respect to ideal gas at low pressure.

(ii) Interpret the behaviour of real gas with respect to ideal gas at high pressure.

(iii) Mark the pressure and volume by drawing a line at the point where real gas behaves as an ideal gas.

30. A perfectly elastic spherical balloon of 0.2 m diameter was filled with hydrogen at sea level. What will be its diameter (in m) when it has risen to an altitude where the pressure is 0.65 atm? (Assume no change in temperature and atmospheric pressure at sea level).

31. (i) Out of CO_2 and He, which gas have higher value of van der Waals' constant 'b'?

(ii) At 27°C density of a gaseous substance at 3 bar is same as that of hydrogen at 9 bar. What is molar mass of the substance?

32. What will be the pressure (in bar) of a gas mixture when 0.5 L of H_2 at 0.8 bar and at 2.0 L of oxygen at 0.7 bar are introduced in a 1 L vessel at 27°C ?

33. (a) 22 g of dry ice is placed in an evacuated bottle of 1 litre capacity and tightly stoppered. What would be the pressure inside the bottle in atm, when it is heated to 37°C ?

(b) 3.12 g of sulphur is vapourised at 427°C and 760 mm pressure, when the vapours occupy a volume of 700 mL. Find the molecular formula of sulphur. (atomic mass of sulphur = 32).

34. An open beaker at 27°C is heated to 477°C . What percentage of air would have been expelled out?

35. Explain the difficulties faced by the mountaineers with respect to the air present around them. How is this difficulty solved?

Long Answer Type Questions (LA)

36. Payload is defined as the difference between the mass of displaced air and the mass of the balloon. Calculate the payload (in kilograms) when a balloon of radius 10 m, mass 100 kg is filled with helium at 1.66 bar at 27°C. (Density of air = 1.2 kg m⁻³ and $R = 0.083 \text{ bar dm}^3 \text{ K}^{-1} \text{ mol}^{-1}$)

37. A liquefied petroleum gas (LPG) cylinder weighs 14.8 kg when empty. When full it weighs 29.0 kg and shows a pressure of 2.5 atm. In the course of use at 27 °C, the mass of the full cylinder is reduced to 23.2 kg. Find out the volume of the gas in cubic metres used up at the normal usage conditions and the final pressure inside the cylinder. Assume LPG to be *n*-butane with normal boiling point of 0 °C.

38. At temperature, T and pressure P , two ideal gases A and B are mixed. Show that the density d of the mixture is given by

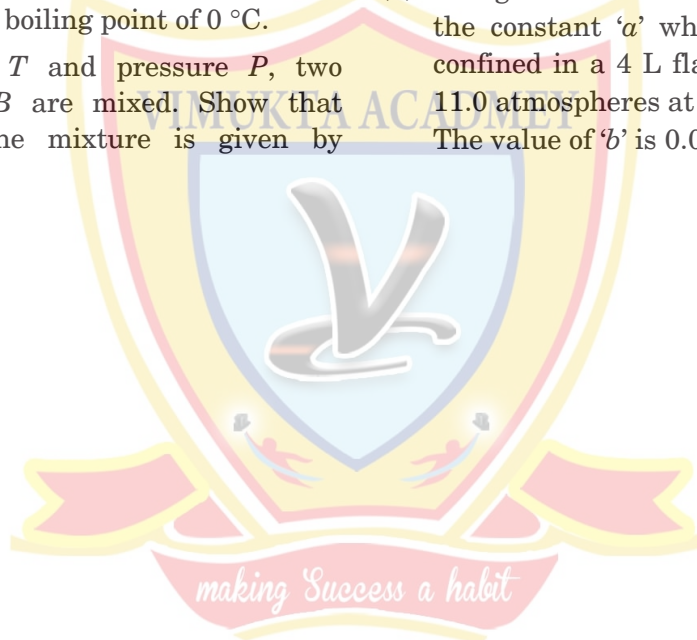
$d = (X_A M_A + X_B M_B) \times \frac{P}{RT}$ where X_A and X_B are the mole fractions and M_A and M_B are the molecular weights of the gases A and B respectively.

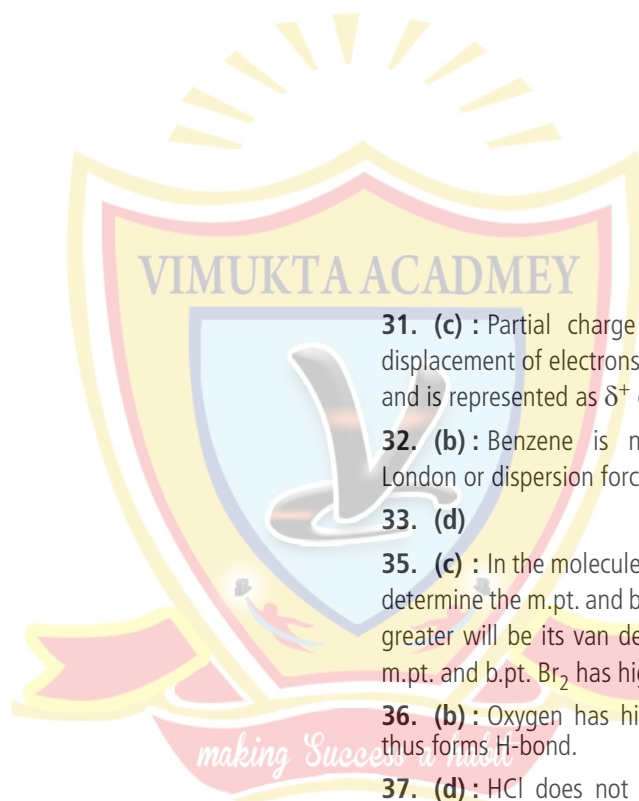
39. The volume of a gas is to be increased by 20% without changing the pressure. To what temperature (in °C) the gas must be heated if the initial temperature of the gas is 27 °C?

40. Answer the following :

(a) How are the van der Waals' constants ' a ' and ' b ' related to the molecular size?

(b) Using van der Waals' equation, calculate the constant ' a ' when two moles of a gas confined in a 4 L flask exerts a pressure of 11.0 atmospheres at a temperature of 300 K. The value of ' b ' is 0.05 L mol⁻¹.





31. (c) : Partial charge is a small charge developed by displacement of electrons. It is less than unit electronic charge and is represented as δ^+ or δ^- .

32. (b) : Benzene is non-polar compound and exhibits London or dispersion forces.

33. (d)

34. (c)

35. (c) : In the molecules the van der Waals force are likely to determine the m.pt. and b.pt. Greater the mass of the molecule greater will be its van der Waals force and higher will be its m.pt. and b.pt. Br_2 has highest m.pt.

36. (b) : Oxygen has high electronegativity and small size, thus forms H-bond.

37. (d) : HCl does not undergo H-bonding and its boiling point is not affected by H-bonding.

38. (a) : Due to intermolecular hydrogen bond in H_2O , its molecules are associated with each other which is responsible for unusual high b.p. of water.

39. (c) : $\text{H}-\text{F}$ has dipole-dipole interaction, London forces and hydrogen bonding due to highest electronegativity of F. Hence, boiling point of $\text{H}-\text{F}$ is highest.

40. (b) : Intramolecular hydrogen bonds are formed within the same molecule.

41. (a) : Applying $P_1V_1 = P_2V_2$

$P_1 = 730 \text{ mm}$, $V_1 = 380 \text{ mL}$, $P_2 = 760 \text{ mm}$, $V_2 = ?$

$$V_2 = \frac{P_1V_1}{P_2} = \frac{730 \times 380}{760} = 365 \text{ mL}$$

42. (a) : Applying, $PV = \frac{m}{M}RT$

$P = 1 \text{ atm}; V = 4.1 \text{ L}; m = 7.0 \text{ g};$

$R = 0.0821 \text{ L atm K}^{-1} \text{ mol}^{-1}; T = 300 \text{ K}$

$$M = \frac{RT}{PV} \times m = \frac{0.0821 \times 300}{1 \times 4.1} \times 7.0 = 42.05 \approx 42 \text{ g mol}^{-1}$$

43. (a) : $P = \frac{\rho RT}{M}$ i.e., $P \propto \rho$

44. (c) : Draw a line at constant pressure parallel to volume-axis. Take volume corresponding to each temperature.

From volume axis, $V_1 > V_2 > V_3$

Hence $T_1 > T_2 > T_3$.

45. (c) : Number of moles, $n = 2$;

Temperature, $T = 540 \text{ K}$

Volume, $V = 44.8 \text{ L}; P = ?$

[R , gas constant = $0.0821 \text{ L atm K}^{-1} \text{ mol}^{-1}$]

According to ideal gas equation $PV = nRT$

$$P = \frac{nRT}{V} = \frac{2 \times 0.0821 \times 540}{44.8} = 2 \text{ atm}$$

46. (c) : The ideal gas equation is $PV = nRT$

When Z (the compressibility factor) is one, the given equation $PV = nZRT$ becomes ideal gas equation.

47. (c) : a and b are expressed in terms of the units of P and V .

$$\text{Pressure correction} = P' = \frac{n^2 a}{V^2}$$

$$a = \frac{P'V^2}{n^2} = \frac{\text{pressure} \times (\text{volume})^2}{(\text{mole})^2}$$

$$\text{Unit of } a = \text{atm} \times (\text{L})^2 \text{mol}^{-2}$$

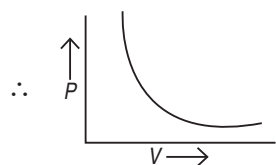
Unit of b is the same as for the volume. i.e., L mol^{-1}

48. (b) : The correction factor 'b' represents excluded volume or covolume.

49. (a)

50. (b) : At constant temperature, plot of PV vs V for real gases is not linear because real gases have intermolecular forces of attraction.

51. (a) : According to Charles's law $V \propto \frac{1}{P}$.



52. (d) : Compressibility factor (Z) can be greater, equal or lesser than 1.

53. (b) : The van der Waals equation is applicable to real gases only, while $PV = nRT$ is applicable to ideal gases.

54. (b) : H - bonding in liquid NH_3 is weaker than $\text{C}_2\text{H}_5\text{OH}$. Thus, escaping tendency is higher and hence, the vapour pressure of NH_3 is also higher.

55. (a)

SUBJECTIVE TYPE QUESTIONS

1. $PV = nRT$

or $V = \frac{nRT}{P} = \frac{4 \times 0.082 \times 273}{1} = 89.54 \text{ L}$

2. Partial pressure of a gas

= Mole fraction of that gas $\times$ Total pressure

3. The volume of a gas decreases if the pressure on the gas is increased keeping the temperature constant.

4. At high pressure and low temperature, the gases deviate from ideal behaviour due to significant intermolecular forces and more than negligible size of the molecules.

5. The chemical composition of water remains same in all the physical states i.e., solid, liquid and gas.

6. Dipole-dipole interactions and hydrogen bonding exist between HF molecules in liquid state.

7. In a mixture of gases, the pressure exerted by the individual gas is called its partial pressure.

8. The extent to which a real gas deviates from ideal behaviour can be conveniently studied in terms of quantity ' Z ' called the compressibility factor, which is defined as $Z = \frac{PV}{nRT}$

For an ideal gas, as $PV = nRT$, $Z = 1$

9. $PV = nRT$

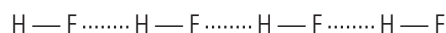
or $P = \frac{wRT}{MV}$ (Since $n = \frac{w}{M}$)

or $P = \frac{dRT}{M}$ [Since $d = \frac{w}{V}$]

or, $P \propto d$

Hence, density (d) of a gas $\propto P$, because R , T and M are constants.

10. Boiling point of HF is higher than HCl due to extensive hydrogen bonding between H — F molecules.



11. Given, $P_1 = 1 \text{ bar}$, $V_1 = 40 \text{ mL} = 0.04 \text{ L} = 0.04 \text{ dm}^3$,
 $T_1 = 373 \text{ K}$, $P_2 = 1.013 \text{ bar}$, $T_2 = 273 \text{ K}$, $V_2 = ?$

$$\frac{P_1 V_1}{T_1} = \frac{P_2 V_2}{T_2}$$

$$\text{or } V_2 = \frac{P_1 V_1 T_2}{T_1 P_2} = \frac{1 \times 0.04 \times 273}{373 \times 1.013} = 0.029 \text{ dm}^3 \text{ or } 29 \text{ mL}$$

12. Given $P_1 = 0.92 - 0.023 = 0.897 \text{ bar}$,
 $V_1 = 0.068 \text{ dm}^3$, $T_1 = 293 \text{ K}$, $P_2 = 1 \text{ bar}$ (Pressure of dry gas),
 $V_2 = ?$, $T_2 = 273 \text{ K}$

$$\frac{P_1 V_1}{T_1} = \frac{P_2 V_2}{T_2} \text{ or } V_2 = \frac{P_1 V_1 T_2}{P_2 T_1}$$

$$= \frac{0.897 \times 0.068 \times 273}{1 \times 293} = 0.057 \text{ dm}^3 \text{ or } 57 \text{ mL}$$

13. According to ideal gas equation, $PV = nRT$

$$\text{or } V = \frac{nRT}{P} = \frac{2 \times 8.314 \times 300}{25 \times 10^5} \approx 2 \times 10^{-3} \text{ m}^3$$

$$14. M_{\text{oxide}} = \frac{d_{\text{oxide}} RT}{P_{\text{oxide}}} \quad \dots(1)$$

$$M_{\text{N}_2} = \frac{d_{\text{N}_2} RT}{P_{\text{N}_2}} \quad \dots(2)$$

Dividing equation (1) by equation (2) gives

$$\frac{M_{\text{oxide}}}{M_{\text{N}_2}} = \frac{d_{\text{oxide}} \times P_{\text{N}_2}}{d_{\text{N}_2} \times P_{\text{oxide}}}$$

But $d_{\text{oxide}} = d_{\text{N}_2}$

$$\therefore M_{\text{oxide}} = \frac{5 \text{ bar}}{2 \text{ bar}} \times 28 \text{ g mol}^{-1}$$

$$M_{\text{oxide}} = 70 \text{ g mol}^{-1}$$

15. Time taken to distribute 10^{10} grains = 1 sec.

Time taken to distribute 6.023×10^{23} grains

$$= \frac{1 \times 6.022 \times 10^{23}}{10^{10}} = 6.022 \times 10^{13} \text{ sec}$$

$$= \frac{6.022 \times 10^{13}}{60 \times 60 \times 24 \times 365} = 1.90956 \times 10^6 \text{ years}$$

16. According to ideal gas equation, $PV = nRT$

$$n = \frac{8.8}{44} \text{ moles, } P = 1 \text{ bar,}$$

$$T = 273 + 31.1 = 304.1 \text{ K,}$$

$$V = ?$$

$$R = 0.083 \text{ bar L K}^{-1} \text{ mol}^{-1}$$

$$\text{Now, } V = \frac{nRT}{P}$$

$$= \frac{\left(\frac{8.8}{44} \text{ moles}\right) \times (0.083 \text{ bar L K}^{-1} \text{ mol}^{-1}) \times (304.1 \text{ K})}{1 \text{ bar}}$$

$$= 5.05 \text{ L}$$

17. (i) Dipole-induced dipole forces

(ii) Hydrogen bonding

(iii) Dispersion forces

(iv) Dipole-dipole forces

18. 'a' measures the intermolecular forces of attraction. The greater the value of 'a', the more will be intermolecular forces of attraction. 'b' measures volume occupied by molecules of a gas.

19. Partial pressure of oxygen gas,

$$P = \frac{nRT}{V}, n = \frac{8}{32} \text{ mol, } V = 1 \text{ dm}^3, T = 300 \text{ K}$$

$$P_{(\text{O}_2)} = \frac{8 \times 0.083 \times 300}{32 \times 1} = 6.225 \text{ bar}$$

Partial pressure of hydrogen gas,

$$P = \frac{nRT}{V}, n = \frac{4}{2} = 2 \text{ mol}$$

$$P_{(\text{H}_2)} = \frac{2 \times 0.083 \times 300}{1} = 49.8 \text{ bar}$$

$$\text{Total pressure} = P_{(\text{O}_2)} + P_{(\text{H}_2)} = 6.225 + 49.8$$

$$= 56.025 \text{ bar}$$

20. Compressibility factor, $Z = \frac{PV}{nRT}$

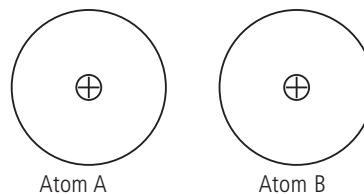
$$n = 1 \text{ mol, } P = 5 \times 10^6 \text{ Pa, } V = 350 \text{ mL} = 0.350 \times 10^{-3} \text{ m}^3$$

$$R = 8.314 \text{ N m K}^{-1} \text{ mol}^{-1}, T = 27 + 273 = 300 \text{ K}$$

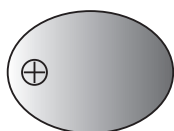
$$\therefore Z = \frac{5 \times 10^6 \times 0.350 \times 10^{-3}}{1.0 \times 8.314 \times 300} = 0.702$$

Thus, SO_2 is more compressible than an ideal gas (which has $Z = 1$).

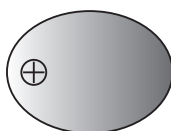
21. London or dispersion forces : This is the weakest intermolecular force. It is a temporary attractive force that results when the electrons in two adjacent atoms occupy positions that make the atoms to form temporary dipoles. This force is sometimes called an *dipole-induced dipole attraction*. Because of the constant motion of the electrons, an atom or molecule can develop a temporary (instantaneous) dipole when its electrons are distributed unsymmetrically about the nucleus.



symmetrical distribution of electronic charge cloud
(a)

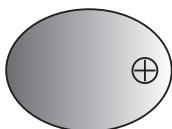


Atom 'A' with instantaneous dipole, more electron density on the right hand side

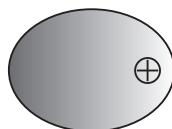


Atom 'B' with induced dipole,

(b)



Atom 'A' more electron density on the left hand side

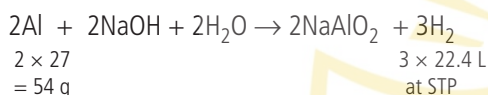


Atom 'B' with induced dipole

(c)

Dispersion forces or London forces between atoms.

22. The reaction between aluminium and caustic soda is



$\therefore$ 54 g of Al produces H_2 at S.T.P. = $3 \times 22.4 \text{ L}$

0.15 g of Al will produce H_2 at S.T.P.

$$= \frac{3 \times 22.4}{54} \times 0.15 = 0.186 \text{ L}$$

At STP

$$P_1 = 1 \text{ atm}$$

$$V_1 = 0.186 \text{ L}$$

$$T_1 = 273 \text{ K}$$

Given conditions

$$P_2 = 1 \text{ bar} = 0.987 \text{ atm}$$

$$V_2 = ?$$

$$T_2 = 273 + 20 = 293 \text{ K}$$

Applying ideal gas equation, $\frac{P_1 V_1}{T_1} = \frac{P_2 V_2}{T_2}$

$$\frac{1 \times 0.186}{273} = \frac{0.987 \times V_2}{293}$$

$$V_2 = \frac{293}{0.987} \times \frac{1 \times 0.186}{273} = 0.2030 \text{ L} = 203 \text{ mL}$$

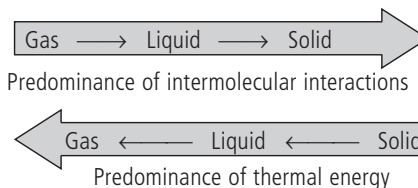
23. The intermolecular forces tend to keep the molecules together but thermal energy tend to keep them apart. Thus, these two compete and the competition between these two (i.e., intermolecular forces and thermal energy) results in three states of matter.

(i) In a solid, the intermolecular forces predominate over the thermal energy and hence, the particles are held together in rigid, highly-oriented and close-packed structure.

(ii) In liquids, the intermolecular forces are no longer strong enough, however, these are still sufficient so that particles remain in each other's environment, hence, liquids have sufficient mobility.

(iii) In gases, the thermal energy dominates the effect of intermolecular forces, thus, the gas molecules acquire the unrestricted and independent mobility in the vapour state.

Predominance of thermal energy and the intermolecular forces in the three state of matter is as follows :



24. Given, $n = 2$ moles, $V = 5$ litres, $T = 27^\circ\text{C} = (27 + 273) \text{ K} = 300 \text{ K}$

$$a = 4.17 \text{ atm litre}^2 \text{ mol}^{-2}, b = 0.037 \text{ litre mol}^{-1}$$

Also, we know that $R = 0.0821 \text{ litre atm K}^{-1} \text{ mol}^{-1}$

(i) If the gas behaves like an ideal gas, we have $PV = nRT$

$$\therefore P = \frac{nRT}{V} = \frac{2 \times 0.0821 \times 300}{5} = 9.85 \text{ atm}$$

(ii) If the gas behaves like a real gas, we apply van der Waals' equation i.e.,

$$\left(P + \frac{an^2}{V^2}\right)(V - nb) = nRT \text{ or } P = \frac{nRT}{V - nb} - \frac{an^2}{V^2}$$

$$= \frac{2 \times 0.0821 \times 300}{5 - 2 \times 0.037} - \frac{4.17 \times (2)^2}{(5)^2} = 9.33 \text{ atm.}$$

25. Case I : Let molar mass of gas be $M \text{ g mol}^{-1}$

Weight of gas = 2.9 g

$$\text{No. of moles} = \frac{\text{Weight}}{\text{Molar mass}} = \frac{2.9}{M}$$

$$T = 273 + 95 = 368 \text{ K}$$

$$PV = \frac{2.9}{M} \times R \times 368$$

...(i)

Case II : Mass of dihydrogen = 0.184 g

$$\text{No. of moles of } \text{H}_2 = \frac{0.184}{2}$$

$$T = 273 + 17 = 290 \text{ K}$$

$$PV = \frac{0.184}{2} \times R \times 290$$

...(ii)

From equations (i) and (ii),

$$\frac{2.9}{M} \times 368 = \frac{0.184}{2} \times 290$$

$$M = \frac{2.9 \times 368 \times 2}{0.184 \times 290} = 40 \text{ g mol}^{-1}$$

26. (i) A solid has high rigidity because thermal motion is too weak to overcome the strong intermolecular forces of attraction.

(ii) In a gas, thermal energy is so high that the molecules cannot come close together. Hence, there are large empty spaces between them.

(iii) In a liquid, there is a reasonable balance between the attractive intermolecular forces and thermal energy. Hence, molecules in a liquid exist together, *i.e.*, it is a condensed state of matter but there is no rigidity. That is why they have no definite shape.

27. (i) Boyle's law : According to Boyle's law, at constant temperature, the volume of a fixed amount of a gas is inversely proportional to its pressure, *i.e.*, if volume increases, the pressure would decrease. This is because as volume increases, the number of molecules striking the walls of a container in a given time decreases leading to decrease in pressure.

$$P \propto \frac{1}{V} ; P = \frac{\text{Constant}}{V}$$

$$\Rightarrow PV = \text{Constant} \Rightarrow P_1 V_1 = P_2 V_2$$

(ii) Avogadro's law : This law states that equal volume of all gases under similar conditions of temperature and pressure contain equal number of molecules.

$$V \propto n(T, P \text{ constant}) = \frac{V_1}{n_1} = \frac{V_2}{n_2}$$

28. (a) Boyle's law

(b) Applying gas equation (combined gas law),

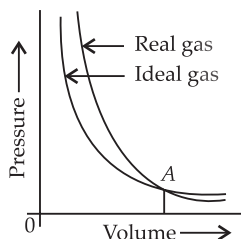
$$\frac{P_1 V_1}{T_1} = \frac{P_2 V_2}{T_2} \Rightarrow \frac{760 \times 600}{298} = \frac{P_2 \times 640}{283}$$

$$\text{or } P_2 = 676.6 \text{ mm of Hg}$$

29. (i) At low pressure real gas starts behaving like an ideal gas.

(ii) At high pressure gases deviate from ideal behaviour.

(iii)



At point A real gas behaves as an ideal gas.

30. If r_1 is the radius of the balloon at sea level, then, volume of the balloon at sea level = $\frac{4}{3} \pi r_1^3 = \frac{4}{3} \pi (0.1 \text{ m})^3$

$$\text{i.e., volume of the gas at sea level } (V_1) = \frac{4}{3} \pi (0.1 \text{ m})^3$$

Pressure at the sea level (P_1) = 1 atm

Suppose the radius of the balloon at altitude = r_2

$$\text{Then, volume of the balloon at altitude } (V_2) = \frac{4}{3} \pi r_2^3$$

Pressure at the altitude (P_2) = 0.65 atm (Given)

As temperature remains constant, applying Boyle's law,

$$\begin{array}{ccc} P_1 V_1 & = & P_2 V_2 \\ \text{(At sea level)} & & \text{(At altitude)} \end{array}$$

$$1 \text{ atm} \times \frac{4}{3} \pi (0.1 \text{ m})^3 = 0.65 \text{ atm} \times \frac{4}{3} \pi r_2^3$$

$$\text{or } r_2^3 = \frac{(0.1 \text{ m})^3}{0.65} = 1.54 \times 10^{-3} \text{ m}^3$$

$$\therefore r_2 = (1.54 \times 10^{-3})^{1/3} \text{ m} = 0.1154 \text{ m}$$

$\therefore$ Diameter of the balloon at altitude

$$= 2 \times 0.1154 \text{ m} = 0.2308 \text{ m}$$

31. (i) Since, CO_2 molecules have larger size than that of He molecules, hence, CO_2 has larger value of van der Waals' constant 'b'.

$$(ii) \quad M_1 = \frac{d_1 R T}{P_1} \quad \dots(i)$$

[where,

M_1 = Molecular weight of the substance

d_1 = Density of the substance

P_1 = Pressure of the substance]

$$M_{H_2} = \frac{d_{H_2} R T}{P_{H_2}} \quad \dots(ii)$$

Dividing equation (i) by equation (ii)

$$\frac{M_1}{M_{H_2}} = \frac{d_1 \times P_{H_2}}{d_{H_2} \times P_1} \text{ or } \frac{M_1}{M_{H_2}} = \frac{P_{H_2}}{P_1} \quad [\text{Given, } d_1 = d_{H_2}]$$

$$\text{or } M_1 = \frac{M_{H_2} \times P_{H_2}}{P_1} = \frac{2 \times 9}{3} = 6$$

32. Partial pressure of H_2 :

$$V_1 = 0.5 \text{ L, } V_2 = 1 \text{ L,}$$

$$P_1 = 0.8 \text{ bar, } P_2 = ?$$

By Boyle's law, $P_1 V_1 = P_2 V_2$

$$P_2 = \frac{0.8 \times 0.5}{1} = 0.4 \text{ bar}$$

Partial pressure of O_2 :

$$V_1 = 2.0 \text{ L, } V_2 = 1 \text{ L, } P_1 = 0.7 \text{ bar, } P_2 = ?$$

By Boyle's law, $P_1 V_1 = P_2 V_2$

$$P_2 = \frac{0.7 \times 2.0}{1} = 1.4 \text{ bar}$$

Pressure of the gas mixture,

$$P_{\text{mix}} = p_{H_2} + p_{O_2} = 0.4 + 1.4 = 1.8 \text{ bar}$$

33. (a) $W = 22 \text{ g CO}_2$, $V = 1 \text{ L}$, $M = 44$, $T = 37 + 273 = 310 \text{ K}$, $P = ?$

Dry ice is solid CO_2 , which when heated in an evacuated bottle it is converted into gaseous CO_2 .

From ideal gas equation,

$$PV = \frac{WRT}{M} \Rightarrow P = \frac{22 \times 0.082 \times 310}{44 \times 1} = 12.71 \text{ atm.}$$

Now, pressure inside the bottle is 12.71 atm.

(b) For sulphur,

$$W = 3.12 \text{ g}, T = 427 + 273 = 700 \text{ K}$$

$$P = 760 \text{ mm} = 1 \text{ atm.}$$

$$V = 700 \text{ ml} = 0.7 \text{ L.}$$

Now molecular mass of sulphur,

$$M = \frac{WRT}{PV} = \frac{3.12 \times 0.082 \times 700}{1 \times 0.7} = 255.84$$

As atomic mass = 32, so no. of atoms in one molecule of sulphur = $\frac{255.84}{32} = 8$

Hence, molecular formula of sulphur is S_8 .

34. Suppose the number of moles of gas present at 27°C in flask of volume V at pressure P is n_1 , then assuming ideal gas behaviour,

$$PV = n_1 R \times 300$$

Suppose n_2 = number of moles at 477°C , then

$$PV = n_2 R \times 750$$

From equation (i) and equation (ii), we get

$$n_2 = \frac{300}{750} \times n_1 = 0.4 n_1$$

So, $(1 - 0.4) \times 100\%$ i.e. 60% is expelled out.

35. At altitude, the atmospheric pressure is low. Hence, air is less dense. As a result, less oxygen is available for breathing. The person feels uneasiness, headache, etc. This is called altitude sickness. This difficulty is solved by carrying oxygen cylinders with them.

$$\text{36. Volume of balloon} = \frac{4}{3} \pi r^3$$

Radius of balloon, $r = 10 \text{ m}$

$$V = \frac{4}{3} \times 3.14 \times (10)^3 = 4186.7 \text{ m}^3$$

$$\begin{aligned} \text{Mass of displaced air} &= 4186.7 \text{ m}^3 \times 1.2 \text{ kg m}^{-3} \\ &= 5024.04 \text{ kg} \end{aligned}$$

$$\begin{aligned} \text{Moles of gas present} &= \frac{PV}{RT} \\ &= \frac{1.66 \times 4186.7 \times 10^3}{0.083 \times 300} = 279.11 \times 10^3 \text{ moles} \end{aligned}$$

$$\begin{aligned} \text{Mass of helium present} &= 279.11 \times 10^3 \times 4 \\ &= 1116.44 \times 10^3 \text{ g} \\ &= 1116.44 \text{ kg} \end{aligned}$$

$$\begin{aligned} \text{Mass of filled balloon} &= 100 + 1116.44 \\ &= 1216.44 \text{ kg} \end{aligned}$$

$$\begin{aligned} \text{Payload} &= \text{Mass of displaced air} - \text{Mass of balloon} \\ &= 5024.04 - 1216.44 = 3807.6 \text{ kg} \end{aligned}$$

37. Weight of LPG originally present

$$= 29.0 - 14.8 = 14.2 \text{ kg}$$

Pressure = 2.5 atm

$$\begin{aligned} \text{Weight of LPG present after use} &= 23.2 - 14.8 \\ &= 8.4 \text{ kg} \end{aligned}$$

Since volume of the cylinder is constant, applying

$$\begin{aligned} PV = nRT \Rightarrow \frac{P_1}{P_2} &= \frac{n_1}{n_2} = \frac{w_1/M}{w_2/M} = \frac{w_1}{w_2} \\ \Rightarrow \frac{2.5}{P_2} &= \frac{14.2}{8.4} \text{ or } P_2 = \frac{2.5 \times 8.4}{14.2} = 1.48 \text{ atm} \end{aligned}$$

$$\therefore \text{Weight of used gas} = 14.2 - 8.4 = 5.8 \text{ kg}$$

$$\text{Moles of gas} = \frac{5.8 \times 10^3}{58} = 100 \text{ mol}$$

Normal conditions : $P = 1 \text{ atm}, T = 273 + 27 = 300 \text{ K}$

Volume of 100 mol of LPG at 1 atm and 300 K

$$V = \frac{nRT}{P} = \frac{100 \times 0.082 \times 300}{1} = 2460 \text{ L} = 2.460 \text{ m}^3$$

38. At temperature T and pressure P , two ideal gases, A and B are mixed.

n_A and n_B are the number of moles of A and B respectively.

n is total moles of A and B present in the mixture.

$$n = n_A + n_B$$

Let the gas mixture has a volume V .

V_A and V_B are volume of A and B respectively.

From ideal gas equation, $PV = nRT$.

$$\text{For gas } A, PV_A = n_A RT. \therefore V_A = \frac{n_A RT}{P}$$

$$\text{For gas } B, PV_B = n_B RT. \therefore V_B = \frac{n_B RT}{P}$$

M_A and M_B are molecular weights of the gases and X_A and X_B are the mole fractions of the gases A and B respectively.

$$\text{Density of a gas, } d = \frac{\text{mass}}{\text{volume}}$$

$$d = \frac{M}{V} = \frac{\frac{n_A M_A}{\frac{n_A RT}{P}} + \frac{n_B M_B}{\frac{n_B RT}{P}}}{\frac{n_A RT}{P} + \frac{n_B RT}{P}} = \frac{P}{RT} \left[\frac{n_A M_A + n_B M_B}{n_A + n_B} \right]$$

$$= \frac{P}{RT} \left\{ \frac{n_A M_A + n_B M_B}{n} \right\} = \frac{P}{RT} \left\{ \frac{n_A}{n} M_A + \frac{n_B}{n} M_B \right\}$$

$$d = \frac{P}{RT} \{X_A M_A + X_B M_B\}$$

39. Suppose volume of gas at $27^\circ\text{C} = V \text{ cm}^3$

$$\text{Increase in volume desired} = 20\% \text{ of } V = \frac{20}{100} \times V = 0.2 V$$

$$\therefore \text{ Final volume} = V + 0.2 V = 1.2 V$$

$$\text{Now, } V_1 = V \text{ cm}^3, T_1 = 300 \text{ K,}$$

$$V_2 = 1.2 V, T_2 = ?$$

At constant P ,

$$\frac{V_1}{T_1} = \frac{V_2}{T_2} \Rightarrow \frac{V}{300} = \frac{1.2 V}{T_2}$$

$$\Rightarrow T_2 = 360 \text{ K} = 360 - 273 = 87^\circ\text{C}$$

40. (a) The van der Waals' constant ' a ' is measure of intermolecular attractions. Therefore, the value of ' a ' reflects the tendency of the gas to liquefy. The gas having larger value of ' a ', will liquefy more easily. The van der Waals' constant ' b ' is a measure of the close-packed molecular volume. Thus the molecule of a gas having greater value of ' b ' has bigger size.

(b) The van der Waals' equation is

$$\left(P + \frac{n^2 a}{V^2} \right) (V - nb) = nRT$$

$$\text{Given, } n = 2 \text{ mol, } V = 4 \text{ L,}$$

$$P = 11.0 \text{ atm, } T = 300 \text{ K}$$

Substituting the values in the above equation,

$$\left(11.0 \text{ atm} + \left(\frac{2 \text{ mol}}{4 \text{ L}} \right)^2 \cdot a \right) (4 \text{ L} - 2 \text{ mol} \times 0.05 \text{ L mol}^{-1})$$

$$= 2 \text{ mol} \times 0.082 \text{ L atm K}^{-1} \text{ mol}^{-1} \times 300 \text{ K}$$

$$\left(11.0 \text{ atm} + \frac{a}{4} \text{ mol}^2 \text{ L}^{-2} \right) (4 \text{ L} - 0.1 \text{ L}) = 49.2 \text{ L atm}$$

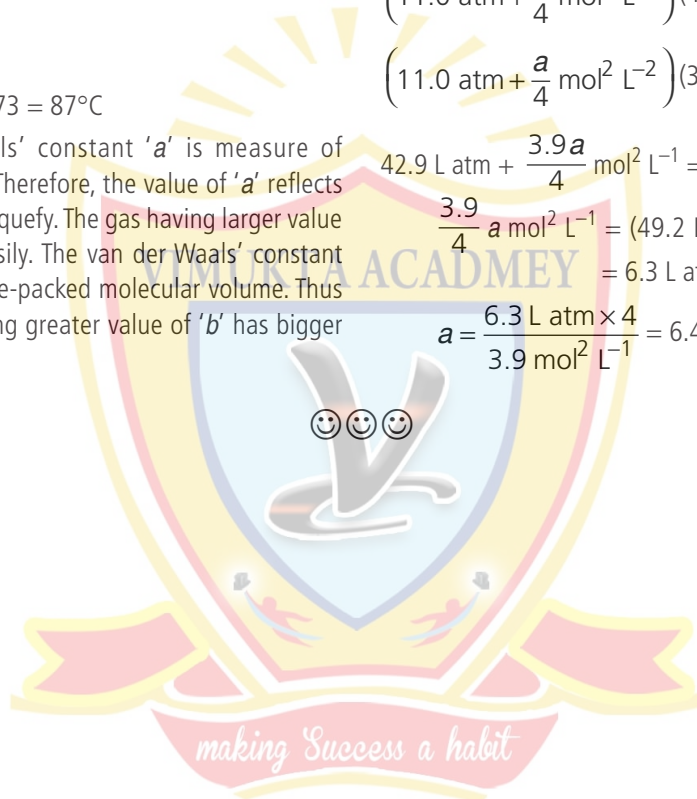
$$\left(11.0 \text{ atm} + \frac{a}{4} \text{ mol}^2 \text{ L}^{-2} \right) (3.9 \text{ L}) = 49.2 \text{ L atm}$$

$$42.9 \text{ L atm} + \frac{3.9 a}{4} \text{ mol}^2 \text{ L}^{-1} = 49.2 \text{ L atm}$$

$$\frac{3.9}{4} a \text{ mol}^2 \text{ L}^{-1} = (49.2 \text{ L atm} - 42.9 \text{ L atm})$$

$$= 6.3 \text{ L atm}$$

$$a = \frac{6.3 \text{ L atm} \times 4}{3.9 \text{ mol}^2 \text{ L}^{-1}} = 6.46 \text{ atm L}^2 \text{ mol}^{-2}$$



CHAPTER

2

Thermodynamics





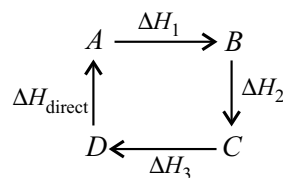
➡ Case Based MCQs

Case I : Read the passage given below and answer the following questions from 31 to 35.

Hess's Law of Constant Heat Summation

This law was presented by Hess in 1840. According to the law, if a chemical reaction can be made to take place in a number of ways in one or in several steps, the total enthalpy change is always the same.

Thus, the total enthalpy change of a chemical reaction depends on the initial and final stages only.



Total enthalpy change from A to D,

$$\Delta H_{\text{total}} = \Delta H_1 + \Delta H_2 + \Delta H_3$$

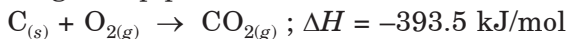
$$\Delta H_{\text{total}} + \Delta H_{\text{direct}} = 0$$

$$\text{i.e., } \Delta H_{\text{total}} = -\Delta H_{\text{direct}}$$

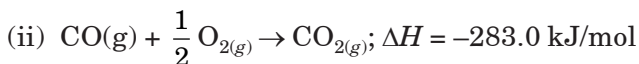
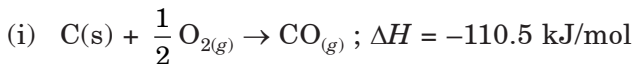
For example, formation of CO_2 from C in two

different manners involves a total heat change of -393.5 kJ/mol

Single step process :

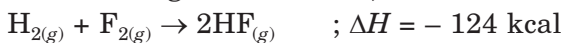


Two step process :



$$\Delta H_{\text{Total}} = -110.5 + (-283.0) = -393.5 \text{ kJ/mol}$$

31. For the given reactions,



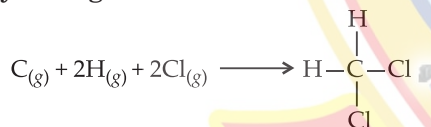
then the value of ΔH for $\text{H}_{(g)} + \text{F}_{(g)} \rightarrow \text{HF}_{(g)}$ is

- (a) 142 kcal (b) -132.9 kcal
(c) 132 kcal (d) 134 kcal.

32. In which of the enlisted cases, Hess's law is not applicable?

- (a) Determination of lattice energy
(b) Determination of resonance energy
(c) Determination of enthalpy of transformation of one allotropic form to another
(d) Determination of entropy

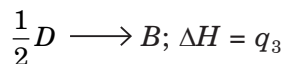
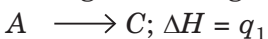
33. Use the bond energy data and calculate the enthalpy change for



The bond energies of C — H and C — Cl are 413 and 328 kJ mol^{-1} respectively.

- (a) -1465 kJ/mol (b) 1465 kJ/mol
(c) -1482 kJ/mol (d) 1482 kJ/mol

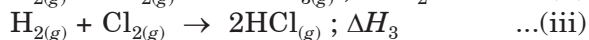
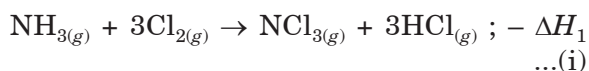
34. A hypothetical reaction, $A \longrightarrow 2B$, proceeds through following sequence of steps :



The heat of reaction is

- (a) $q_1 - q_2 + 2q_3$ (b) $q_1 + q_2 - 2q_3$
(c) $q_1 + q_2 + 2q_3$ (d) $q_1 + 2q_2 - 2q_3$

35. Given :



The heat of formation of $\text{NCl}_{3(g)}$ in the terms of ΔH_1 , ΔH_2 and ΔH_3 is

$$(a) \Delta H_f = -\Delta H_1 - \frac{\Delta H_2}{2} - \frac{3}{2}\Delta H_3$$

$$(b) \Delta H_f = \Delta H_1 + \frac{\Delta H_2}{2} - \frac{3}{2}\Delta H_3$$

$$(c) \Delta H_f = \Delta H_1 - \frac{\Delta H_2}{2} + \frac{3}{2}\Delta H_3$$

(d) none of these.

Case II : Read the passage given below and answer the following questions from 36 to 40.

A thermodynamically irreversible process is always accompanied by an increase in the entropy of the system and its surroundings taken together while in a thermodynamically reversible process, the entropy of the system and its surrounding taken together remain unchanged i.e. for reversible process

$$\Delta S_{\text{sys}} + \Delta S_{\text{surr}} = 0$$

and for irreversible process

$$\Delta S_{\text{sys}} + \Delta S_{\text{surr}} > 0$$

combining the two we have

$$\Delta S_{\text{sys}} + \Delta S_{\text{surr}} = 0$$

where 'equal' to sign refers to a reversible process while the 'greater than' sign refers to an irreversible process.

Change in entropy for an ideal gas under different conditions may be calculated as

$$\Delta S = 2.303 nR \log_{10} \frac{V_2}{V_1} = 2.303 nR \log_{10} \frac{P_1}{P_2}$$

(for isothermal process)

$$\Delta S = 2.303 nC_p \log_{10} \frac{T_2}{T_1} = 2.303 nC_p \log_{10} \frac{V_2}{V_1}$$

(for isobaric process)

$$\Delta S = 2.303 nC_v \log_{10} \frac{T_2}{T_1} = 2.303 nC_v \log_{10} \frac{P_2}{P_1}$$

36. Calculate entropy change when 5 moles of an ideal gas expands reversibly and isothermally from an initial volume of 5 litre to 50 litre at 27°C .

- (a) 190.15 JK^{-1} (b) 95.74 JK^{-1}
(c) 87.25 JK^{-1} (d) 90.13 JK^{-1} .

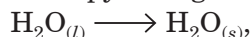
37. The entropy of the universe

- (a) increasing and tending towards maximum value
(b) decreasing and tending to be zero
(c) remains constant
(d) decreasing and increasing with a periodic rate.

38. The total entropy change for a system and its surroundings increases, if the process is
 (a) reversible (b) irreversible
 (c) exothermic (d) endothermic.

39. ΔS is positive for the change
 (a) mixing of two gases (b) boiling of liquid
 (c) melting of solid (d) all of these.

40. Entropy changes for the process,



at normal pressure and 274 K are given below

$$\Delta S_{\text{system}} = -22.13, \Delta S_{\text{surr}} = +22.05,$$

the process is non-spontaneous because ('surr' stands for surrounding and 'u' stands for universe)

- (a) ΔS_{system} is -ve (b) ΔS_{surr} is +ve
 (c) ΔS_u is -ve (d) $\Delta S_{\text{system}} \neq \Delta S_{\text{surr}}$

Case III : Read the passage given below and answer the following questions from 41 to 45.

The enthalpy of a system is defined as the sum of the internal energy of the system and the energy that arises due to its pressure and volume. Mathematically, the enthalpy is defined by the equation,

$$H = U + PV$$

Enthalpy change (ΔH) of a system is the heat absorbed or evolved by the system at constant pressure.

$$\Delta H = q_p, \Delta H = \Delta U + P\Delta V$$

41. Which of the following is not correct about enthalpy?

- (a) It is an extensive property.
 (b) It is not a state function.
 (c) Its absolute value cannot be determined.
 (d) Enthalpy of a compound is equal to enthalpy of formation of that compound.

42. In which of the following reactions will ΔU be equal to ΔH ?

- (a) $\text{H}_{2(g)} + \frac{1}{2} \text{O}_{2(g)} \longrightarrow \text{H}_2\text{O}_{(l)}$
 (b) $\text{H}_{2(g)} + \text{I}_{2(g)} \longrightarrow 2\text{HI}_{(g)}$
 (c) $\text{N}_2\text{O}_{4(g)} \longrightarrow 2\text{NO}_{2(g)}$
 (d) $2\text{SO}_{2(g)} + \text{O}_{2(g)} \longrightarrow 2\text{SO}_{3(g)}$

43. Lattice enthalpies are determined by

- (a) Born-Haber cycle (b) Hess's law
 (c) lattice cycle (d) none of these.

44. In which of the following thermochemical changes ΔH is always negative?

- (a) Enthalpy of solution
 (b) Enthalpy of hydrogenation
 (c) Enthalpy of reaction
 (d) Enthalpy of transition

45. The heat of a chemical reaction is given by the following expression

- (a) $\Delta H = \Sigma H_R - \Sigma H_P$ (b) $\Delta H = \Sigma H_R$
 (c) $\Delta H = \Sigma H_P - \Sigma H_R$ (d) $\Delta H = \Sigma H_P$

Case IV : Read the passage given below and answer the following questions from 46 to 50.

In chemical thermodynamics, the type of work involved is mostly pressure-volume work that is the work done when a system (gas) expands or contracts against an external opposing pressure. Hence, work is also defined as the transfer of energy that can be used to change the height of a mass in the surroundings.

Pressure-volume work : The work W , that is done due to the expansion or compression of a gas against an external opposing pressure P is called pressure-volume work.

$$PV = \frac{f}{A} \times V = \frac{f}{d^2} \times d^3 = f \cdot d = W$$

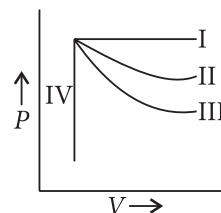
46. Which of the following statement is correct?

- (a) $\Delta U = W_{\text{adiabatic}}$
 (b) $W_{ad} = +ve$, when work is done on the system
 (c) $W_{ad} = -ve$, when work is done by the system
 (d) All of the above.

47. A process in which the system does not exchange heat with the surroundings is known as

- (a) isothermal (b) isobaric
 (c) isochoric (d) adiabatic.

48. Which of the following is correct match as far as the thermodynamic processes are involved?



Curve	Thermodynamic Process
1. I	i. Adiabatic
2. II	ii. Isochoric
3. III	iii. Isobaric
4. IV	iv. Isothermal

The correct match is

- | 1 | 2 | 3 | 4 |
|---------|----|-----|-----|
| (a) i | ii | iii | iv |
| (b) iii | iv | i | ii |
| (c) ii | i | iv | iii |
| (d) iii | iv | ii | i |

49. 3 moles of an ideal gas are expanded isothermally and reversibly from 10 m^3 to

20 m^3 at 300 K. The work done is ($R = 8.314 \text{ J K}^{-1} \text{ mol}^{-1}$).

- | | |
|---------------|---------------|
| (a) +5.187 kJ | (b) -5.187 kJ |
| (c) -2.175 kJ | (d) +3.750 kJ |

50. A gas expands in vacuum. The work done by the gas is

- | | |
|-------------|--------------------------|
| (a) zero | (b) minimum |
| (c) maximum | (d) cannot be predicted. |

Assertion & Reasoning Based MCQs

For question numbers 51-60, a statement of assertion followed by a statement of reason is given. Choose the correct answer out of the following choices.

- (a) Assertion and reason both are correct statements and reason is correct explanation for assertion.
(b) Assertion and reason both are correct statements but reason is not correct explanation for assertion.
(c) Assertion is correct statement but reason is wrong statement.
(d) Assertion is wrong statement but reason is correct statement.

51. **Assertion :** The heat absorbed during the isothermal expansion of an ideal gas against vacuum is zero.

Reason : The volume occupied by the molecules of an ideal gas is zero.

52. **Assertion :** U is a state function.

Reason : T is an intensive property.

53. **Assertion :** Sublimation of the solid is non-spontaneous.

Reason : Sublimation is endothermic process.

54. **Assertion :** The sum of $q + w$ is a state function.

Reason : Work and heat are state functions.

55. **Assertion :** Enthalpy of formation of graphite is zero but of diamond it is not zero.

Reason : Enthalpy of formation of the most stable allotrope is taken as zero.

56. **Assertion :** Decrease in free energy causes spontaneous reaction.

Reason : Spontaneous reactions are invariable exothermic reactions.

57. **Assertion :** Heat of neutralisation for both H_2SO_4 and HCl with NaOH is 53.7 kJ mol^{-1} .

Reason : Both HCl and H_2SO_4 are strong acids.

58. **Assertion :** Some salts are sparingly soluble in water at room temperature.

Reason : The entropy increases on dissolving the salts.

59. **Assertion :** Spontaneous process is an irreversible process and may be reversed by some external agency.

Reason : Decrease in enthalpy is a contributory factor for spontaneity.

60. **Assertion :** Entropy of system increases for a spontaneous reaction.

Reason : Enthalpy of reaction always decreases for spontaneous reaction.

SUBJECTIVE TYPE QUESTIONS

Very Short Answer Type Questions (VSA)

- | | |
|---|---|
| 1. State the difference between adiabatic and isothermal processes. | 3. Define enthalpy of formation. |
| 2. State first law of thermodynamics. | 4. When is bond energy equal to bond dissociation energy? |

- Under what conditions will a reaction be spontaneous if both ΔH and ΔS are negative?
- Give an example of a spontaneous process which is endothermic.
- State the second law of thermodynamics.
- State two ways by which the internal energy

of a system may be changed.

- Give one point of difference : Extensive and intensive properties.

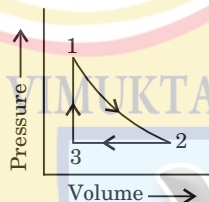
- What will be the sign of ΔS for the following reaction?



➡ Short Answer Type Questions (SA-I)

- 18.0 g of water completely vapourises at 100°C and 1 bar pressure and the enthalpy change in the process is $40.79 \text{ kJ mol}^{-1}$. What will be the enthalpy change for vapourising two moles of water under the same conditions? What is the standard enthalpy of vaporisation for water?

- A sample of 1.0 mol of a monoatomic ideal gas is taken through a cyclic process of expansion and compression as shown in figure. What will be the value of ΔH for the cycle as a whole?



- If the combustion of 1 g of graphite produces 20.7 kJ of heat, what will be molar enthalpy change? Give the significance of sign also.

- The enthalpy of reaction for the reaction : $2\text{H}_{2(g)} + \text{O}_{2(g)} \rightarrow 2\text{H}_2\text{O}_{(l)}$ is $\Delta_r H = -572 \text{ kJ mol}^{-1}$. What will be standard enthalpy of formation of $\text{H}_2\text{O}_{(l)}$?

- 10 g of argon gas is compressed isothermally and reversibly at a temperature of 27°C from

10 litre to 5 litre. Calculates q , w and ΔU for this process.

($R = 2.0 \text{ cal K}^{-1} \text{ mol}^{-1}$, atomic weight of Argon = 40)

- Calculate the entropy change in surrounding when 1.00 mol of $\text{H}_2\text{O}_{(l)}$ is formed under standard conditions.

$$\Delta_f H = -286 \text{ kJ mol}^{-1}.$$

- Justify the following statements :

- Many thermodynamically feasible reactions do not occur under ordinary conditions.
- At low temperature, enthalpy change dominates the value of ΔG and at high temperature it is the entropy which dominates the value of ΔG .

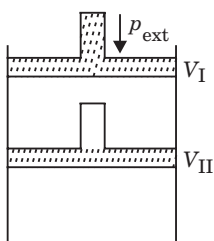
- (a) What is a thermochemical equation?
(b) Write one application of Hess's law.

- In a process, 701 J of heat is absorbed by a system and 394 J of work is done by the system. What is the change in internal energy for the process?

- Derive the relationship, $\Delta H = \Delta U + \Delta n_g RT$.

➡ Short Answer Type Questions (SA-II)

- What will be the work done on an ideal gas enclosed in a cylinder, when it is compressed by a constant external pressure, p_{ext} in a single step as shown in Figure. Explain graphically.



- Although heat is a path function but heat absorbed by the system under certain specific conditions is independent of path. What are those conditions? Explain.

- Comment on the spontaneity of a reaction at constant temperature and pressure in the following cases :

- $\Delta H < 0$ and $\Delta S > 0$
- $\Delta H > 0$ and $\Delta S < 0$
- $\Delta H > 0$ and $\Delta S > 0$

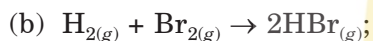
24. (i) Classify the following processes as reversible or irreversible :

- Dissolution of sodium chloride.
 - Evaporation of water at 373 K and 1 atm pressure.
 - Mixing of two gases by diffusion.
 - Melting of ice without rise in temperature.
- (ii) When an ideal gas expands in vacuum, there is neither absorption nor evolution of heat. Why?

25. (i) Explain why the enthalpy changes for the given reactions are not enthalpies of formation of CaCO_3 and HBr .

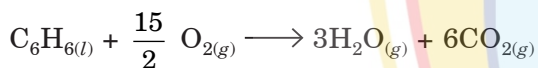


$$\Delta_r H = -178.3 \text{ kJ mol}^{-1}$$



$$\Delta_r H = -72.8 \text{ kJ mol}^{-1}$$

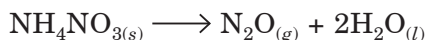
26. Benzene burns in O_2 according to the equation:



If enthalpy of formation of $\text{C}_6\text{H}_{6(l)}$, $\text{H}_2\text{O}_{(g)}$ and $\text{CO}_{2(g)}$ are 11.7, -68.3, and -94 k cal respectively, calculate the amount of heat liberated by burning 1 kg benzene.

27. The heat of solution of anhydrous CuSO_4 is -15.9 kcal and that of $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$ is 2.8 kcal. Calculate the heat of hydration of CuSO_4 .

28. The molar heat of formation of $\text{NH}_4\text{NO}_{3(s)}$ is -367.54 kJ and those of $\text{N}_2\text{O}_{(g)}$ and $\text{H}_2\text{O}_{(l)}$ are +81.46 kJ and -285.78 kJ respectively at 25 °C and 1.0 atmospheric pressure. Calculate ΔH and ΔU for the reaction,

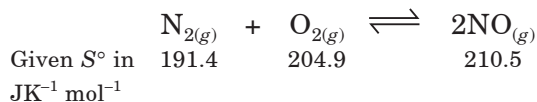


29. A heated copper block at 130 °C loses 340 J of heat to the surroundings which are at room temperature of 32 °C. Calculate

- the entropy change of the system (copper block)
- the entropy change in the surroundings
- the total entropy change in the universe due to this process.

Assume that the temperature of the block and the surroundings remains constant.

30. Assume ΔH° and ΔS° to be independent of temperature, at what temperature will the reaction given below become spontaneous?



and $\Delta H^\circ = 180.8 \text{ kJ mol}^{-1}$

31. The following data is known about ZnSO_4 : $\Delta H = 7.25 \text{ kJ mol}^{-1}$ and $\Delta S = 7.0 \text{ JK}^{-1} \text{mol}^{-1}$. Calculate its melting point.

32. Calculate the free energy change when 1 mole of NaCl is dissolved in water at 298 K.

Given:

(a) Lattice energy of $\text{NaCl} = 778 \text{ kJ mol}^{-1}$

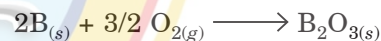
(b) Hydration energy of $\text{NaCl} = -774.3 \text{ kJ mol}^{-1}$

(c) Entropy change at 298 K = 43 J mol^{-1}

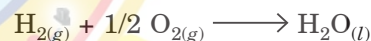
33. Diborane is a potential rocket fuel which undergoes combustion according to the reaction



From the following data, calculate the enthalpy change for the combustion of diborane.



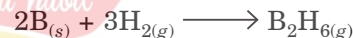
$$\Delta H = -1273 \text{ kJ mol}^{-1}$$



$$\Delta H = -286 \text{ kJ mol}^{-1}$$



$$\Delta H = 44 \text{ kJ mol}^{-1}$$



$$\Delta H = 36 \text{ kJ mol}^{-1}$$

34. The enthalpy change involved in the oxidation of glucose is -2880 kJ/mol. Twenty five percent of this energy is available for muscular work. If 100 kJ of muscular work is needed to walk one kilometer, what is the maximum distance that a person will be able to walk after consuming 125 g of glucose?

35. 10 g of argon gas is compressed isothermally and reversibly at a temperature of 27 °C from 10 L to 5 L. Calculate the value of q , w , ΔU and ΔH .

($R = 2.0 \text{ cal K}^{-1} \text{mol}^{-1}$, at. wt. of Ar = 40)

➡ Long Answer Type Questions (LA)

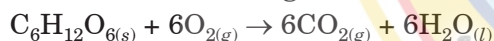
36. Answer the following :

- Why does entropy of a solid increases on fusion?
- State the thermodynamic conditions of spontaneous occurrence of a process.
- Why a non-spontaneous reaction becomes spontaneous when coupled with a suitable spontaneous reaction?
- For an isolated system, $\Delta U = 0$, what will be ΔS ?

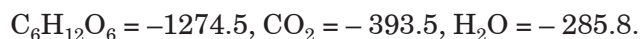
37. (a) The melting point of ice is 273 K. At this temperature the enthalpy of fusion of ice is 6.025 kJ mol⁻¹. Calculate the entropy change for melting of 1 mole of ice. State giving reason whether the entropy change for vaporization of one mole of water will be more or less than entropy change per mole for fusion of ice.

(b) Calculate the entropy change involved in the conversion of one mole of liquid water at 373 K to vapour at the same temperature (Latent heat of vaporization, $\Delta H_{\text{vap}} = 2.257$ kJ/g).

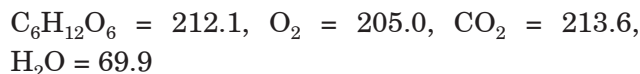
38. Calculate the standard Gibbs energy change for the combustion of α -D glucose at 298 K



Given that standard enthalpies of formation (kJ mol⁻¹)



Entropies (J K⁻¹ mol⁻¹)



39. A gas absorbs 120 J of heat and expands against the external pressure of 1.10 atm from a volume of 0.5 L to 2.0 L. What is the change in internal energy? (1 L atm = 101.3 J)

40. Whenever an acid is neutralised by a base, the net reaction is



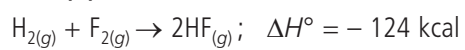
Calculate the heat evolved for the following experiments:

- 0.50 mole of HCl solution is neutralised by 0.50 mole of NaOH solution.
- 0.50 mole of HNO₃ solution is mixed with 0.30 mole of KOH solution.
- 100 mL of 0.2 M HCl is mixed with 100 mL of 0.3 M NaOH solution.
- 400 mL of 0.2 M H₂SO₄ is mixed with 600 mL of 0.1 M KOH solution.





31. (b): For the reaction,



$$\Delta H^\circ = \Sigma B.E. (\text{reactants}) - \Sigma B.E. (\text{products})$$

$$\begin{aligned} \text{or } -124 &= \Delta H_{\text{H-H}} + \Delta H_{\text{F-F}} - 2\Delta H_{\text{H-F}} \\ &= 104 + 37.8 - 2\Delta H_{\text{H-F}} \end{aligned}$$

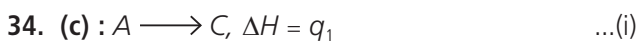
$$\therefore 2\Delta H^\circ_{\text{H-F}} = 104 + 37.8 + 124 = 265.8 \text{ kcal}$$

$$\text{Bond energy of H-F} = \frac{265.8}{2} = 132.9 \text{ kcal}$$

$$\therefore \Delta H^\circ \text{ for the given reaction} = -132.9 \text{ kcal}$$

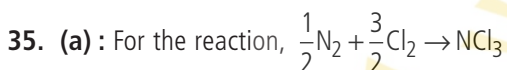
32. (d) : Hess's law cannot be used for the determination of entropy.

$$\begin{aligned} \mathbf{33. (c) :} \Delta H &= [-2 \times 413 + (-2 \times 328)] \\ &= [-826 - 656] = -1482 \text{ kJ/mol} \end{aligned}$$



Now adding (i), (ii) and (iii), we get

i.e., $q_1 + q_2 + 2q_3$, is the heat of reaction for the reaction $A \rightarrow 2B$.



$$\text{Applying: } \frac{1}{2}\text{Eqn. (ii)} + \text{Eqn. (i)} - \frac{3}{2}\text{Eqn. (iii)}$$

$$\text{We get } \Delta H_f = \frac{1}{2}(-\Delta H_2) + (-\Delta H_1) - \frac{3}{2}\Delta H_3$$

$$= -\Delta H_1 - \frac{\Delta H_2}{2} - \frac{3}{2}\Delta H_3$$

$$\begin{aligned} \mathbf{36. (b) :} \Delta S &= 2.303nR \log_{10} \frac{V_2}{V_1} \\ &= 2.303 \times 5 \times 8.314 \log_{10} \frac{50}{5} = 95.74 \text{ JK}^{-1} \end{aligned}$$

37. (a) : Entropy of universe is tending towards maximum.

38. (b) : $\Delta S > 0$ for irreversible process.

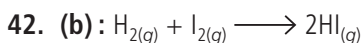
39. (d) : In every process randomness increases.

$$\begin{aligned} \mathbf{40. (c) :} \Delta S &= \Delta S_{\text{system}} + \Delta S_{\text{surrounding}} \\ &= -22.13 + 22.05 = -0.08 \end{aligned}$$

For a spontaneous process, ΔS must be positive i.e.,

$$\Delta S = \Delta S_{\text{system}} + \Delta S_{\text{surrounding}} \geq 0$$

41. (b) : Enthalpy is a state function.



$$\Delta n = 2 - (1 + 1) = 0$$

$$\Delta H = \Delta U + \Delta nRT$$

$$\Delta H = \Delta U + (0)RT$$

$$\therefore \Delta H = \Delta U$$

43. (a) : An enthalpy diagram called Born-Haber cycle is used to determine the lattice enthalpies.

44. (b) **45. (c)**

46. (d) : For an adiabatic process, $dq = 0$

$$\Delta U = W_{\text{adia}}$$

The positive sign expresses that W_{ad} is positive when work

is done on the system. If the work is done by the system, W_{ad} will be negative.

47. (d) : If a process is carried out under such condition that no exchange of heat takes place between the system and surroundings, the process is termed as adiabatic. The system is thermally isolated, i.e., $dq = 0$ and the temperature of the system varies.

48. (b)

$$\mathbf{49. (b) :} W_{\text{max}} = -2.303 nRT \log \frac{V_2}{V_1}$$

$$n = 3 \text{ mol}, R = 8.314 \text{ J K}^{-1} \text{ mol}^{-1}, T = 300 \text{ K}$$

$$V_1 = 10 \text{ m}^3, V_2 = 20 \text{ m}^3$$

$$\text{Hence, } W_{\text{max}} = -2.303 \times 3 \text{ mol} \times 8.314 \text{ J K}^{-1} \text{ mol}^{-1}$$

$$\times 300 \text{ K} \times \log \frac{20 \text{ m}^3}{10 \text{ m}^3}$$

$$= -2.303 \times 3 \times 8.314 \text{ J} \times 300 \times \log 2$$

$$= -2.303 \times 3 \times 8.314 \text{ J} \times 300 \times 0.301$$

$$= -5187 \text{ J} = -5.187 \text{ kJ}$$

The amount of work done is -5.187 kJ .

50. (a) : $W = 0$ because $P_{\text{ex}} = 0$.

51. (b)

52. (b)

53. (d) : Sublimation of the solid is spontaneous due to increase in entropy.

54. (c) : $\Delta E = q + w$

Internal energy is a state function, but not q or w .

55. (a) : Graphite is the most stable form.

56. (c) : Exothermic reactions are spontaneous at low temperature but becomes non-spontaneous at high temperature.

57. (a)

58. (b)

59. (b)

60. (a) : ΔS is +ve and ΔH is -ve for a spontaneous reaction at all temperatures.

SUBJECTIVE TYPE QUESTIONS

1. In adiabatic process no heat can flow from the system to the surroundings or vice versa, whereas, in isothermal process temperature remains constant throughout the process.

2. According to first law of thermodynamics, energy can neither be created nor be destroyed although it may be changed from one form to another.

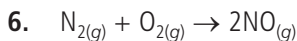
3. The enthalpy change accompanying the formation of one mole of a compound from its elements is called enthalpy of formation.

4. When bond energy is measured in isolated gaseous state, it becomes equal to bond dissociation energy.

5. $\Delta G = \Delta H - T\Delta S$

Given, $\Delta H = -ve$ and $\Delta S = -ve$

For a reaction to be spontaneous, ΔG should be negative, reaction takes place spontaneously only at lower temperature.



It is endothermic as well as spontaneous.

7. Whenever a spontaneous process takes place, it is always accompanied by an increase in total entropy of the universe.

$$\Delta S_{univ} = \Delta S_{sys} + \Delta S_{surr} > 0$$

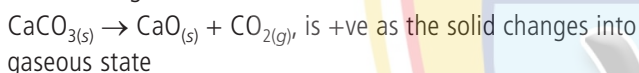
8. (i) If the work is done by or on the system.

(ii) Heat is absorbed by the system or heat is evolved from the system.

9. Extensive properties: Properties that depend on the quantity of matter contained in the system, e.g., mass, volume, etc.

Intensive properties: Properties which depend on the nature of the substance and not on the amount of substance, e.g., viscosity, etc.

10. The sign of ΔS for the reaction,



11. Enthalpy of a reaction is the energy change per mole for the process.

18 g of $H_2O = 1$ mole ($\Delta H_{vap} = 40.79 \text{ kJ mol}^{-1}$)

For 2 moles of H_2O , enthalpy of vaporisation = 2×40.79
= 81.58 kJ

$$\Delta H_{vap}^\circ = 40.79 \text{ kJ mol}^{-1}$$

12. For a cyclic process $\Delta H = 0$

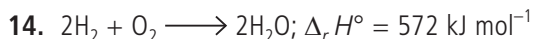
13. Molar enthalpy change for graphite (ΔH)

= enthalpy change for 1 g $\times$ molar mass of C

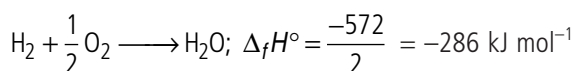
$$= -20.7 \times 12$$

$$= -2.48 \times 10^2 \text{ kJ mol}^{-1}$$

Since the sign of $\Delta H = -ve$, it is an exothermic reaction.



$\Delta_f H^\circ$ will be half of the enthalpy of the given equation as enthalpy of formation is the enthalpy change of the reaction when 1 mole of the compound is formed from its elements.



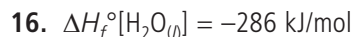
15. $W_{iso, rev} = -2.303nRT \log \frac{V_2}{V_1} = -2.303nRT \log \frac{5}{10}$

$$= -2.303 \times \frac{10}{40} \times 2 \times 300 \log \frac{5}{10} = 103.9 \text{ cal}$$

For isothermal process $\Delta U = 0$

From 1st law of thermodynamics, $\Delta U = q + W$

$$\therefore q = -W = -103.9 \text{ cal}$$



$$\therefore q_{surroundings} = +286 \text{ kJ/mol}$$

$$\Delta S_{(surr)} = \frac{q}{298} = \frac{286 \times 1000}{298} \text{ JK}^{-1} \text{ mol}^{-1} = 959.73 \text{ JK}^{-1} \text{ mol}^{-1}$$

17. (a) Under ordinary conditions, the average energy of the reactants may be less than the threshold energy. They require some activation energy to initiate the reaction.

(b) $\Delta G = \Delta H - T\Delta S$

At lower temperature, if value of ΔH is negative ΔG will be -ve and if ΔH is positive, ΔG will be positive. While at higher temperature $-T\Delta S$ will be high, thus, sign of ΔS will decide whether ΔG will be positive or negative.

18. (a) The balanced chemical equation which includes the amount of heat evolved or absorbed during the reaction is called a thermochemical equation.

(b) Hess's law is found very useful in calculating the enthalpy change for the reaction for which experimental determination is not possible.

19. Heat absorbed by the system (q) = 701 J

Work done by the system (w) = -394 J

According to first law of thermodynamics,

$$\Delta U = q + w = 701 + (-394) = 701 - 394 = 307 \text{ J}$$

20. The enthalpy H can be written as :

$$H = U + pV \quad \dots(i)$$

For finite changes at constant pressure, we can write equation (i) as

$$\Delta H = \Delta U + \Delta(pV)$$

Since p is constant, we can write

$$\Delta H = \Delta U + p\Delta V \quad \dots(ii)$$

Let us consider a reaction involving gases. If V_A is the total volume of the gaseous reactants, V_B is the total volume of the gaseous products, n_A is the number of moles of gaseous reactants and n_B is the number of moles of gaseous products, all at constant pressure and temperature, then using the ideal gas law, we write,

$$pV_A = n_A RT$$

and $pV_B = n_B RT$

$$\text{Thus, } pV_B - pV_A = n_B RT - n_A RT = (n_B - n_A) RT$$

$$\text{or } p(V_B - V_A) = (n_B - n_A) RT$$

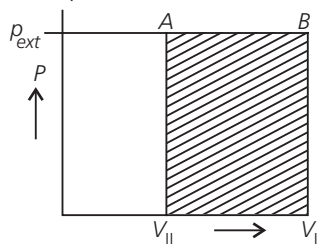
$$\text{or } p\Delta V = \Delta n_g RT \quad \dots(iii)$$

where, $\Delta n_g = n_{\text{gaseous products}} - n_{\text{gaseous reactants}}$

Substituting the value of $p\Delta V$ from equation (iii) in equation (ii), we get

$$\Delta H = \Delta U + \Delta n_g RT$$

21. Work done is equal to the shaded area $ABV_{II}V_I$



22. $q = \Delta U + (-w)$

$-w = p\Delta V$

$\therefore q = \Delta U + p\Delta V$

At constant volumes $\Delta V = 0$

Hence, $q_V = \Delta U + 0 = \Delta U$

At constant pressure,

$q_p = \Delta U + p\Delta V$

Since $\Delta U + p\Delta V = \Delta H$

$\Rightarrow q_p = \Delta H$

Hence, at constant volume and at constant pressure heat change is a state function because it is equal to ΔU and ΔH respectively which are state functions.

23. (i) Both energy factor and randomness factor favour the process. Hence, reaction will always be spontaneous.

(ii) Both factor opposes the process. Hence, reaction would always be non-spontaneous.

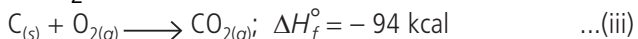
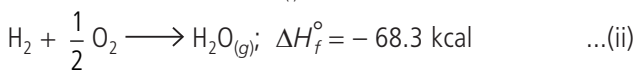
(iii) Energy factor opposes but randomness factor favours. For spontaneity, $T\Delta S > \Delta H$. Hence, reaction is spontaneous at high temperature and non-spontaneous at low temperature.

24. (i) (a) Reversible (b) Reversible
(c) Irreversible (d) Reversible.

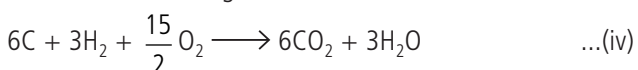
(ii) In an ideal gas, there are no intermolecular forces of attraction. Hence, no energy is required to overcome these forces. Moreover, when a gas expands against vacuum, work done is zero as $P_{\text{ext}} = 0$. Hence, internal energy of the system does not change i.e., there is no absorption or evolution of heat.

25. (i) (a) Given enthalpy change is not enthalpy of formation of CaCO_3 because it is not being formed from constituting elements.

(b) Given enthalpy change is not enthalpy of formation of HBr because 2 moles of HBr are being formed.

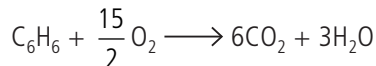


To get required equation, multiply eq. (ii) by 3 and eq. (iii) by 6 and then add; we get:



$\Delta H^\circ = -768.9 \text{ k cal}$

Subtract eq. (i) from (iv), we get:



$\Delta H^\circ = -780.6 \text{ k cal}$

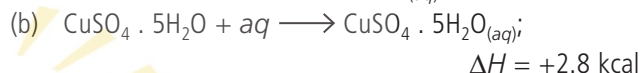
78 g C_6H_6 gives heat = 780.6 k cal

$\therefore 1000 \text{ g } \text{C}_6\text{H}_6 \text{ will give heat} = \frac{780.6 \times 1000}{78}$

$= 10067.7 \text{ kcal/mol}$

Heat liberated by burning 1 kg benzene = 10067.7 k cal/mole

27. Given:

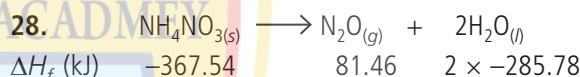


subtracting eqn. (b) from (a), we get:



$\Delta H = -15.9 - 2.8 = -18.7 \text{ k cal}$

Heat of hydration of $\text{CuSO}_4 = -18.7 \text{ k cal}$



$\Delta H_f^\circ (\text{kJ}) \quad -367.54 \quad 81.46 \quad 2 \times -285.78$

$\Delta H = \Delta H_f^\circ (\text{Products}) - \Delta H_f^\circ (\text{Reactants})$

$= [2 \times (-285.78) + 81.46] - (-367.54)$

$= -122.56 \text{ kJ} = -122.56 \times 10^3 \text{ J}$

$\Delta n_g = n_p - n_r = 1 - 0 = 1$

$\Delta H = \Delta U + \Delta n_g RT \longrightarrow \Delta U = \Delta H - \Delta n_g RT$

$\Delta U = -122.56 \times 10^3 - 1 \times 8.314 \times 298$

$= -125.04 \times 10^3 \text{ J} = -125.04 \text{ kJ}$

29. $T_{\text{system}} = 130^\circ\text{C} = 130 + 273 \text{ K} = 403 \text{ K}$

$T_{\text{surr}} = 32^\circ\text{C} = 32 + 273 \text{ K} = 305 \text{ K}$

$q_{\text{system}} = -340 \text{ J}, q_{\text{surr}} = +340 \text{ J}$

(i) $\Delta S_{\text{system}} = \frac{q_{\text{system}}}{T_{\text{system}}} = \frac{-340 \text{ J}}{403 \text{ K}} = -0.84 \text{ J K}^{-1}$

(ii) $\Delta S_{\text{surr}} = \frac{q_{\text{surr}}}{T_{\text{surr}}} = \frac{+340 \text{ J}}{305 \text{ K}} = +1.11 \text{ J K}^{-1}$

(iii) $\Delta S_{\text{total}} \text{ or } \Delta S_{\text{universe}} = \Delta S_{\text{system}} + \Delta S_{\text{surr}}$
 $= -0.84 + (+1.11) \text{ J K}^{-1} = 0.27 \text{ J K}^{-1}$

30. As $\Delta S^\circ = S^\circ_{\text{Product}} - S^\circ_{\text{Reactant}}$

$\Delta S^\circ = (2S^\circ_{\text{NO}}) - (S^\circ_{\text{N}_2} + S^\circ_{\text{O}_2})$

$= 2 \times 210.5 - (191.4 + 204.9)$

$= 24.7 \text{ J K}^{-1} \text{ mol}^{-1} = 24.7 \times 10^{-3} \text{ kJ K}^{-1} \text{ mol}^{-1}$

Since $\Delta G^\circ = \Delta H^\circ - T\Delta S^\circ$

$\Delta G^\circ = 180.8 - (T \times 24.7 \times 10^{-3}) \text{ kJ mol}^{-1}$

for a spontaneous process ΔG° should be -ve which is possible if

$$T\Delta S^\circ > \Delta H^\circ; T > \frac{\Delta H^\circ}{\Delta S^\circ}$$

$$T > \frac{180.8 \times 10^3}{24.7} = 7319 \text{ K}$$

The reaction becomes spontaneous above the temperature of 7320 K.

$$31. \Delta S_{(\text{fus})} = \frac{\Delta H_{(\text{fus})}}{T_f} \text{ or } T_f = \frac{\Delta H_{(\text{fus})}}{\Delta S_{(\text{fus})}}$$

$$\Delta H_{(\text{fus})} = 7.25 \text{ kJ mol}^{-1} = 7.25 \times 1000 = 7250 \text{ J mol}^{-1}$$

$$\Delta S_{(\text{fus})} = 7.0 \text{ J K}^{-1} \text{ mol}^{-1}$$

$$\therefore T_f = \frac{(7250 \text{ J mol}^{-1})}{(7.0 \text{ J K}^{-1} \text{ mol}^{-1})} = 1035.7 \text{ K}$$

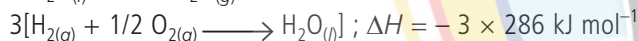
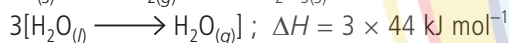
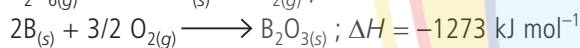
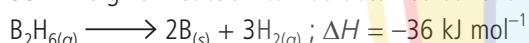
$$32. \Delta H_{\text{dissolution}} = \Delta H_{\text{lattice}} + \Delta H_{\text{hydration}} = 778 - 774.3 \\ = 3.7 \text{ kJ mol}^{-1} = 3700 \text{ J mol}^{-1}$$

$$\Delta S_{\text{dissolution}} = 43 \text{ J mol}^{-1}$$

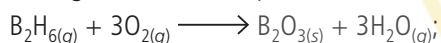
$$\therefore \Delta G_{\text{dissolution}} = \Delta H - T\Delta S \\ = 3700 - 298 \times 43 = -9114 \text{ J}$$

$$\text{or } \Delta G_{\text{dissolution}} = -9.114 \text{ kJ.}$$

33. The given reaction can be obtained as follows:



Adding all the above equations,



$$\Delta H = -2035 \text{ kJ mol}^{-1}$$

34. Energy available for muscular work from

$$1 \text{ mole of glucose} = \frac{2880 \times 25}{100} = 720 \text{ kJ/mole}$$

Thus 180 g of glucose (mol. wt. of glucose) supplies 720 kJ energy.

$$125 \text{ g of glucose will supply} = \frac{720}{180} \times 125 = 500 \text{ kJ}$$

100 kJ is needed to walk 1 km

$$500 \text{ kJ is needed to walk } \frac{1}{100} \times 500 = 5.0 \text{ km}$$

$$35. w = -2.303nRT \log \frac{V_2}{V_1}$$

$$= -2.303 \times \frac{10}{40} \times 2 \times 300 \times \log \frac{5}{10}$$

$$w = 103.991 \text{ cal}$$

$$\Delta U = 0, q = \Delta U - w$$

$$\therefore q = -w = -103.991 \text{ cal}$$

When temperature is constant,

$$P_1V_1 = P_2V_2 \text{ or } PV = \text{constant}$$

$$\Delta H = \Delta U + \Delta(PV) = 0 + 0 = 0.$$

36. (i) In a solid, the constituent particles are fixed. On melting, they fall apart and are free to move, i.e., their randomness increases.

(ii) For spontaneous occurrence, ΔG of the process must be < 0 , i.e., -ve. This can be so under the following conditions :

(a) ΔH is negative and ΔS is positive (at any temperature).

(b) If ΔH and ΔS both are positive, then T should be so high that $T\Delta S < \Delta H$.

(c) If both ΔH and ΔS are negative, then T should be so low that $T\Delta S < \Delta H$.

(iii) The overall free energy change of the coupled reaction is negative.

(iv) When energy factor has no role to play, for the process to be spontaneous ΔS must be +ve i.e., $\Delta S > 0$.

$$37. (a) \Delta S_{(\text{fusion})} = \frac{\Delta H_{(\text{fusion})}}{T} = \frac{6025}{273} = 22.07 \text{ J K}^{-1} \text{ mol}^{-1}.$$

The entropy change for the conversion of 1 mole of liquid water to steam at the boiling point will be more as compared to the value at the freezing point, because in the vaporisation randomness increases (due to the conversion of liquid into vapour) much more than in fusion.

(b) Latent heat of vaporisation per mole

$$= 2.257 \times 10^3 \times 18 = 40,626 \text{ J mol}^{-1}$$

$$\Delta S_{(\text{vap})} = \frac{\Delta H_{(\text{vap})}}{T} = \frac{40626}{373} = 108.9 \text{ J K}^{-1} \text{ mol}^{-1}$$

$$38. \text{As } \Delta G^\circ = \Delta H^\circ - T\Delta S^\circ$$

$$\Delta H^\circ_f = \Delta H^\circ_{f(\text{product})} - \Delta H^\circ_{f(\text{reactant})}$$

$$\therefore \Delta H^\circ = [6\Delta H^\circ_f(\text{CO}_2) + 6\Delta H^\circ_f(\text{H}_2\text{O})] \\ - [\Delta H^\circ_f(\text{C}_6\text{H}_{12}\text{O}_6) + 6\Delta H^\circ_f(\text{O}_2)]$$

$$= [6 \times (-393.5) + 6 \times (-285.8)] - [(-1274.5) + 6 \times 0] \\ [\Delta H^\circ_f(\text{O}_2) = 0]$$

$$= -2361.0 - 1714.8 + 1274.5 = -2801.3 \text{ kJ mol}^{-1}$$

$$\text{Similarly } \Delta S^\circ = S^\circ_{(\text{product})} - S^\circ_{(\text{reactant})}$$

$$\Delta S^\circ = [6S^\circ(\text{CO}_2) + 6S^\circ(\text{H}_2\text{O})] - [S^\circ(\text{C}_6\text{H}_{12}\text{O}_6) + S^\circ(\text{O}_2)] \\ = [6 \times (213.6) + 6 \times (69.9)] - [(212.1) + 6 \times (205.0)] \\ = [1281.6 + 419.4] - [212.1 + 1230] \\ = 258.9 \text{ J K}^{-1} \text{ mol}^{-1}$$

$$\Delta G^\circ = \Delta H^\circ - T\Delta S^\circ$$

$$\Delta H^\circ = -2801.3 \text{ kJ} = -2801300 \text{ J mol}^{-1}$$

$$\Delta S^\circ = 258.9 \text{ J K}^{-1} \text{ mol}^{-1}$$

$$T = 298 \text{ K}$$

$$\therefore \Delta G^\circ = -2801300 - 298 \times (258.9) \\ = -2878.4 \text{ kJ mol}^{-1}$$

39. Work of expansion,

$$W = -P\Delta V$$

$$P\Delta V = -1.10 \times (2.0 - 0.5) = -1.10 \times 1.5$$

$$= -1.650 \text{ L atm}$$

$$= -1.650 \times 101.3 \text{ J} \quad (1 \text{ L atm} = 101.3 \text{ J})$$

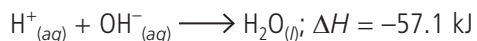
$$= -167.1 \text{ J}$$

Since work is done by the system, $w = -167.1 \text{ J}$

Heat absorbed by the system = 120 J or $q = +120 \text{ J}$

Now, $\Delta E = q + w = 120 + (-167.1) = -47.1 \text{ J}$

40. According to reaction :



When 1 mole of H^+ ions and 1 mole of OH^- ions are neutralised, 1 mole of water is formed and 57.1 kJ of energy is released.

(i) 0.50 mole HCl $\equiv$ 0.50 mole H^+ ions

0.50 mole NaOH $\equiv$ 0.50 mole OH^- ions

On mixing, 0.50 mole of water is formed.

Heat evolved for the formation of 0.50 mole of water

$$= 57.1 \times 0.5 = 28.55 \text{ kJ}$$

(ii) 0.50 mole $\text{HNO}_3 \equiv$ 0.50 mole H^+ ions

0.30 mole KOH $\equiv$ 0.30 mole OH^- ions

i.e., 0.30 mole of H^+ ions react with 0.30 mole of OH^- ions to form 0.30 mole of water molecules.

Heat evolved in the formation of 0.3 mole of water

$$= 57.1 \times 0.3 = 17.13 \text{ kJ}$$

(iii) 100 mL of 0.2 M HCl will give

$$100 \times \frac{0.2}{1000} = 0.02 \text{ mole of } \text{H}^+ \text{ ions and 100 mL of 0.3 M}$$

$$\text{NaOH will give } \frac{0.3}{1000} \times 100 = 0.03 \text{ mole of } \text{OH}^- \text{ ions}$$

i.e., 0.02 mole of H^+ ions react with 0.02 mole of OH^- ions to form 0.02 mole of water molecules.

Heat evolved in the formation of 0.02 mole of water

$$= 0.02 \times 57.1 = 1.142 \text{ kJ}$$

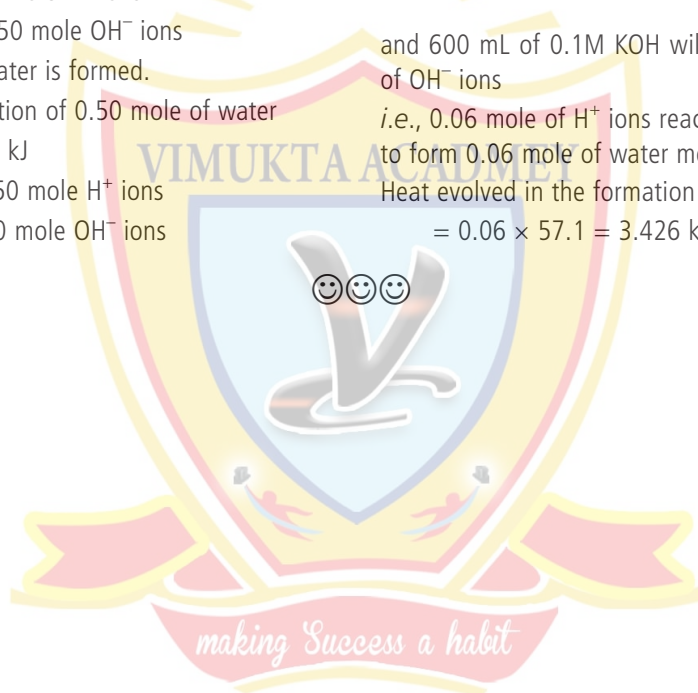
(iv) 400 mL of 0.2M H_2SO_4 will give $\frac{400 \times 0.2}{1000} = 0.08$ mole of H^+ ions

and 600 mL of 0.1M KOH will give $\frac{600 \times 0.1}{1000} = 0.06$ mole of OH^- ions

i.e., 0.06 mole of H^+ ions react with 0.06 mole of OH^- ions to form 0.06 mole of water molecules.

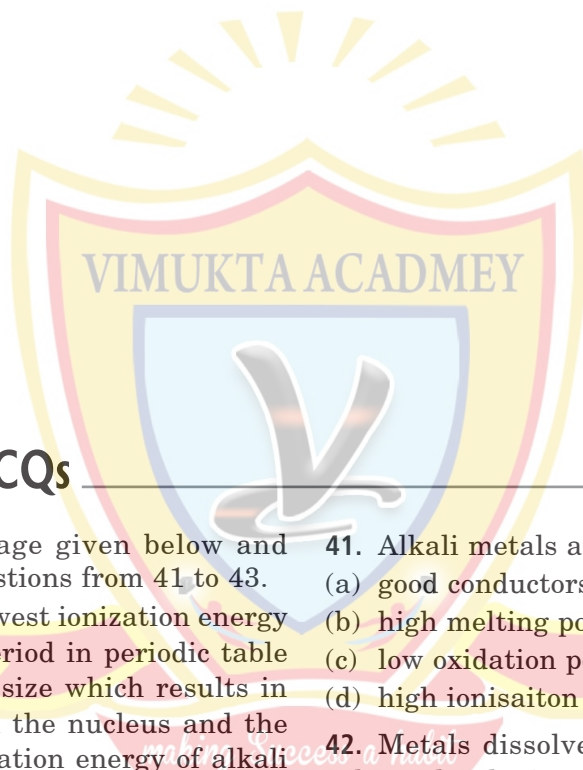
Heat evolved in the formation of 0.06 mole of water

$$= 0.06 \times 57.1 = 3.426 \text{ kJ.}$$



The s-Block Elements

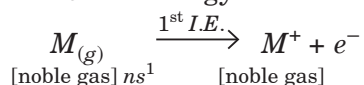




➔ Case Based MCQs

Case I : Read the passage given below and answer the following questions from 41 to 43.

Alkali metals have the lowest ionization energy in their corresponding period in periodic table because they have large size which results in a large distance between the nucleus and the outermost electron. Ionization energy of alkali metals decreases from Li to Cs due to increase in atomic size. First ionization energy of alkali metals is very low but they have very high value of second ionization energy.



Metal	Ionization Energy (kJ mol ⁻¹)	
	IE ₁	IE ₂
Li	520.1	7296
Na	495.7	4563
K	418.6	3051
Rb	402.9	2633
Cs	375.6	2230

41. Alkali metals are characterised by
- good conductors of heat and electricity
 - high melting points
 - low oxidation potentials
 - high ionisation potentials.

42. Metals dissolve in liquid ammonia giving coloured solutions which are conducting in nature. The colour of the solution and reason of its conductance is
- yellow, NH₄⁺
 - blue, ammoniated metals
 - orange, [M(NH₃)_x]⁺
 - blue, ammoniated electron.

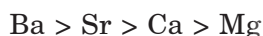
43. Alkali metals displace hydrogen from water forming bases due to the reason that
- they are far above the hydrogen in electrochemical series based on oxidation potential
 - they are far below the hydrogen in electrochemical series based on oxidation potential

- (c) their ionization potential is less than that of other elements.
- (d) they contain only one electron in their outermost shell.

Case II : Read the passage given below and answer the following questions from 44 to 46.

Alkaline earth metals are less reactive with water as compared to alkali metals. Their reactivity with water increases down the group. Be does not react with water at all, magnesium reacts only with hot water while other metals Ca, Sr and Ba react with cold water.

Order of the reactivity with water :



$\text{Be}(\text{OH})_2$ is amphoteric, but the hydroxides of Mg, Ca, Sr and Ba are basic. The basic strength increases from Mg to Ba.

$\text{Be}(\text{OH})_2$	$\text{Mg}(\text{OH})_2$	$\text{Ca}(\text{OH})_2$	$\text{Sr}(\text{OH})_2$	$\text{Ba}(\text{OH})_2$
Amphoteric	Weakly basic	Basic	Strongly basic	Strongly basic

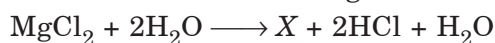
44. Which of the following statements is false?

- (a) Strontium decomposes water readily than beryllium.
- (b) Barium carbonate melts at a higher temperature than calcium carbonate.
- (c) Barium hydroxide is more soluble in water than magnesium hydroxide.
- (d) Beryllium hydroxide is more basic than barium hydroxide.

45. Chemical compound 'A' is used as flocculant in water and sewage treatment. It reacts with Na_2CO_3 to generate caustic soda. When CO_2 is bubbled through 'A', it turns cloudy. What is the chemical formula of 'A'?

- (a) CaCO_3 (b) CaO
- (c) $\text{Ca}(\text{OH})_2$ (d) $\text{Ca}(\text{HCO}_3)_2$

46. What is X in the following reaction?

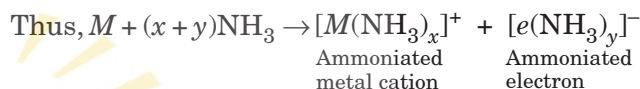
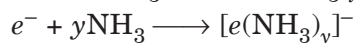
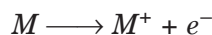


- (a) MgO
- (b) Mg
- (c) $\text{Mg}(\text{OH})_2$
- (d) $\text{Mg}(\text{OH})\text{Cl}$

Case III : Read the passage given below and answer the following questions from 47 to 50.

All alkali metals dissolve and form blue solution in liquid ammonia. When alkali metals

are dissolved in liquid ammonia, there is a considerable expansion in total volume hence such solutions are called expanded metals. The blue solution of an alkali metal in ammonia shows certain characteristic properties which are explained on the basis of formation of ammoniated (solvated) metal cations and ammoniated electrons in the metal ammonia solution in the following way :



The blue solution is paramagnetic and has high electrical conductivity due to the presence of unpaired electron in the cavities in ammoniacal solution.

47. A metal M reacts with N_2 to give a compound 'A' (M_3N). 'A' on heating at high temperature gives back 'M' and 'A' on reacting with H_2O gives a gas 'B'. 'B' intensifies colour of CuSO_4 solution when passes through it. M and B can be

- (a) Mg and NH_3
- (b) Na and NH_3
- (c) Li and NH_3
- (d) Al and NH_3

48. Sodium dissolves in liquid NH_3 to give a deep blue solution. This is due to

- (a) ammoniated Na^+
- (b) ammoniated Na^-
- (c) formation of Na^+/Na^- pair
- (d) ammoniated electrons.

49. The increasing order of the density of alkali metals is

- (a) $\text{Li} < \text{K} < \text{Na} < \text{Rb} < \text{Cs}$
- (b) $\text{Li} < \text{Na} < \text{K} < \text{Rb} < \text{Cs}$
- (c) $\text{Cs} < \text{Rb} < \text{Na} < \text{K} < \text{Li}$
- (d) $\text{Cs} < \text{Rb} < \text{K} < \text{Na} < \text{Li}$

50. The reaction between sodium and water can be made less vigorous by

- (a) lowering the temperature
- (b) adding a little alcohol
- (c) amalgamating sodium
- (d) adding a little acetic acid.

➡ Assertion & Reasoning Based MCQs

For question numbers 51-60, a statement of assertion followed by a statement of reason is given. Choose the correct answer out of the following choices.

- (a) Assertion and reason both are correct statements and reason is correct explanation for assertion.
- (b) Assertion and reason both are correct statements but reason is not correct explanation for assertion.
- (c) Assertion is correct statement but reason is wrong statement.
- (d) Assertion is wrong statement but reason is correct statement.

51. Assertion : The carbonate of lithium decomposes easily on heating to form lithium oxide and CO_2 .

Reason : Lithium being very small in size polarises large carbonate ion leading to the formation of more stable Li_2O and CO_2 .

52. Assertion : Generally alkali and alkaline earth metals form superoxides.

Reason : There is a single bond between O and O in superoxides.

53. Assertion : BeSO_4 is soluble in water while BaSO_4 is not.

Reason : Hydration energy decreases down the group from Be to Ba and lattice energy remains almost constant.

54. Assertion : Lithium salts are hydrated.

Reason : Lithium has higher polarising power than other alkali metal group members.

55. Assertion : Beryllium carbonate is kept in the atmosphere of carbon dioxide.

Reason : Beryllium carbonate is unstable and decomposes to give beryllium oxide and carbon dioxide.

56. Assertion : s -block elements are highly electropositive.

Reason : The valence electrons present in s -orbital are loosely held.

57. Assertion : The ionization enthalpies of alkali metals increase progressively as we move down the group from Li to Cs.

Reason : The ionic radii increase as we move down the group from Li to Cs.

58. Assertion : The solubility of the alkaline earth metal hydroxides in water increases with increase in atomic number down the group.

Reason : The alkaline earth metal hydroxides are basic in character. The only exception is $\text{Be}(\text{OH})_2$ which is amphoteric.

59. Assertion : Electrode potential is a measure of the tendency of an element to lose electrons in the aqueous solution. So, the reducing property can be correlated in terms of electrode potentials (E°) of alkali metals.

Reason : More negative is the electrode potential, higher is the tendency of the element to lose electrons and hence, stronger is the reducing agent.

60. Assertion : The first element (Li) of group I differs considerably from the rest of the elements of the same group.

Reason : Lithium has small ionic and atomic radii, high electronegativity and ionization enthalpy, high polarising power of its cation and absence of d -electrons in its valence shell compared to rest of the elements of the same group.

SUBJECTIVE TYPE QUESTIONS

➡ Very Short Answer Type Questions (VSA)

1. Which has a higher melting point, sodium or potassium?
2. Why are the first ionization enthalpies of group 1 elements low?
3. Why are group 2 elements harder than group 1 elements?
4. BeCl_2 in aqueous solution exists as $[\text{Be}(\text{OH})_4]^{2+}$. Explain.

5. Find out the oxidation state of sodium in Na_2O_2 .
6. Halides of Be dissolve in organic solvents while those of Ba do not. Why is it so?
7. What will be formed when lithium nitrate is heated?
8. Why are lithium salts commonly hydrated and those of the other alkali ions are usually anhydrous?
9. Explain why Cs is used in photoelectric cell?
10. Potassium carbonate cannot be prepared by Solvay process. Why ?

➡ Short Answer Type Questions (SA-I)

11. On combustion Li forms Li_2O ; sodium gives the peroxide, Na_2O_2 ; and potassium, rubidium and caesium give superoxides, MO_2 . Why Li does not form a peroxide?
12. Complete the reactions :
(a) $\text{Be}(\text{OH})_2 + 2\text{OH}^- \longrightarrow$
(b) $\text{BaO} + \text{H}_2\text{O} \longrightarrow$
13. (i) Why are alkali metal used in photo-electric cells?
(ii) What happens when K burns in air? Write equation.
14. What happens when :
(a) sodium metal is dropped in water?
(b) sodium metal is dissolved in liquid ammonia?
15. Why is Li_2CO_3 decomposed at a lower temperature whereas Na_2CO_3 at higher temperature?
16. What happens when :
(i) sodium peroxide dissolved in hot water?
(ii) sodium metal is heated in free supply of air?
17. Draw the structure of (i) BeCl_2 (vapour) (ii) BeCl_2 (solid).
18. (i) Name the chief factors responsible for the anomalous behaviour of lithium.
(ii) Complete the following reactions :
(a) $4\text{LiNO}_3 \xrightarrow{\Delta}$ (b) $2\text{NaNO}_3 \xrightarrow{\Delta}$
19. Account for the following:
(i) Magnesium does not show any flame colouration.
(ii) Group 1 elements have low melting and boiling points.
20. Alkaline earth metals always form divalent cations even though their second ionization enthalpies are almost double than their first ionization enthalpies. Explain.

➡ Short Answer Type Questions (SA-II)

21. Compare the solubility and thermal stability of the following compounds of the alkali metals with those of the alkaline earth metals.
(a) Nitrates (b) Carbonates
(c) Sulphates
22. (a) Write balanced equations for the reactions:
(i) alkali metal and water
(ii) alkali metal and dihydrogen.
(b) The E for Cl_2/Cl^- is +1.36, for I_2/I^- is +0.53, for Ag^+/Ag is +0.79, for Na^+/Na is -2.71 and for Li^+/Li is -3.04. Arrange the following ionic species in decreasing order of reducing strength:
 I^- , Ag , Cl^- , Li , Na
23. Compare the alkali metals and alkaline earth metals with respect to (i) ionisation enthalpy (ii) basicity of oxides and (iii) solubility of hydroxides.
24. Give reason :
(i) Alkali metals are strong reducing agents
(ii) Alkali metals are soft metals
(iii) Alkali metals tarnish in air easily.
25. Complete the reactions:
(i) $\text{Na} + \text{O}_2 \longrightarrow$
(ii) $\text{BeCl}_2 + \text{LiAlH}_4 \longrightarrow$
(iii) $\text{NaNO}_3 \xrightarrow{\Delta}$
26. Explain term Diagonal Relationship in context of Be and Al. Give two points of similarities between them.
27. Give reason:
(a) Sodium is less reactive than potassium.
(b) Be and Mg do not give colour to the flame whereas other alkaline earth metals do so.
(c) Sodium is stored under kerosene oil.

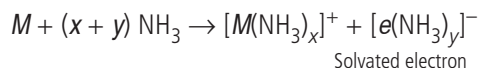
28. An element of group 2 forms covalent oxide which is amphoteric in nature and dissolves in water to give an amphoteric hydroxide. Identify the element and write chemical reactions of the hydroxide of the element with an alkali and an acid.
29. In what ways lithium shows similarities to magnesium in its chemical behaviour?
30. Arrange the following in order of property mentioned against each :
- BeCl_2 , MgCl_2 , CaCl_2 , BaCl_2 [Increasing ionic character]
 - $\text{Mg}(\text{OH})_2$, $\text{Ca}(\text{OH})_2$, $\text{Ba}(\text{OH})_2$, $\text{Sr}(\text{OH})_2$ [Increasing solubility in water]
31. Given reason:
- Alkali metals impart colours to Bunsen Burner flame.
 - KO_2 is paramagnetic.
 - Alkaline earth metals give blue solutions, when dissolved in liquid ammonia.
32. (a) Among the given alkali metals, which has least melting point? Na, K, Rb, Cs
(b) Which one of the following alkali metals gives hydrated salts? Li, Na, K, Cs
33. How would you explain the following observations?
- BeO is almost insoluble but BeSO_4 is soluble in water,
 - LiI is more soluble than KI in ethanol.
34. Comment on the following observations :
- Why are alkali metals not found in nature?
 - The mobilities of the alkali metal ions in aqueous solution are $\text{Li}^+ < \text{Na}^+ < \text{K}^+ < \text{Rb}^+ < \text{Cs}^+$
35. Discuss the general characteristics and gradation in properties of alkaline earth metals.

➡ Long Answer Type Questions (LA)

36. Present a comparative account of the alkali and alkaline earth metals with respect to the following characteristics :
- Tendency to form ionic/ covalent compounds
 - Nature of oxides and their solubility in water
 - Formation of oxosalts
 - Solubility of oxosalts
 - Thermal stability of oxosalts
37. (a) (i) What are the products formed when alkali metal oxide (M_2O), peroxide (M_2O_2) and superoxide (MO_2) hydrolysed by water?
(ii) Why sodium forms peroxide but potassium forms superoxide?
- (b) (i) Can we store sodium in water or not? Why.
(ii) Write balanced equations for the reactions between :
- Na_2O_2 and H_2O
 - KO_2 and H_2O .
38. (a) Explain why can alkali and alkaline earth metals not be obtained by chemical reduction methods.
(b) Explain the following :
- Be and Mg do not impart any colour to the flame.
 - What happens when Mg is burnt in air?
 - Account for the following:
 - Magnesium does not show any flame colouration.
 - Group 1 elements have low melting and boiling points.
39. The s-block elements are characterised by their larger atomic sizes, lower ionisation enthalpies, invariable +1 oxidation state and solubilities of their oxosalts. In the light of these features describe the nature of their oxides, halides and oxosalts.
40. (a) When a metal of group 1 was dissolved in liquid ammonia, the following observations were obtained:
- Blue solution was obtained initially.
 - On concentrating the solution, blue colour changed to bronze colour.
- How do you account for the blue colour of the solution? Give the name of the product formed on keeping the solution for some time.
(b) The stability of peroxide and superoxide of alkali metals increase as we go down the group. Explain giving reason.

41. (a) : Alkali metals are good conductors of heat and electricity but have low melting points and ionisation potentials.

42. (d) : The metal atom loses electron and combines with ammonia molecule.

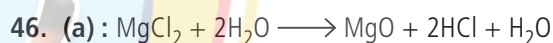
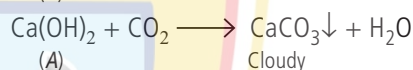


These ammoniated electrons are responsible for blue colour and conductivity of the solution.

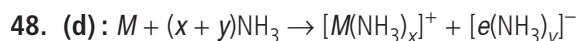
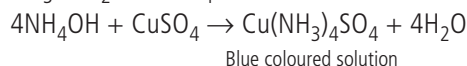
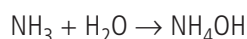
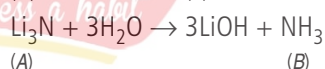
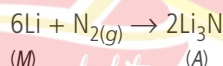
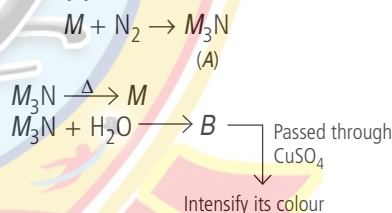
43. (b) : In electrochemical series based on oxidation potential alkali metals are far below hydrogen, thus they displace hydrogen from water.

44. (d) : Basic strength of alkali metal hydroxides increases down the group. Hence $\text{Ba}(\text{OH})_2$ is more basic than $\text{Be}(\text{OH})_2$.

45. (c) : $\text{Ca}(\text{OH})_2$ is used as flocculant.



47. (c) : Reaction can be summarized as :



The deep blue colour of the solution is due to the ammoniated electrons which absorb energy in the visible region of light and thus impart blue colour to the solution.

49. (a) : Density gradually increases in moving down from Li to Cs. K is, however, lighter than Na.

Densities of alkali metals are as follows :

Element	Li	Na	K	Rb	Cs
Density (g/mL)	0.54	0.97	0.86	1.53	1.90

i.e., $\text{Li} < \text{K} < \text{Na} < \text{Rb} < \text{Cs}$

50. (c)

51. (a)

52. (c) : Only K, Rb and Cs from alkali metals form superoxides and superoxides possess three electron bond $(:\ddot{\text{O}} \text{---} \ddot{\text{O}}:)^-$.

53. (a)

54. (c) : Li^+ has maximum degree of hydration due to its small size and for this reason lithium salts are mostly hydrated.

55. (a)

56. (a) : The loosely held *s*-electron in the outermost valence shell of these elements makes them the most electropositive metals which readily give ions, M^+ or M^{2+} .

57. (d) : The ionization enthalpies of alkali metals decrease progressively as we move down the group from Li to Cs.

58. (b) : The solubility of the alkaline earth metal hydroxides in H_2O increases with increase in atomic number down the group. This is due to the reason that both lattice energy and hydration energy decrease down the group as the size of the cation increases but lattice energy decreases more rapidly than the hydration energy and hence, their solubility increases down the group.

59. (a)

60. (a)

SUBJECTIVE TYPE QUESTIONS

1. Sodium has higher melting point than potassium because of stronger metallic bonding.

2. The ionization enthalpy, is the energy required to completely remove an electron from an isolated gaseous atom or ion. The closer and more tightly an electron is bound to its nucleus, the more difficult it will be to remove the electron and higher will be the ionization enthalpy. Group 1 elements have low first ionization enthalpies because the loss of an electron takes place readily to form a stable octet.

3. This is because the smaller atomic size causes the electrons to be packed more closely, thereby forming strong metallic bonds.

4. Due to small size and high ionisation enthalpy, Be forms coordination compound.

5. Let oxidation state of Na = x

The oxidation state of O = -1 (present as peroxide)

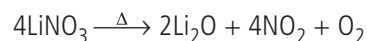
$$\therefore 2x + 2(-1) = 0 \Rightarrow 2x - 2 = 0$$

$$\Rightarrow 2x = 2 \Rightarrow x = +1$$

$\therefore$ Oxidation state of Na in Na_2O_2 is $+1$.

6. This is because halides of Be are covalent while those of Ba are ionic.

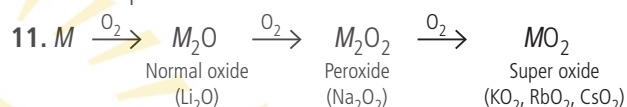
7. Lithium nitrate on heating decomposes to give Li_2O , NO_2 and O_2



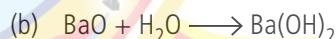
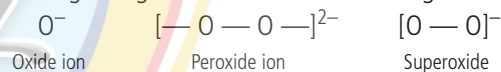
8. Because of smallest size among alkali metals, Li^+ can polarise water molecules more easily than the other alkali metal ions and hence get attached to lithium salts as water of crystallisation. For example, lithium chloride crystallises as $\text{LiCl} \cdot 2\text{H}_2\text{O}$ but sodium chloride as NaCl .

9. Calcium is used in photoelectric cell due to low value of ionization potential.

10. Solvay process cannot be extended to the manufacture of K_2CO_3 because KHCO_3 is too soluble to be precipitated by the addition of ammonium hydrogen carbonate to a saturated solution of potassium chloride.

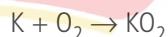


The normal oxides contain O^{2-} ion, the peroxides contain O_2^{2-} ion and superoxides contain O_2^- ion. The peroxides and superoxides become more stable with increase in atomic number of the alkali metal. The formation and stability of these oxides can be explained on the basis of lattice energy effects. Li^+ ion being a small ion has a strong positive field around it and can stabilise only a small anion, O^{2-} whereas Na^+ being a large cation can stabilise a large ion and so on.

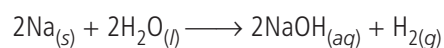


13. (i) Alkali metals are used in photoelectric cells and because of their low ionization energies they lose electrons very easily on irradiation.

(ii) Potassium reacts with oxygen to form superoxide.



14. (a) Sodium hydroxide and hydrogen gas will be formed which will catch fire.



(b) Sodium metal dissolves in liquid ammonia and gives deep blue solution which is conducting in nature.

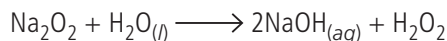


The blue colour of the solution is due to the ammoniated electron which absorbs energy in the visible region of light and thus imparts blue colour to the solution.

15. Li_2CO_3 decomposes on heating because the Li^+ ion exerts a strong polarising action and distorts the electron cloud of the nearby oxygen atom of the large CO_3^{2-} ion. This results in the weakening of the C—O bond and strengthening of the Li—O bond. This ultimately facilitates the decomposition

of Li_2CO_3 into Li_2O and CO_2 . The lattice energy of Li_2O is higher than the lattice energy of Li_2CO_3 . This also favours decomposition of Li_2CO_3 . Na^+ ion being bigger in size, the lattice energy of Na_2O is less stable than that of Na_2CO_3 . Therefore, Na_2CO_3 does not decompose on heating.

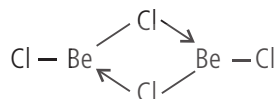
16. (i) Sodium hydroxide and hydrogen peroxide will be formed.



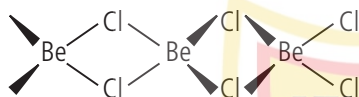
(ii) Sodium peroxide is formed.



17. (i) BeCl_2 exists as a dimer in vapour phase, which dissolves into the linear monomer $\text{Cl}-\text{Be}-\text{Cl}$ at 1200 K.

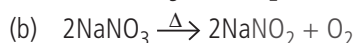
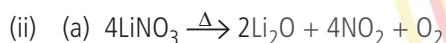


(ii) $\text{BeCl}_2(\text{solid})$: Beryllium chloride has a chain structure in solid state :



18. (i) Chief factors responsible for the anomalous behaviour of lithium are :

- (I) its very small size
- (II) high electronegativity
- (III) high ionization enthalpy and
- (IV) absence of vacant d -atomic orbital in the valence shell.

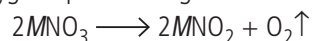


19. (i) In Mg, the electrons are tightly held and excitation by mere flame is rather difficult, thus, Mg does not show flame colouration.

(ii) As group 1 elements have large size, the intermetallic bonds in them are quite weak. Hence they have low melting and boiling points.

20. In the solid state, higher enthalpy of lattice formation by M^{2+} ions (as compared to M^+ ions) more than compensates the higher second ionization enthalpies of metals and in aqueous solution, higher enthalpy of hydration of M^{2+} ions (as compared to M^+ ions) more than compensate at the higher second ionization enthalpy of metals.

21. (a) **Nitrates** : Nitrates of alkali metals are of the type MNO_3 , they are soluble in water and do not undergo hydrolysis. Except LiNO_3 , other nitrates of this group decompose to nitrites and oxygen upon heating.



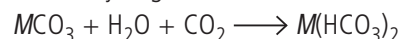
Nitrates of alkaline earth metals are of the type $\text{M}(\text{NO}_3)_2$. They are soluble in water and upon heating they decompose into their corresponding oxides with evolution of a mixture of NO_2 and oxygen.



(b) **Carbonates** : All alkali metals form carbonates of the type M_2CO_3 . Their carbonates are highly stable towards heat and readily soluble in water. The stability and solubility of the carbonates increases in the same order:



Li_2CO_3 decomposes on heating and is insoluble in water. Carbonates of alkaline earth metals (MCO_3) are insoluble in water, they dissolve in water only in presence of CO_2 due to the formation of a hydrogen carbonate.



Solubility of carbonates decreases as we descend the group and stability increases due to increase in electropositive character of the metal.

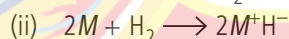
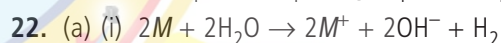
(c) **Sulphates** : Alkali metal sulphates are of the type M_2SO_4 . Except Li_2SO_4 , all other sulphates are soluble in water. Alkaline earth metal sulphates are of the type MSO_4 . The solubility of sulphates decreases on moving down the group. CaSO_4 is sparingly soluble, SrSO_4 , BaSO_4 and RaSO_4 are almost insoluble. The solubilities of BeSO_4 and MgSO_4 are due to high energy of solvation of smaller Be^{2+} and Mg^{2+} ions. The order of solubility is :



The sulphates decompose on heating to give the corresponding oxide (MO).



The stability increases as the basic nature of the metal increases.



(b) The order is : $\text{Li} > \text{Na} > \text{I}^- > \text{Ag} > \text{Cl}^-$

23. (i) **Ionisation Enthalpy** : The ionisation energies of alkaline earth elements are higher than those of alkali metals due to higher nuclear charge and smaller radii.

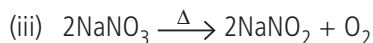
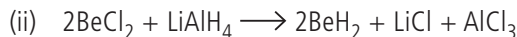
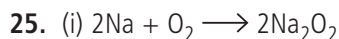
(ii) **Basicity of oxides** : Oxides of alkali metals are stronger bases as compared to those of alkaline earth metals present in the same period. e.g., when Na_2O is dissolved in water, NaOH formed is a stronger base than when MgO is dissolved in water to form $\text{Mg}(\text{OH})_2$. This is due to higher ionisation energies of alkaline earth metals.

(iii) **Solubility of hydroxides** : Alkali metal hydroxides are more soluble in water as compared to the hydroxides of alkaline earth metals present in the same period. This is due to higher lattice energy of the hydroxides of alkaline earth elements as compared to those of alkali metals.

24. (i) The alkali metals have only one electron in their valence shell which they lose easily, owing to their low ionization enthalpies, hence they are strong reducing agents.

(ii) Due to the presence of weak metallic bonding, alkali metals are soft and can be cut with a knife.

(iii) Lithium, sodium and potassium are all soft metals that are easily cut with a knife. The freshly cut surface is a shiny, silver colour, but this tarnishes quickly to a dull grey as the metal reacts with oxygen and water present in the air.



26. The charge/radius ratio of Be^{2+} is nearly the same as that of Al^{3+} ion. Hence, beryllium resembles aluminium. This is known as diagonal relationship.

(i) Like aluminium, beryllium is not readily attacked by acids because of the presence of an oxide film on the surface of the metal.

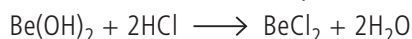
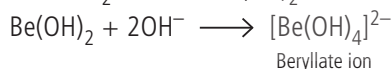
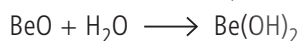
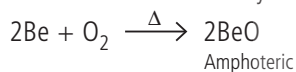
(ii) Beryllium hydroxide dissolves in excess of alkali to give a beryllate ion, $[\text{Be}(\text{OH})_4]^{2-}$ just as aluminium hydroxide gives aluminate ion, $[\text{Al}(\text{OH})_4]^-$.

27. (a) This is mainly due to high ionisation enthalpy of sodium as compared to potassium. Therefore, potassium is more electropositive, reactive and a stronger reducing agent than sodium.

(b) In Be and Mg, the electrons are tightly held and excitation by mere flame is rather difficult, thus, they do not show flame colouration.

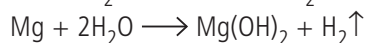
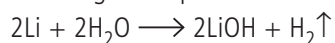
(c) Sodium is stored in kerosene oil because in air, sodium is easily oxidised to oxide which may dissolve in the moisture to form hydroxide.

28. The element is beryllium.

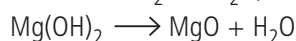


29. (a) Both Li and Mg are harder and lighter than the other metals in their respective groups.

(b) Li like Mg decomposes water slowly to liberate hydrogen.



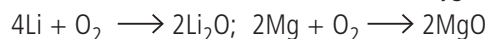
The hydroxides so formed are weak bases and decompose on heating.



(c) Both form nitrides upon direct combination with nitrogen gas.

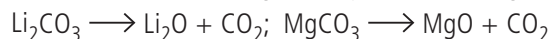


(d) Both form monoxides when heated in oxygen.

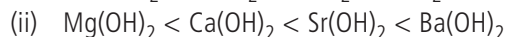


The oxides do not combine with excess oxygen to give superoxides.

(e) Carbonates of Li and Mg decompose on heating.



(f) Both LiCl and MgCl_2 are soluble in ethanol, they are deliquescent and crystallise from aqueous solution as hydrates, $\text{LiCl} \cdot 2\text{H}_2\text{O}$ and $\text{MgCl}_2 \cdot 6\text{H}_2\text{O}$.



31. (i) Alkali metals and their salts impart colour to flame. It is because their loosely held valence electrons get excited to higher energy level. When they return back they release visible light of characteristic colour to flame.

(ii) In KO_2 , superoxide ion O_2^- is present. O_2^- ion has one unpaired electron in its antibonding molecular orbital. Hence, it is paramagnetic in nature.

(iii) Like alkali metals the alkaline earth metals dissolve in liquid ammonia to give deep blue black solution forming ammoniated ions.



32. (a) Atomic size increases as we move down the group. As a result, the strength of metallic bonding decreases on moving down a group in the periodic table. This causes a decrease in the melting point. Among the given metals, the size of Cs is the largest and thus it has the least melting point.

(b) Smaller the size of an ion, the more highly it is hydrated. Among alkali metal ions, Li^+ ion is smallest in size. Also, it has the highest charge density and highest polarizing power. Hence, it attracts water molecules more strongly than the other alkali metal ions. As a result, it forms hydrated salts such as $\text{LiCl} \cdot 2\text{H}_2\text{O}$. The other alkali metal ions are larger in size than Li^+ and have weaker charge densities. Hence, they usually do not form hydrated salts.

33. (i) BeO is almost insoluble in water as it is covalent in nature and tightly held together in the solid state while BeSO_4 is highly soluble in water due to high energy of solvation of smaller Be^{2+} ion.

(ii) LiI having much more covalent character than KI because of small size of Li^+ as compared to K^+ . Hence, LiI is more soluble in ethanol.

34. (a) All the alkali metals have one valence electron, ns^1 , outside the noble gas core. The loosely held s-electron in the outermost valence shell of these elements makes them the most electropositive metals. They readily lose an electron to give monovalent M^+ ions. Thus, due to the reason cited above, alkali metals are never found free in nature but are always found in combined state.

(b) This is attributed to the hydration of the cation in water. As a result, size of the cation increases and its mobility decreases. Due to the smallest size, Li^+ ion is hydrated to the maximum and has least mobility while Cs^+ ion due to least hydration has maximum mobility.

35. (i) Electronic configuration : The valence electronic configuration of atoms of the group II A elements is ns^2 , where 'n' is the period number.

(ii) **Atomic and ionic sizes :** The size of the atom increases gradually from Be to Ra. Their ions are also large and size of the ion increases from Be^{2+} to Ra^{2+} .

(iii) **Ionisation enthalpy :** The 1st and 2nd ionisation energies of these metals decrease from Be to Ba as size increases.

36. (i) Tendency to form ionic/covalent compounds :

(a) All common compounds of alkali metals are generally ionic in nature. Halides and oxides are ionic in nature with exception of lithium compounds which are generally covalent in nature.

(b) Alkaline earth metals form ionic oxides and halides except Be which forms covalent compounds.

(ii) **Nature of oxides and their solubility in water :**

(a) Alkali metals form oxides, peroxides and superoxides. These oxides are basic in nature and basic character increases down the group. Oxides dissolve in water to give hydroxides. These hydroxides are strong bases.

(b) Alkaline earth metal oxides are basic in nature but less basic than alkali metal oxides. BeO is amphoteric while other oxides are basic and form sparingly soluble hydroxides.

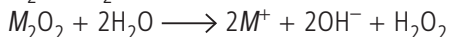
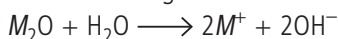
(iii) **Formation of oxosalts :**

(a) Alkali metals form sulphates, carbonates and bicarbonates. (b) Alkaline earth metals form sulphates, carbonates and nitrates.

(iv) **Solubility of oxosalts :** (a) Alkali metal oxosalts are generally soluble in water. Solubility increases down the group. (b) Carbonates and sulphates of alkaline earth metals become insoluble as we move down the group.

(v) **Thermal stability of oxosalts :** (a) Alkali metal oxosalts are thermally stable and stability increases down the group. Li_2CO_3 and Li_2SO_4 decompose on heating. (b) Alkaline earth metals carbonates decompose on heating while their thermal stability increases down the group.

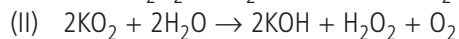
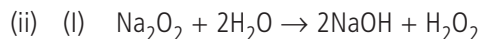
37. (a) (i) Alkali metal oxide, peroxide and superoxide are easily hydrolysed by water to form the hydroxides according to the following reactions:



(ii) Sodium ion is larger in size and has comparatively weaker +ve field around it which can not prevent oxide ion to combine with another oxygen atom to form peroxide ion.

While K^+ is larger than Na^+ and has still weaker +ve field which can not prevent even peroxide ion to combine with another oxygen atom to form superoxide.

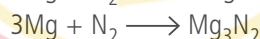
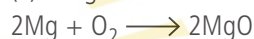
(b) (i) Sodium will react with water to form sodium hydroxide and hydrogen gas which will catch fire. For this reason, sodium cannot be stored in water.



38. (a) Alkali metals cannot be extracted by the reduction of their oxides and other compounds as they are strong reducing agents themselves and no such reducing agents are there which can reduce them to get pure metal.

(b) (i) In Be and Mg, the electrons are tightly held and excitation by mere flame is rather difficult, thus, they do not show flame colouration.

(ii) Mg burns in air to form MgO and Mg_3N_2 .



(c) (i) In Mg, the electrons are tightly held and excitation by mere flame is rather difficult, thus, Mg does not show flame colouration.

(ii) As group 1 elements have large size, the intermetallic bonds in them are quite weak. Hence they have low melting and boiling points.

39. (i) Nature of oxides – Alkali metals form M_2O , M_2O_2 and MO_2 types of oxides. The stability of the peroxide or superoxide increases as the size of metal cation increases, this is due to stabilisation of large anions by larger cations.

(ii) Nature of halides – Alkali metal halides have general formula MX . All halides are soluble in water. LiF is very less soluble in water due to its high lattice energy. Their m.pt. and b.pt. follow the trend – fluoride > chloride > bromide > iodide due to increase in size of halide ion, the lattice energy increases.

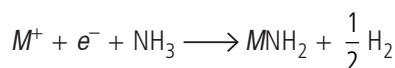
(iii) Oxosalts – Oxosalts of alkali metals are generally soluble in water and thermally stable. As electropositive character increases down the group, stability of carbonates and bicarbonates increases.

40. (a) (i) Alkali metals dissolve in liquid ammonia giving deep blue solutions which are conducting in nature.



Blue colour is due to ammoniated electron.

(ii) On standing, hydrogen is slowly liberated and amide is formed. The colour changes to bronze. On keeping for sometime.



(b) The increase in stability of peroxides and superoxides of alkali metals from Li to Cs can be explained by stabilisation of larger anions by larger cations through higher lattice energy.



The p -Block Elements (Group 13 and 14)



The *p*-Block Elements (Group 13 and 14)



Recap Notes

- The elements in which the last electron enters into any of the outermost *p*-orbitals are called *p-block elements*.
 - The general outer electronic configuration of the *p*-block elements is ns^2np^{1-6} .
 - The elements belonging to the group 13 to 18 of the long form of periodic table are *p-block elements*. The *p*-block elements include *metals*, *non-metals* and *metalloids*.
 - The *p*-block elements enter into chemical combination by losing, gaining or sharing the valence electrons.
 - Nature of compounds** : The *p*-block elements mostly form covalent compounds. The halogens, however, form ionic compounds with alkali and alkaline earth metals.
 - Oxidation states** : The highest oxidation state shown by a *p*-block element is equal to the total number of valence electrons.
 - The first element of a group differs from the heavier elements in their ability to form $p\pi-p\pi$ bonds.
- Electronegativity** : First decreases from B to Al and then increases.
- Metallic or electropositive character** : First increases from B to Al and then decreases.
- Density** : Increases down the group.
- Melting and boiling points** : Melting points decreases sharply on moving down the group from B to Ga and then increases from Ga to Tl while boiling points decreases from B to Tl.
- Reducing character** : $\text{Al} > \text{Ga} > \text{In} > \text{Tl}$.

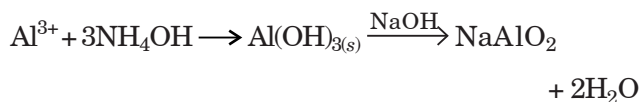
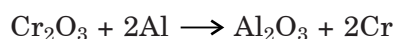
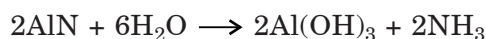
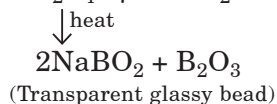
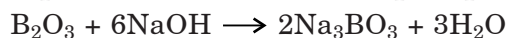
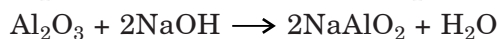
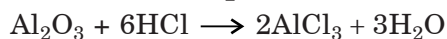
Chemical properties :

Hydrides (MH_3)	$\text{MH}_3 + \text{H}^- \rightarrow [\text{MH}_4]^-$ Electron acceptor Electron donor Anionic complex (where $M = \text{B, Al and Ga}$) Thermal stability : $\text{BH}_3 > \text{AlH}_3 > \text{GaH}_3 > \text{InH}_3 > \text{TlH}_3$
Oxides (M_2O_3) and Hydroxides $\text{M}(\text{OH})_3$	Basic strength : $\text{B}_2\text{O}_3 < \text{Al}_2\text{O}_3 < \text{Ga}_2\text{O}_3 < \text{In}_2\text{O}_3 < \text{Tl}_2\text{O}_3$ $\text{B}(\text{OH})_3 < \text{Al}(\text{OH})_3 < \text{Ga}(\text{OH})_3$ Acidic $\underbrace{\hspace{1cm}}$ Amphoteric $\underbrace{\hspace{1cm}}$ Basic Basic $< \text{In}(\text{OH})_3 < \text{Tl}(\text{OH})_3$
Halides (MX_3)	$2\text{M} + 3\text{X}_2 \rightarrow 2\text{MX}_3$ (where, $M = \text{B, Al, Ga, In and}$ $\text{X} = \text{F, Cl, Br, I}$)
	Lewis acid strength of boron trihalides : $\text{BF}_3 < \text{BCl}_3 < \text{BBr}_3 < \text{BI}_3$
	Lewis acid strength of trihalides of group 13 elements : $\text{BX}_3 > \text{AlX}_3 > \text{GaX}_3 > \text{InX}_3$ Tl does not form trihalides.

GROUP 13 ELEMENTS (BORON FAMILY)

- Group 13 of periodic table consists of following elements : ${}_5\text{B}$, ${}_{13}\text{Al}$, ${}_{31}\text{Ga}$, ${}_{49}\text{In}$, ${}_{81}\text{Tl}$ and ${}_{113}\text{Nh}$.
- Physical properties** :
 - Physical state** : All are solids, B is non-metallic, Al, Ga, In and Tl are silvery white metals.
 - Atomic and ionic radii** : Increases from B to Al then decreases from Al to Ga and then again increases.
 - Ionisation enthalpy** : Less than the corresponding members of alkaline earth metals and shows no regular trend down the group : $\text{B} > \text{Tl} > \text{Ga} > \text{Al} > \text{In}$

► **Some important chemical reactions :**



● **Preparation, properties and uses of boron:**

Preparation	Physical properties	Chemical properties	Uses
$\text{B}_2\text{O}_3 + 3\text{Mg} \xrightarrow{\text{Heat}} 3\text{MgO} + 2\text{B}_{(s)}$	It is extremely hard and black coloured solid.	$2\text{B} + 3\text{X}_2 \longrightarrow 2\text{BX}_3$	In making filaments which are used in making light composite materials for aircraft.
$2\text{BX}_3 + 2\text{H}_2 \xrightarrow[2\text{B} + 6\text{HX}]{1270^\circ\text{C Ta or W}}$	It has low electrical conductivity and is non-metallic.	$\text{B} + 3\text{HNO}_3 \xrightarrow{(\text{conc.})} \text{H}_3\text{BO}_3 + 3\text{NO}_2\uparrow$	As semiconductor in making electronic devices.
$\text{KBF}_4 \xrightarrow{\text{Electrolysis}} \text{K}^+ + \text{B}^{3+} + 4\text{F}^-$	- It has two allotropes : Crystalline boron : Black and chemically inert. It is very hard in nature. Amorphous boron: Brown and chemically active. - The transition between these forms is extremely slow process.	$2\text{B} + 3\text{H}_2\text{SO}_4 \xrightarrow{(\text{conc.})} 2\text{H}_3\text{BO}_3 + 3\text{SO}_2\uparrow$	In preparation of metal borides which are used as protective shields and control rods in nuclear reactors because $^{10}_5\text{B}$ has high ability to absorb neutrons.
$\text{B}_2\text{H}_6 \xrightarrow[1773\text{ K}]{\text{Heat}} 2\text{B} + 3\text{H}_2$	- It is poor conductor of heat and electricity. - It has two isotopes : $^{10}_5\text{B}$(20%) and $^{11}_5\text{B}$(80%).	$4\text{B} + 3\text{O}_2 \longrightarrow 2\text{B}_2\text{O}_3$ $2\text{B} + \text{N}_{2(g)} \longrightarrow 2\text{BN}_{(s)}$ $2\text{B} + 6\text{NaOH} \xrightarrow{>773\text{ K}} 2\text{Na}_3\text{BO}_3 + 3\text{H}_2$ $2\text{B} + 3\text{H}_2\text{O} \longrightarrow \text{B}_2\text{O}_3 + 3\text{H}_2$ (Red (Steam) hot)	In steel industry for increasing the hardness of steel.

● **Anomalous behaviour of boron :**

Property	Boron	Other elements of group 13
Metallic behaviour	Non-metal	Metals
Maximum covalency	4	6
Allotropy	Exhibits	Do not exhibit
Oxidation states	Only +3	+1, +3
Compounds	Only covalent	Both ionic and covalent
Halides	Monomeric	Polymeric
Aqueous solution	No ionisation	Form cations
Oxides and hydroxides	Acidic	Mainly basic
Action of non-oxidising acids	No action	React
Combination with metals	Forms boride	Do not combine (form alloy)

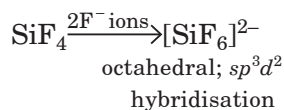
GROUP 14 ELEMENTS (CARBON FAMILY)

- Group 14 of periodic table consists of following elements : ${}_6\text{C}$, ${}_{14}\text{Si}$, ${}_{32}\text{Ge}$, ${}_{50}\text{Sn}$, ${}_{82}\text{Pb}$ and ${}_{114}\text{Fl}$.

Property	C	Si	Ge	Sn	Pb	Fl
Atomic number(Z)	6	14	32	50	82	114
Covalent radius (pm)	77	118	122	140	144	180
Electro-negativity	2.5	1.8	1.8	1.8	1.8	—
Oxidation states	+4	+4	+2, +4	+2, +4	+2, +4	0, +1, +2, +4, +6

● **General characteristics and properties:**

- ▶ The variable oxidation states are seen due to *inert pair effect*.
- ▶ C and Si → non-metals, Ge → metalloid, Sn and Pb → metals. Metallic character increases down the group.
- ▶ The covalent nature in the compounds of Sn^{4+} and Pb^{4+} ions is due to high polarisation produced by them.
- ▶ As we move down the group, stability of lower oxidation state increases due to inert pair effect.
- ▶ Except C, all other elements form complexes due to presence of vacant *d*-orbitals in them *i.e.*, these behave as Lewis acids *e.g.*,



- ▶ Semiconductor grade Si is prepared by the reduction of highly pure $\text{SiCl}_4/\text{SiHCl}_3$ with H_2 or by the pyrolysis of SiH_4 mainly.

▶ **Nature of oxides :**

Oxides	Nature
CO , CO_2 , SiO_2	acidic
GeO , GeO_2 , SnO , SnO_2 , Pb_3O_4	amphoteric
PbO , PbO_2	basic

▶ **Thermal stability of tetrahalides :**



- ▶ Elements in this group are relatively unreactive but reactivity increases down the group. Pb often appears more noble than expected due to a surface coating of oxide and partly due to high over potential for the reduction of H^+ to H_2 at a lead surface.

▶ **Reactivity of the elements of group 14 :**

Reagent	Reactivity
H_2O	C, Si, Ge, Pb are unaffected by H_2O .
	$\text{Sn} + 2\text{H}_2\text{O} \longrightarrow \text{SnO}_2 + 2\text{H}_2$ (steam)
Dilute acids	C, Si, Ge are unaffected by dilute acids.
	Pb does not dissolve in dilute H_2SO_4 due to formation of PbSO_4 coating.
Concentrated acids	Diamond is unaffected by concentrated acids, but graphite is oxidised by concentrated HNO_3 to give graphitic acid ($\text{C}_{11}\text{H}_4\text{O}_5$) which is an insoluble yellowish green substance and to graphite oxide with hot concentrated HF/HNO_3 .
	Si is oxidised and changes to SiF_4 by hot concentrated HNO_3/HF .
	Pb does not dissolve in concentrated HCl due to formation of PbCl_2 coating.

Alkalies	Carbon is unaffected by alkalies. Sn and Pb are slowly attacked by cold alkali, and rapidly by hot alkali, giving stannates $\text{Na}_2[\text{Sn}(\text{OH})_6]$ and plumbates $\text{Na}_2[\text{Pb}(\text{OH})_6]$.
Complex formation	Si, Ge, Sn and Pb can show coordination number more than 4. <i>e.g.</i> , : Si, Ge (6), Sn, Pb (8)
Halogens	Diamond is unreactive, but graphite reacts forming $(\text{CF})_n$. Si and Ge form volatile SiX_4 and GeX_4 respectively. Sn and Pb are less reactive. Sn reacts with Cl_2 and Br_2 in cold, and with F_2 and I_2 on warming. Lead reacts with F_2 in cold and with Cl_2 on heating forming PbX_2 .

● **Anomalous behaviour of carbon :**

Property	Carbon	Other elements
Hardness	hardest	less hard
M.pt. and B.pt.	high	low
Maximum covalency	4	6
Multiple bonds	$p\pi$ - $p\pi$ (high extent)	$p\pi$ - $d\pi$ (low extent)
Catenation	very high tendency	very low
Tetrahalides	does not undergo hydrolysis	undergo hydrolysis

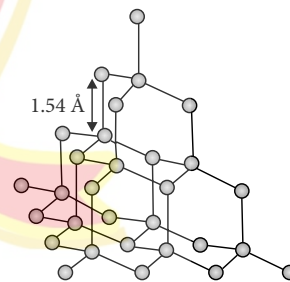
- **Multiple bonding:** C has strong tendency to form $p\pi$ - $p\pi$ multiple bonds either with itself ($\text{C}=\text{C}$, $\text{C}\equiv\text{C}$) or with atoms like N, O ($\text{C}\equiv\text{N}$, $\text{C}=\text{O}$). Carbon does not have d -orbitals and never forms $d\pi$ - $p\pi$ or $d\pi$ - $d\pi$ bonds. Silicon on the other hand forms $d\pi$ - $d\pi$ bonds.
- In case of $\text{N}(\text{CH}_3)_3$, geometry is pyramidal but in case of $\text{N}(\text{SiH}_3)_3$ it is planar because the lone pair on N-atom is transferred to the empty d -orbitals of silicon. ($p\pi$ - $d\pi$ overlapping).

● **Allotropes of carbon :**

- **Crystalline form :** It includes diamond, graphite and fullerene.
- **Amorphous form :** It includes coal, charcoal, lampblack, etc.

● **Diamond**

- It is a transparent, crystalline substance with very high refractive index.
- It is the purest form of carbon found naturally and can also be made artificially.
- Diamond is the hardest natural substance known and is a bad conductor of heat and electricity.
- **Structure :**
 - All important properties of diamond are attributed to its structure.
 - Each carbon atom of diamond is bonded to four other carbon atoms, through sp^3 hybridised orbitals, situated at the corners of a regular tetrahedron, with C-C bond length 1.54 Å and bond angle of 109.5°.
 - This gives diamond a three-dimensional network structure which explains its hardness and poor conductivity due to absence of any free electron.

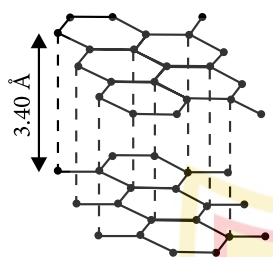


Structure of diamond

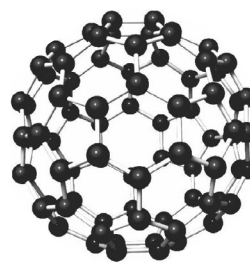
● **Graphite**

- It is a dark grey, crystalline solid which is soft and greasy to touch. It possesses a metallic lustre.
- It is also known as 'plumbago' (black lead) as it leaves a black mark on paper.
- It is a good conductor of electricity and its conductivity increases with rise in temperature.
- **Structure :**
 - It has a two dimensional sheet like structure with each carbon atom being covalently bonded to three carbon atoms through sp^2 hybridised orbitals, forming a planar hexagonal structure.

- The fourth electron of each carbon forms a $\pi(\pi)$ bond with partial overlap with that of the neighbouring carbon.
- The C – C bond length is 1.42 Å, shorter than that in diamond. The π -electrons are free to move (mobile electrons) and account for the electrical conductivity.
- The adjacent hexagonal layers (sheets) are held by weak van der Waals' forces thus, making it easy for the layers to slide over one another, accounting for its soft and greasy texture.



Structure of graphite



Structure of Buckminsterfullerene

► Uses of carbon :

- Diamond is used as previous decorative stones in jewellery because of its unusual brilliant shine.
- Graphite is used as a lubricant at higher temperature and as a refractory material in making crucibles and electrodes for high temperature work.
- Coal and charcoal are used as fuel.
- Coal is **used** for manufacturing producer gas and water gas.
- Coke is used as a reducing agent in metallurgical operations.

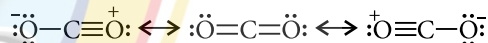
● Fullerenes

- Fullerenes are made by heating graphite in an electric arc in the presence of inert gases such as helium or argon. The sooty material formed by condensation of vapourised C_n small molecules consists of mainly C_{60} with smaller quantity of C_{70} and traces of fullerenes consisting of even number of carbon atoms up to 350 or above. Fullerenes are the only pure form of carbon because they have smooth structure without having 'dangling' bonds. Fullerenes are cage like molecules. C_{60} molecule has a shape like soccer ball and called Buckminsterfullerene.
- It contains twenty, six-membered rings and twelve, five-membered rings. A six membered ring is fused with six or five membered ring but a five membered ring can only be fused with six membered ring. All the carbon atoms are equal and they undergo sp^2 hybridisation. This ball shaped molecule has 60 vertices and each one is occupied by one carbon atom and it also contains both single and double bonds with C–C distances of 143.5 pm and 138.3 pm, respectively. Spherical fullerenes are also called *bucky balls* in short.

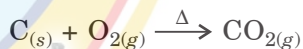
● Some important compounds of carbon and silicon :

● Carbon dioxide (CO_2)

► Structure :



► Preparation :



► Properties :

- It is an acidic oxide, and reacts with bases, forming salts.
- $CO_2 + Ca(OH)_2 \rightarrow CaCO_3 + H_2O$
- $CO_2 + K_2CO_3 + H_2O \rightarrow KHCO_3$
- Solid CO_2 is called '*dry ice*' and is used to freeze foods and ice-cream.
- It is consumed during photosynthesis.



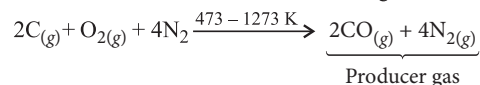
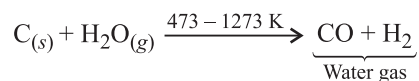
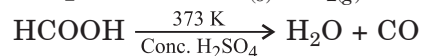
► Uses :

- In aerated water, e.g., in soda water, etc.
- In extinguishing fire.
- Solid carbon dioxide (dry ice) is used as refrigerant.
- As *carbogen* [mixture of $O_2 + CO_2$ (5-10%)] in artificial respiration especially for pneumonia patients and victims of CO poisoning.

- **Carbon monoxide (CO)**

► **Structure** : $\text{C}=\ddot{\text{O}}: \leftrightarrow \text{:C}\equiv\ddot{\text{O}}^+ \text{ or } \text{:C}\equiv\ddot{\text{O}}:$

► **Preparation** : $2\text{C}_{(s)} + \text{O}_{2(g)} \xrightarrow{\Delta} 2\text{CO}_{(g)}$



► **Properties :**

- Highly poisonous due to the ability to form a complex with haemoglobin (Hb)

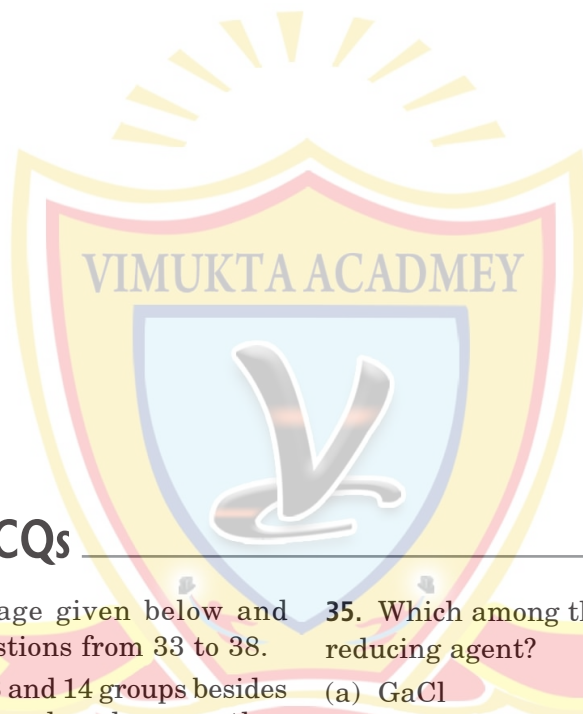
which is 300 times more stable than $\text{O}_2\text{-Hb}$ complex thus, prevents Hb in the RBCs from carrying O_2 around the body.

- It is a powerful reducing agent and reduces many metal oxides to the metal.
- CO molecule acts as a donor and reacts with metals to form metal carbonyls.

► **Uses :**

- As an important constituent of two industrial fuels, *i.e.*, water gas and producer gas.
- In Mond's process for the purification of nickel.





➡ Case Based MCQs

Case I : Read the passage given below and answer the following questions from 33 to 38.

The heavier members of 13 and 14 groups besides the group oxidation state also show another oxidation state which is two units less than the group oxidation state. Down the group ($\downarrow$), the stability of higher oxidation state decreases and that of lower oxidation state increases. This concept which is commonly called inert pair effect has been used to explain many physical and chemical properties of the element of these groups.

33. Heavier members of groups 13 exhibit oxidation state

- (a) +3 only (b) +1 only
(c) +1 and +3 both (d) +1, +2, +3

34. Which among the following is the strongest oxidising agent?

- (a) SiO_2 (b) GeO_2
(c) SnO_2 (d) PbO_2

35. Which among the following is the strongest reducing agent?

- (a) GaCl (b) InCl
(c) BCl_3 (d) TlCl

36. The strongest reductant among the following is

- (a) SnCl_2 (b) SnCl_4
(c) PbCl_2 (d) GeCl_2

37. Which of the following statement is wrong?

- (a) Tl(III) salt undergo disproportionation.
(b) CO is used as a reducing agent.
(c) CO_2 is a greenhouse gas.
(d) SiO_2 is a covalent solid.

38. Which of the following act as the strongest acid?

- (a) Tl_2O_3 (b) SnO_2
(c) PbO_2 (d) CO_2

Case II : Read the passage given below and answer the following questions from 39 to 43.

Allotropy : The phenomenon of existence of the same substance (element or compound) in two or more forms, in the same physical state, having different properties. Different forms are called allotropes or allotropic modifications.

Except lead, all other elements of group 14 show allotropy.

Element	Allotropic form
C	Crystalline : graphite and diamond Amorphous : coal, coke and charcoal
Si	Crystalline and amorphous
Ge	Two crystalline forms
Sn	Three forms : grey tin, white tin, rhombic tin

39. Wood charcoal is used in gas masks because it

- (a) is poisonous
- (b) liquefies gases
- (c) is porous
- (d) adsorbs poisonous gases.

40. Which of the following is not sp^2 hybridised?

- (a) Graphite
- (b) Graphene
- (c) Fullerene
- (d) Dry ice

41. Fullerene with formula C_{60} has a structure where every carbon atom is

- (a) sp -hybridized
- (b) sp^2 -hybridized
- (c) sp^3 -hybridized
- (d) not hybridized.

42. Thermodynamically the most stable form of carbon is

- (a) diamond
- (b) graphite
- (c) fullerene
- (d) coal.

43. The element that does not show catenation among the following p -block elements is

- (a) carbon
- (b) silicon
- (c) germanium
- (d) lead.

Case III : Read the passage given below and answer the following questions from 44 to 47.

The high charge and small size of Al^{3+} ion gives it a high charge density which is responsible for its tendency to show

- (a) covalency in its compounds in the gaseous state
- (b) high hydration enthalpy which stabilizes its compounds in solution, and

(c) high lattice enthalpy of its compounds in the solid state.

Hence, aluminium can form both covalent and ionic bond. Like halides of boron, halides of aluminium do not show backbonding because of increase in size of aluminium. In fact, aluminium atoms complete their octets by forming dimers. Thus, chloride and bromide of aluminium exist as dimers. Thus, chloride and bromide of aluminium exist as dimers, both in the vapor state and in polar solvents like benzene, while the corresponding boron halides exist as monomer. In boron trihalides, the extent of back bonding decreases with increase in size of halogens and thus Lewis acid character increases. All BX_3 are hydrolysed by water but BF_3 shows as different behavior.

44. Which of the following statements about anhydrous aluminium chloride is correct?

- (a) It exists as Al_2Cl_6 dimer in vapour form.
- (b) It is not easily hydrolysed.
- (c) It sublimes at $100^\circ C$ under vacuum.
- (d) It is a strong Lewis base.

45. Which one of the following statements is correct?

- (a) All boron trihalides form back bonding.
- (b) Anhydrous aluminium chloride is an ionic compound.
- (c) Aluminium bromide make up the electron deficiency by bridging with other aluminium bromide.
- (d) None of these.

46. The dimeric structure of aluminium chloride disappear when

- (a) it dissolves in water
- (b) it reacts with donor molecules like R_3N
- (c) it dissolves in benzene
- (d) both (a) and (b).

47. Which of the following reaction is incorrect?

- (a) $BF_{3(g)} + F^-_{(aq)} \longrightarrow BF_4^-$
- (b) $BF_{3(g)} + 2H_2O \longrightarrow [BF_3OH]^- + H_3O^+$
- (c) $BCl_{3(g)} + 3C_2H_5OH_{(l)} \longrightarrow B(OC_2H_5)_{3(l)} + 3HCl$
- (d) $BCl_{3(g)} + 2C_5H_5N_{(l)} \longrightarrow Cl_3B(C_5H_5N)_{2(s)}$

➡ Assertion & Reasoning Based MCQs

For question numbers 48-55, a statement of assertion followed by a statement of reason is given. Choose the correct answer out of the following choices.

- (a) Assertion and reason both are correct statements and reason is correct explanation for assertion.
(b) Assertion and reason both are correct statements but reason is not correct explanation for assertion.
(c) Assertion is correct statement but reason is wrong statement.
(d) Assertion is wrong statement but reason is correct statement.

48. Assertion : Compounds formed by non-metals with metals are covalent in nature.

Reason : Compounds formed between non-metals themselves are largely covalent.

49. Assertion : Anhydrous AlCl_3 is covalent but hydrated AlCl_3 is ionic.

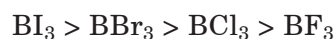
Reason : In water, Al_2Cl_6 dissociates into hydrated Al^{3+} and Cl^- ions due to high heat of hydration of these ions.

50. Assertion : The tetrahalides of carbon are not hydrolysed by water under normal conditions.

Reason : Carbon cannot expand its coordination number beyond 4 because of the absence of d -orbitals.

51. Assertion : All the trihalides of boron act as Lewis acids.

Reason : The relative strength of boron trihalides is of the order



52. Assertion : Al forms $[\text{AlF}_6]^{3-}$ but B does not form $[\text{BF}_6]^{3-}$.

Reason : B does not react with F_2 .

53. Assertion : The tendency for catenation decreases in the order $\text{C} > \text{Si} > \text{Ge} > \text{Sn}$.

Reason : The catenation depends on the strength of the element-element bond.

54. Assertion : Boron differs from aluminium and other members of group 13 in a number of properties.

Reason : Boron shows anomalous behaviour.

55. Assertion : In carbon dioxide, the carbon is sp^3 hybridized.

Reason : CO is a linear monomeric covalent compound.

SUBJECTIVE TYPE QUESTIONS

➡ Very Short Answer Type Questions (VSA)

- Why does graphite act as a good lubricant?
- What is inert pair effect?
- Write reactions to justify amphoteric nature of aluminium.
- Describe the shapes of BF_3 and BH_4^- . Assign the hybridisation of boron in these species.
- How would you explain the lower atomic radius of Ga as compared to Al?
- Explain why is there a phenomenal decrease in ionization enthalpy from carbon to silicon?
- Suggest reasons why the B—F bond lengths in BF_3 (130 pm) and BF_4^- (143 pm) differ.
- What are electron deficient compounds? Are BCl_3 and SiCl_4 electron deficient species? Explain.
- What are fullerenes? How are they prepared?
- If B—Cl bond has a dipole moment, explain why BCl_3 molecule has zero dipole moment.

➡ Short Answer Type Questions (SA-I)

11. What are allotropes? Sketch the structure of two allotropes of carbon namely diamond and graphite. What is the impact of structure on physical properties of two allotropes?
12. Draw the structures of $\text{BCl}_3 \cdot \text{NH}_3$ and AlCl_3 (dimer).
13. Explain why the following compounds behave as Lewis acids?
(i) BCl_3 (ii) AlCl_3
14. Describe the general trends in the metallic character of the elements in groups 13 and 14.
15. The +1 oxidation state in group 13 and +2 oxidation state in group 14 becomes more and more stable with increasing atomic number. Explain.
16. What do you understand by (a) inert pair effect (b) allotropy and (c) catenation?
17. Carbon and silicon both belong to the group 14, but inspite of the stoichiometric similarity, the dioxides, (*i.e.*, carbon dioxide and silicon dioxide), differ in their structures. Comment.
18. Though fluorine is more electronegative than chlorine yet BF_3 is a weaker Lewis acid than BCl_3 . Comment.
19. Explain, why CO_2 is a gas whereas SiO_2 is a solid?
20. Give reason why CCl_4 is immiscible in water, whereas SiCl_4 is easily hydrolysed.

➡ Short Answer Type Questions (SA-II)

21. (a) Classify following oxides as neutral, acidic, basic or amphoteric:
 CO , B_2O_3 , SiO_2 , CO_2 , Al_2O_3 , PbO_2 , Tl_2O_3
(b) Write suitable chemical equations to show their nature.
22. Silicon forms SiF_6^{2-} ion whereas corresponding fluoro compound of carbon is not known. Explain.
23. Arrange the following in increasing order of the property indicated :
(a) SiCl_2 , GeCl_2 , SnCl_2 and PbCl_2 (stability)
(b) CO , SiO , SnO , GeO , PbO (basicity)
(c) SiF_4 , SiCl_4 , SiI_4 , SiBr_4 (stability)
24. Explain the difference in properties of diamond and graphite on the basis of their structures.
25. Describe the general trends in the following properties of the elements in Groups 13 and 14.
(i) Oxidation states
(ii) Atomic size
(iii) Nature of halide
26. Explain the following :
(i) Boron does not exist as B^{3+} ion.
(ii) Discuss the Lewis acid nature of boron halides.
27. How does AlCl_3 act as a Lewis acid?
28. Explain the following :
(a) Electron gain enthalpy of chlorine is more negative as compared to fluorine.
(b) Pb^{4+} acts as an oxidising agent but Sn^{2+} acts as a reducing agent.
29. BCl_3 exists as monomer whereas AlCl_3 is dimerised through halogen bridging. Give reason. Explain the structure of the dimer of AlCl_3 also.
30. Complete the following chemical equations and identify X, Y and Z.
$$\text{Z} + 3 \text{LiAlH}_4 \longrightarrow \text{X} + 3 \text{LiF} + 3 \text{AlF}_3$$
$$\text{X} + 6 \text{H}_2\text{O} \longrightarrow \text{Y} + 6 \text{H}_2$$
$$\text{X} + 3 \text{O}_2 \xrightarrow{\Delta} \text{B}_2\text{O}_3 + 3 \text{H}_2\text{O}$$

➡ Long Answer Type Questions (LA)

31. (a) Boron fluoride exists as BF_3 but boron hydride doesn't exist as BH_3 . Give reason. In which form does it exist? Explain its structure.
- (b) A tetravalent element forms monoxide and dioxide with oxygen. When air is passed over heated element (1273 K), producer gas is

obtained. Monoxide of the element is a powerful reducing agent and reduces ferric oxide to iron. Identify the element and write formulas of its monoxide and dioxide. Write chemical equations for the formation of producer gas and reduction of ferric oxide with the monoxide.

32. Explain the following :

- (a) Carbon shows catenation property but lead does not.
- (b) Lead does not form PbI_4 .
- (c) Pb^{4+} acts as an oxidising agent but Sn^{2+} acts as a reducing agent.

33. Explain the following:

- (i) Why PbO_2 is a stronger oxidising agent than SnO_2 ?
- (ii) Why ionisation enthalpy of Ga is higher than that of Al?

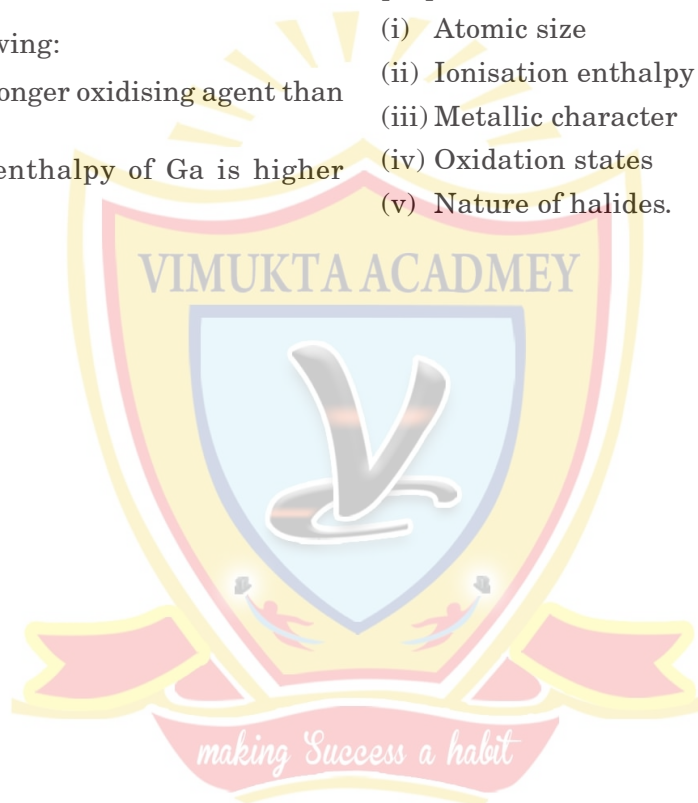
(iii) Thallous compounds (Tl^+) are more stable than thallic (Tl^{3+}) compounds. Why?

34. Three pairs of compounds are given below. Identify that compound in each of the pairs which has group 13 element in more stable oxidation state. Give reason for your choice. State the nature of bonding also.

- (i) TlCl_3 , TlCl (ii) AlCl_3 , AlCl
- (iii) InCl_3 , InCl

35. Describe the general trends in the following properties of the elements of groups 13.

- (i) Atomic size
- (ii) Ionisation enthalpy
- (iii) Metallic character
- (iv) Oxidation states
- (v) Nature of halides.



40. (d) : Solid CO_2 is dry ice in which carbon atom undergoes sp -hybridisation.

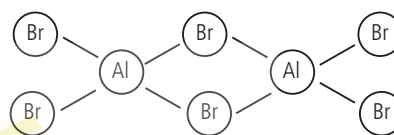
41. (b) : In fullerene with formula C_{60} , all the carbon atoms are equal and they undergo sp^2 hybridisation.

42. (b)

43. (d)

44. (a) : Anhydrous AlCl_3 dimerise to form Al_2Cl_6 in vapour form.

45. (c) : AlX_3 ($\text{X} = \text{Cl}, \text{Br}, \text{I}$) make up the electron deficiency by forming the dimer Al_2X_6



46. (a) : $\text{AlCl}_3 + \text{H}_2\text{O} \rightarrow \text{Al}(\text{OH})_3$

47. (b) : $\text{BF}_3 + \text{H}_2\text{O} \rightarrow \text{BF}_4^-$

48. (d) : Compounds formed by non-metals with metals are generally ionic in view of larger differences in their electronegativities. Compounds formed between non-metals themselves are largely covalent in character because of small differences in their electronegativities.

49. (a)

50. (a)

51. (b) : BF_3 is weakest Lewis acid.

52. (c) : B does not have vacant d -orbitals as for B, second shell is the outermost shell.

53. (a) : As we move down the group 14, the element-element bond energies decrease rapidly, viz. $\text{C}-\text{C}$ (355 kJ mol^{-1}), $\text{Si}-\text{Si}$ (222 kJ mol^{-1}), $\text{Ge}-\text{Ge}$ (167 kJ mol^{-1}) and $\text{Sn}-\text{Sn}$ (155 kJ mol^{-1}), so the tendency for catenation decreases in the order $\text{C} > \text{Si} > \text{Ge} > \text{Sn}$.

54. (b) : This is due to small atomic size, high electronegativity, high ionization energy and absence of d -orbital of B.

55. (d) : In CO_2 , C is sp hybridized. It forms two σ bonds with two oxygen atoms and two $p\pi-p\pi$ multiple bonds. So CO_2 is a linear, monomeric and covalent compound.

33. (c) : In heavier members due to inert pair effect both +1, +3 oxidation state are possible.

34. (d)

35. (c) : +1, +3 oxidation state are possible in Ga.

Ga^{3+} is more stable than Ga^{1+} hence. GaCl act as reducing agent.

36. (d) : GeCl_2 is the strongest reductant because Ge in +4 oxidation state is more stable than Ge in +2 oxidation state.

37. (a) : $\text{Tl}(\text{III})$ salt doesn't undergo disproportionation reaction because +3 oxidation state of Tl is more stable due to inert pair effect.

38. (d) : CO_2 is most acidic as acidic character of oxide decrease down the group.

39. (d) : Wood charcoal adsorbs large volume of poisonous gases from atmosphere.

SUBJECTIVE TYPE QUESTIONS

1. Graphite has sheet like structure and it is slippery so, it can act as lubricant.

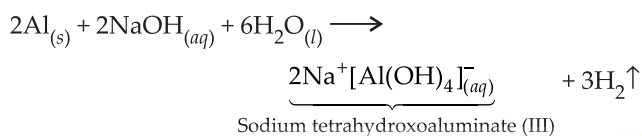
2. The tendency of s -electrons of the valence shell to participate in bond formation decreases down the group. This

reluctance of the s -electrons to participate in bond formation is called inert pair effect.

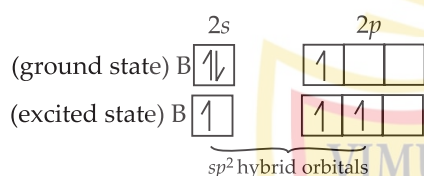
3. Amphoteric substances are those that can react with both acids and bases. Aluminium reacts with HCl to liberate H_2 gas as :



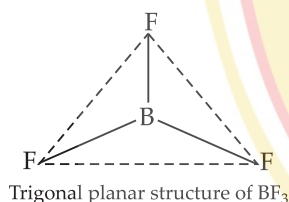
Aluminium can react with aqueous alkali and liberate hydrogen gas.



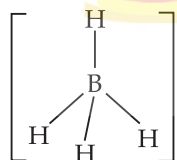
4. BF_3 has a planar triangular structure which arises from the sp^2 hybrid orbitals.



These three sp^2 hybrid orbitals are directed towards the corners of triangle and BF_3 has a trigonal structure.



BH_4^- may be assumed to be made of a central B atom, 3H atoms and one hydride ion H^- .



In order to accommodate the 3H atoms and one H^- ion, B undergoes sp^3 hybridisation yielding four orbitals, 3 of which contain one e^- each and one is empty. The fourth, empty orbital accommodates the H^- ion. Thus, the structure of BH_4^- is tetrahedral.

5. This can be understood from the variation in the inner core of the electronic configuration. The presence of additional 10 d -electrons offer only poor screening effect for the outer electron from the increased nuclear charge in gallium. Consequently, the atomic radius of gallium (135 pm) is less than that of aluminium (143 pm).

6. Large decrease in ionisation potential from C to Si is due to increase in size of the atom and shielding effect.

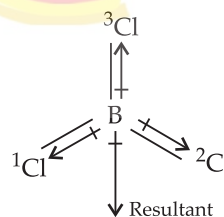
7. The bond length in any compound is dependent on the hybridisation of the central atom. Boron in BF_3 is sp^2 hybridised which means that the s -character is 33% and therefore, the bond length is shorter. Also due to similar size of both atoms and vacant p -orbital of B, a $p\pi-p\pi$ back bonding from F to B occurs causes partial double bond character. This further decreases the bond length of B — F. In BF_4^- , the hybridisation of B is sp^3 which means that the s -character is 25% and therefore, a longer bond length.

8. Electron deficient compounds are those where the central atom has less than 8 electrons in its outermost shell. Out of BCl_3 and $SiCl_4$, the former is an electron deficient compound since it has only 6 electrons in the outermost shell. $SiCl_4$ is not an electron deficient compound.

However, it can accept electrons by expanding its octet due to presence of empty d -orbitals. Thus, it may form species like $SiCl_6^{2-}$.

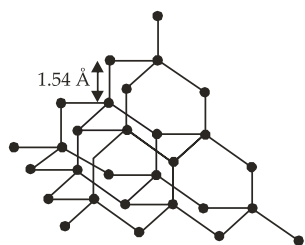
9. Fullerenes are the purest form of carbon, consisting of mainly C_{60} units. C_{60} unit has a shape of football, called Buckminsterfullerene. Fullerenes are prepared by heating graphite in an electric arc in the presence of inert gas such as helium or argon.

10. The dipole moment of any molecule is the vector sum total of each of the dipole moments. In BCl_3 molecule, although the B—Cl bonds individually are polar, the resultant dipole moment becomes zero.

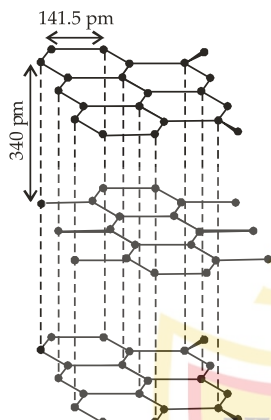


We can see that the dipole moments of B—¹Cl and B—²Cl produce a resultant which is equal in magnitude but opposite in direction to B—³Cl and hence cancels it out. That is why the net dipole moment of BCl_3 is zero.

11. The property due to which an element exists in two or more forms which differ in their physical and some of the chemical properties is known as allotropy and the various forms are called allotropes or allotropic modifications. Carbon exists in two allotropic forms crystalline and amorphous. The crystalline forms are diamond and graphite.



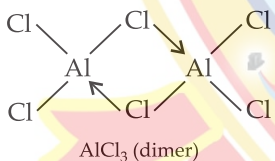
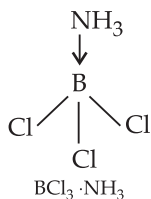
Structure of diamond



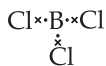
Structure of graphite

Diamond due to extended covalent bonding is the hardest natural substance on the earth. Graphite has layer of sheets which are held by weak van der Waals' forces thus, it can be cleaved easily between layers which makes it soft and slippery.

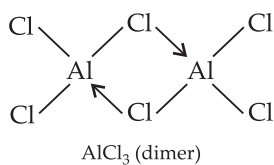
12.



13. (i) BCl_3 – Boron has 6 electrons in its outermost orbital and has a vacant p -orbital. Thus, it is an electron deficient compound hence acts as Lewis acid and accepts a lone pair of electrons.



(ii) AlCl_3 is also an electron deficient compound and acts as Lewis acid. It generally forms a dimer to achieve stability.



14. Metallic character : Metallic character increases from boron to aluminium then decreases down the group for group

13 elements. Due to smaller size group 14 elements are less metallic. Metallic character increases gradually down the group. C (non-metal), Si, Ge (metalloid) Sn, Pb (metals).

15. In group 13 and 14, as we move down the group, the tendency of s -electrons of the valence shell to participate in bond formation decreases. This is due to ineffective shielding of ns^1 and ns^2 electrons of the valence shell by intervening d - and f -electrons. This is called inert pair effect.

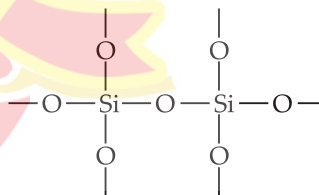
Due to this, ns^1 and ns^2 electrons of valence shell of group 13 and 14 are unable to participate in bonding. Hence, +1 and +2 oxidation states become more stable with increasing atomic number.

16. (a) Inert pair effect : The reluctance of ns^2 pair in p -block elements having higher atomic number to take part in bond formation is called inert pair effect.

(b) Allotropy : The existence of an element in more than one form having different physical properties but same or slightly different chemical properties is called allotropy.

(c) Catenation : The property by virtue of which a large number of atoms of the same element get linked together through covalent bonds resulting in the formation of long chains, branched chains and rings of different sizes is called catenation.

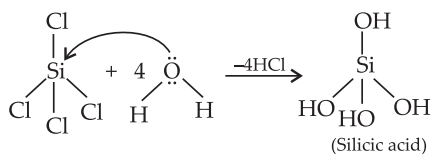
17. Due to absence of d -orbitals multiple $p\pi-p\pi$ bonding is present in carbon dioxide hence CO_2 is linear ($\text{O}=\text{C}=\text{O}$) with sp hybridisation. SiO_2 has discrete single bonded structure in a tetrahedral manner.



18. Due to back bonding in $\text{B} - \text{F}$, electron deficiency is compensated which makes it a weaker Lewis acid than BCl_3 . However, in $\text{B} - \text{Cl}$, back bonding is not significant due to much bigger size of $3p$ -orbital of Cl than vacant $2p$ -orbital of B.

19. Silicon dioxide is a covalent three dimensional network solid due to absence of $p\pi-p\pi$ bonding in SiO_2 and very high $\text{Si} - \text{O}$ bond enthalpy but in CO_2 due to $p\pi-p\pi$ bonding gives discrete molecules unlike SiO_2 . Thus, CO_2 is a gas.

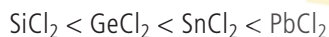
20. CCl_4 cannot be hydrolysed by water because carbon atom can not accommodate lone pair of electrons from oxygen atom of water due to absence of d -orbital. SiCl_4 can be hydrolysed to give Si(OH)_4 due to presence of d -orbitals.



21. (a) Neutral oxides : CO
 Acidic oxides : B_2O_3 , SiO_2 , CO_2
 Basic oxides : Ti_2O_3
 Amphoteric oxides : Al_2O_3 , PbO_2
- (b) (i) CO Neutral to litmus [Neutral]
 (ii) B_2O_3 reacts with basic (metallic) oxides forming meta-borates.
 $\text{B}_2\text{O}_3 + \text{CuO} \longrightarrow \text{Cu(BO}_2)_2$ [Acidic]
 (iii) $\text{SiO}_2 + 2\text{NaOH} \longrightarrow \text{Na}_2\text{SiO}_3 + \text{H}_2\text{O}$ [Acidic]
 (iv) $\text{CO}_2 + 2\text{NaOH} \longrightarrow \text{Na}_2\text{CO}_3 + \text{H}_2\text{O}$ [Acidic]
 (v) $\text{Ti}_2\text{O}_3 + 3\text{H}_2\text{SO}_4 \longrightarrow \text{Ti}_2(\text{SO}_4)_3 + 3\text{H}_2\text{O}$ [Basic]
 (vi) $\text{Al}_2\text{O}_3 + 6\text{HCl} \longrightarrow 2\text{AlCl}_3 + 3\text{H}_2\text{O}$ } [Amphoteric]
 $\text{Al}_2\text{O}_3 + 2\text{NaOH} \longrightarrow 2\text{NaAlO}_2 + \text{H}_2\text{O}$ }
 (vii) $\text{PbO}_2 + 2\text{NaOH} \longrightarrow \text{Na}_2\text{PbO}_3 + \text{H}_2\text{O}$ } [Amphoteric]
 $\text{PbO}_2 + 4\text{HCl} \longrightarrow \text{PbCl}_2 + \text{Cl}_2 + 2\text{H}_2\text{O}$ }

22. SiF_6^{2-} ion exists because of presence of d -orbitals. Silicon expands its octet to give sp^3d^2 hybridisation and forms complexes or ions by accepting electron pairs from donor species like SiF_6^{2-} . Carbon cannot exceed its covalency more than 4. Thus, CF_6^{2-} is not known.

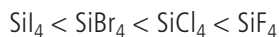
23. (a) The stability of dihalides increases down the group because divalent state becomes more and more stable as we move down the group.



(b) Basicity of oxides increases down the group as metallic character increases.



(c) Si—X bond strength decreases as the size of the halogen increases. The correct order is



24.

Criterion	Diamond	Graphite
Hybridisation	sp^3	sp^2
Structure	Tetrahedral carbon which gives rise to a 3-dimensional structure.	Planar trigonal which gives rise to a 2-dimensional sheet like structure of carbon.
C — C	154 pm	141.5 pm

Hardness	Due to 3-D structure, diamond is the hardest natural element on the earth.	It is made up of 2-D sheets of carbon which slip over each other. This gives graphite a slippery surface.
Electrical Conductivity	Diamond is an insulator.	Graphite is a good conductor of electricity due to presence of delocalised π -electrons.

25. (i) Oxidation states : For group 13 both +1 and +3 oxidation state are observed. +1 oxidation state becomes more stable as we move down the group due to inert pair effect. Boron does not show +3 oxidation state.

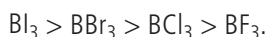
For group 14 common oxidation state are +4 and +2. Tendency to show +2 oxidation state increases down the group.

(ii) Atomic size : Atomic radii of group 13 elements increases down the group with exception $\text{Ga} < \text{Al}$ due to presence of 10 d -electrons which offer poor screening effect. In group 14, there is a considerable increase in radius from C to Si, thereafter from Si $\rightarrow$ Pb a small increase is seen due to presence of completely filled d and f -orbitals.

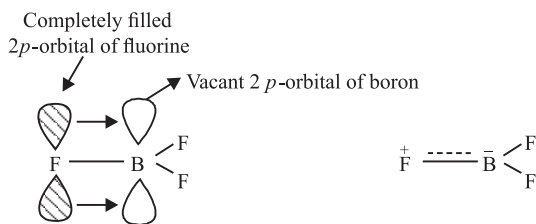
(iii) Nature of halides : Group 13 elements form trihalides (except TlI_3). Due to electron deficient nature BCl_3 accepts electrons and forms adducts. AlCl_3 achieves stability by forming a dimer. Group 14 elements form halides with formula MX_2 and MX_4 . Except CCl_4 other halides are easily hydrolysed. Stability of dihalides increases down the group.

26. (i) Due to small size of boron, the sum of its first three ionisation enthalpies is very high, hence, it does not exist in +3 form.

(ii) The Lewis acid character of boron trihalides follows the order :



The above order is just the reverse of the expected order on the basis of relative electronegativities of the halogens. This can be explained on the basis of the tendency of the halogen atom to back-donate its electrons to the boron atoms resulting in the formation of an additional $p\pi-p\pi$ bond. This type of bond formation is known as dative or back bonding.



Formation of back bonding between boron and fluorine in BF_3 molecule.

As a result of back donation of electrons from fluorine to boron, the electron deficiency of boron atom gets compensated and therefore, the Lewis acid character of BF_3 decreases.

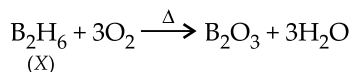
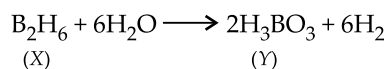
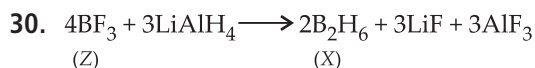
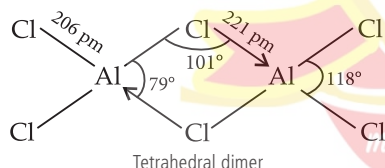
The tendency to form $p\pi-p\pi$ bond is maximum in the case of BF_3 and falls rapidly as we move to BCl_3 and BBr_3 .

27. AlCl_3 is a Lewis acid since it is an electron deficient halide. It has only six electrons in its outermost shell therefore, to complete its octet it accepts a lone pair of electrons and acts as a Lewis acid.

28. (a) Due to small size of fluorine there is inter-electronic repulsion which reduces its tendency to accept electron.

(b) Pb^{4+} acts as oxidising agent because it has a tendency to exist in Pb^{2+} form which is more stable. Sn^{2+} is a reducing agent due to tendency to form Sn^{4+} compounds.

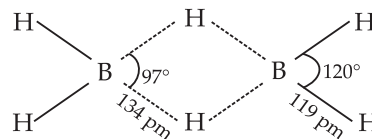
29. Due to absence of d -orbitals in boron, it exists as an electron deficient monomer and achieves stability through accepting electrons from a base like NH_3 . It cannot exist as a dimer due to small size of B which cannot accommodate bigger size 4 Cl atoms around it. AlCl_3 achieves stability by forming a dimer.



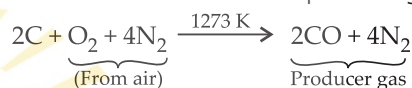
31. (a) Due to non-availability of d -orbitals, boron is unable to expand its octet hence, it exists as BF_3 and is an electron deficient compound. Due to back bonding, electron deficiency of BF_3 is compensated. But in boron hydride, hydrogen atoms do not have lone pairs for back bonding thus, to compensate electron deficiency it exists in the form of diborane.

In the structure of diborane, four terminal hydrogen

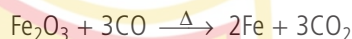
atoms and two boron atoms are in one plane. Above and below this plane, there are two bridging hydrogen atoms. The four terminal B — H bonds are regular two centre two electron bonds while the two bridge B — H — B bonds are different and are three centre — two electron bonds.



(b) The tetravalent element is carbon which forms CO and CO_2 . When heated in air it forms producer gas.



CO is a powerful reducing agent and reduces ferric oxide to iron.



32. (a) Property of catenation is maximum in carbon because C — C bonds are very strong due to smaller size. The tendency of catenation decreases down the group due to increase in size and decrease in electronegativity.

(b) $\text{Pb} + 2\text{I}_2 \not\rightarrow \text{PbI}_4$

I^- is a good reducing agent and therefore, reduces Pb(IV) to Pb(II) easily. That is why, PbI_4 does not exist.

(c) Pb^{4+} acts as an oxidising agent because it has a tendency to exist in Pb^{2+} form which is more stable. Sn^{2+} is a reducing agent due to tendency to form Sn^{4+} compounds.

33. (i) Lead compounds in +2 oxidation state are more stable than +4 oxidation state hence are stronger oxidising agents. Due to stronger inert pair effect Pb^{2+} is more stable than Sn^{2+} .

(ii) As we move from Al to Ga, due to poor shielding of the nucleus by $3d$ -electrons, the effective nuclear charge acting on Ga is slightly higher than that on Al. As a result, ionisation enthalpy of Ga is higher than that of Al.

(ii) TlCl is more stable than TlCl_3 due to inert pair effect.

34. (i) TlCl_3 , $\text{TlCl} - \text{TlCl}$ is in more stable oxidation state (+1 O.S. more stable). It is ionic in nature.

(ii) AlCl_3 , $\text{AlCl} - \text{AlCl}_3$ is more stable (+3 oxidation state). It is covalent in nature.

(iii) InCl_3 , $\text{InCl} - \text{InCl}_3$ is relatively more stable than InCl due to higher stability of +3 oxidation state. It is covalent in nature.

35. (i) Atomic size : Atomic radii of group 13 elements increase down the group with the exception that atomic radius of Ga is less than that of Al due to the presence of 10 *d*-electrons which offer poor screening effect for the outer electrons from the increased nuclear charge in Ga.

(ii) Ionisation enthalpy : For group 13 elements, the trend of ionisation enthalpy is

$B > Al < Ga > In < Tl$. This is due to increase in size and low screening effect of *d*- and *f*- electrons.

(iii) Metallic character : Metallic character increases from boron to aluminium then decreases down the group.

(iv) Oxidation states : For group 13 elements, both +1 and +3 oxidation states are observed. The +1 oxidation state becomes more stable as we move down the group due to inert pair effect. Boron does not show +3 oxidation state.

(v) Nature of halides : Group 13 elements form trihalides (except TlI_3). Due to electron deficient nature, BCl_3 accepts electrons and forms adducts. $AlCl_3$ achieves stability by forming a dimer.



Hydrocarbons

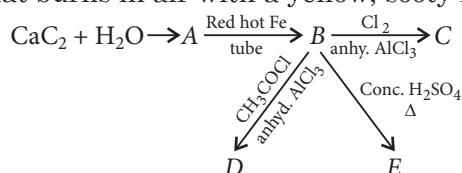




➡ Case Based MCQs

Case I : Read the following and answer the questions from 36 to 40 given below.

Compound (A) is an important industrial feed stocks, but it's largest use as the fuel for the oxyacetylene torch. It is a colourless, foul smelling gas that burns in air with a yellow, sooty flame.



36. Identify the product A.

- (a) ethane (b) ethyne
(c) ethene (d) methane

37. The compound (B) formed is

- (a) cyclohexane (b) benzene
(c) hexane (d) cyclopentane.

38. The reaction $\text{B} \xrightarrow[\text{Anhy. AlCl}_3]{\text{Cl}_2} \text{C}$ follows

- (a) nucleophilic substitution mechanism

- (b) electrophilic addition mechanism
(c) electrophilic substitution mechanism
(d) elimination reaction mechanism.

39. The product E is

- (a) nitrobenzene
(b) benzene sulphonic acid
(c) both (a) and b (d) none of these.

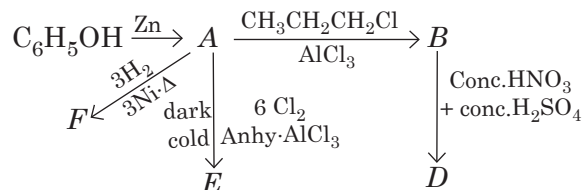
40. Identify the product 'D'.

- (a) Chlorobenzene (b) Bromobenzene
(c) Toluene (d) Acetophenone

Case II : Read the following and answer the questions from 41 to 45 given below.

Compound 'A' is the simplest and ideal aromatic compound. It is also one of the most basic petrochemicals which is mainly used to prepare a number of important chemicals such as toluene, phenol, aniline, biphenyl etc, which are used in manufacture of dyes, detergents, drugs, explosives, pesticides etc. But it is carcinogenic.

In the given sequence of reaction, compound A undergoes a number of changes



41. The name of compound A is

- (a) cyclohexane
- (b) benzene
- (b) cyclohexene
- (d) none of these.

42. The reaction $\text{A} \xrightarrow[\text{AlCl}_3]{\text{CH}_3\text{CH}_2\text{CH}_2\text{Cl}} \text{B}$ is an example of

- (a) elimination reaction
- (b) addition reaction
- (c) electrophilic substitution reaction
- (d) nucleophilic substitutional reaction.

43. The name of product F is

- (a) benzene
- (b) cyclohexene
- (c) cyclohexane
- (d) cyclohex-1 4-diene

44. The major product when 'A' reacts with propyl chloride in the presence of AlCl_3 is

- (a) propyl benzene
- (b) isopropyl benzene
- (c) chloro benzene
- (d) ethyl benzene.

45. The major product 'D' in the given series of reaction is

- (a) 4-nitro isopropyl benzene
- (b) 2-nitro isopropyl benzene
- (c) 3-nitro propyl benzene
- (d) none of these.

➡ Assertion & Reasoning Based MCQs

For question numbers 46-55, a statement of assertion followed by a statement of reason is given. Choose the correct answer out of the following choices.

- (a) Assertion and reason both are correct statements and reason is correct explanation for assertion.
- (b) Assertion and reason both are correct statements but reason is not correct explanation for assertion.
- (c) Assertion is correct statement but reason is wrong statement.
- (d) Assertion is wrong statement but reason is correct statement.

46. **Assertion :** Boiling point of alkanes increases with increase in molecular weight.

Reason : van der Waal's forces increase with increase in molecular weight.

47. **Assertion :** Acetylene is acidic in nature.

Reason : Acetylene is sp hybridised.

48. **Assertion :** *Trans*-pent-2-ene is polar but *trans*-but-2-ene is non-polar.

Reason : The polarity of *cis*-isomer is more than *trans* which are either non-polar or less polar.

49. **Assertion :** Benzene on heating with conc. H_2SO_4 gives benzenesulphonic acid which when heated with superheated steam under pressure gives benzene.

Reason : Sulphonation is a reversible process.

50. **Assertion :** Sodium acetate on Kolbe's electrolysis gives methane.

Reason : Methyl free radical is formed at anode.

51. **Assertion :** $\text{HC} \equiv \text{C}^-$ is more stable than $\text{H}_2\text{C} = \text{CH}^-$.

Reason : $\text{HC} \equiv \text{C}^-$ has more s -character than $\text{H}_2\text{C} = \text{CH}^-$.

52. **Assertion :** All the hydrogen atoms in $\text{CH}_2 = \text{C} = \text{CH}_2$ are attached to sp^2 hybridised carbon atom.

Reason : All the carbon atoms in its are sp^2 hybridized.

53. **Assertion :** Methane cannot be obtained by Wurtz reaction.

Reason : Wurtz reaction leads to the formation of symmetrical alkane having an even number of carbon atoms.

54. **Assertion :** Saturated hydrocarbons are chemically less reactive.

Reason : All isomeric paraffins have same parent name.

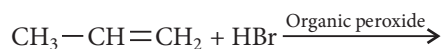
55. **Assertion :** Nitrobenzene does not undergo Friedel Crafts reaction.

Reason : Nitrobenzene is a m -director.

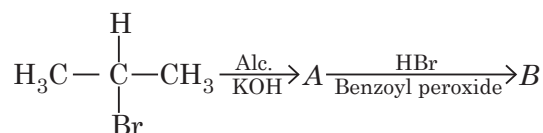
SUBJECTIVE TYPE QUESTIONS

➡ Very Short Answer Type Questions (VSA)

- When alkyne is treated with bromine water then what will be the colour of product?
- What product would you get from acid catalysed hydration of 1-methylcyclohexene? Explain.
- Explain why the branching of an alkane chain lowers its boiling point.
- Give a brief account for the following statement :
CH₄ cannot be synthesized by Wurtz reaction.
- Acetylene is acidic but it does not react with NaOH or KOH. Give reason.
- Complete the following reaction :

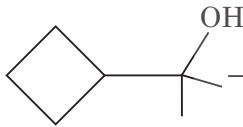


- Write structures of A and B in the following reaction :



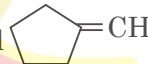
- Name the chain isomer of C₅H₁₂ which has a tertiary hydrogen atom.
- Explain why dry ether is used as a solvent in Wurtz reaction.
- Draw the Newmann projection formula for staggered and eclipsed conformation of ethane.

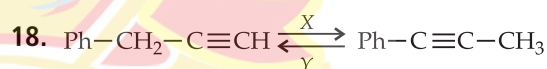
➡ Short Answer Type Questions (SA-I)

- 
 $\xrightarrow{\text{H}_2\text{SO}_4} \text{(A)}$
Major
Identify A
- How will you convert methyl bromide to ethane?
- Why is Wurtz reaction not preferred for the preparation of alkanes containing odd number of carbon atoms?
- Ethyne reacts with dil. H₂SO₄ in presence of mercury salt to give acetaldehyde but with dil. HCl under similar conditions, it gives vinyl chloride. Explain why.
- Which alkyne would you start with and what reagents would you use to prepare :
(i) *cis*-but-2-ene (ii) *trans*-pent-2-ene

- What do the following compounds produce when passed through Cr₂O₃ supported over alumina at 600°C?

- n*-Hexane
- n*-Heptane

- What will  produce on ozonolysis?

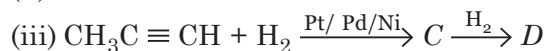
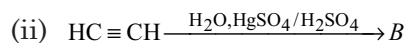


Identify X and Y.

- Arrange benzene, *n*-hexane and ethyne in decreasing order of acidic behaviour. Also, give reason for this behaviour.
- Explain Friedel-Crafts alkylation reaction with chemical equation.

➡ Short Answer Type Questions (SA-II)

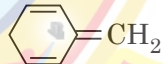
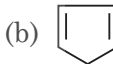
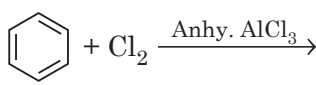
- An alkyne (X) has molecular formula C₅H₈. It reacts neither with sodamide nor with ammoniacal cuprous chloride. Identify X.
- Complete the following reactions :
(i) $\text{HC}\equiv\text{CH} \xrightarrow{\text{NaNH}_2, \text{CH}_3\text{Br}} \text{A}$

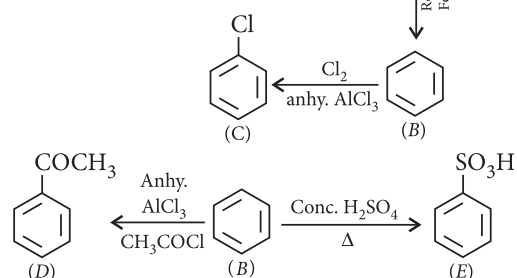
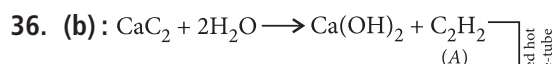


- An alkene 'A' contains three C—C, eight C—H σ bonds and one C—C π bond. 'A' on ozonolysis gives two moles of an aldehyde of molar mass 44 u. Write IUPAC name of 'A'.

24. (a) Explain the order of stability of carbocations giving reason.
 (b) Addition of HBr to propene in the presence of benzoyl peroxide yields 1-bromopropane. Explain with suitable mechanism.
25. (a) Write chemical reactions to illustrate the Kolbe's reaction
 (b) Name the compound that will be required to obtain butane using Kolbe's electrolysis process.
26. Why *cis*-but-2-ene has higher boiling point than *trans*-but-2-ene?
27. Despite their $-I$ effect, halogens are *o*- and *p*-directing in haloarenes. Explain.
28. An organic compound *A* with molecular formula C_3H_8O reacts with conc. H_2SO_4 to give *B*, which on reaction with HCl gives *C*. Compound *C* reacts with metallic sodium to give *D*. Identify compounds *A*, *B*, *C* and *D*.
29. Explain *ortho*- and *para*-directing influence of monosubstituted benzene giving suitable example.
30. Draw the Newman projections of the eclipsed and staggered conformers of ethane. Which of the two is stable and why?
31. Explain anti-Markovnikov addition or peroxide effect or Kharash effect with example.
32. What does ozonolysis of benzene yield?
33. Give two reactions to show acidic character of alkynes.
34. Identify a reagent which can easily distinguish between 1-butyne and 2-butyne.
35. A hydrocarbon (*Z*) has molecular formula C_8H_{10} . It does not decolourise bromine water and is oxidised to benzoic acid on heating with $K_2Cr_2O_7$. It can also have three other isomers *A*, *B* and *C*. Write the structures of *Z*, *A*, *B* and *C*.

➡ Long Answer Type Questions (LA)

36. An alkyl halide $C_5H_{11}Br$ (*A*) reacts with ethanolic KOH to give an alkene '*B*', which reacts with Br_2 to give a compound '*C*', which on dehydrobromination gives an alkyne '*D*'. On treatment with sodium metal in liquid ammonia one mole of '*D*' gives one mole of the sodium salt of '*D*' and half a mole of hydrogen gas. Complete hydrogenation of '*D*' yields a straight chain alkane. Identify *A*, *B*, *C* and *D*. Give the reactions involved.
37. A hydrocarbon '*Y*' decolourises bromine water. On ozonolysis it gives 3-methylbutanal and formaldehyde. Give the name of the compound. Identify *Y*.
38. Explain why the following systems are not aromatic.
- (a)  (b) 
- (c) 
39. Draw the resonating structure of C_6H_5OH (phenol) and C_6H_5CHO (benzaldehyde).
40. Give mechanism for the following reaction.
- 



37. (c)

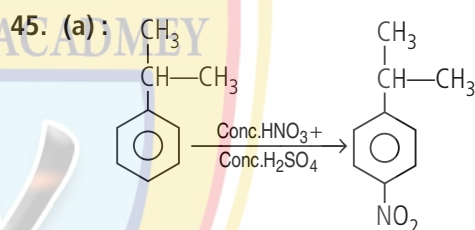
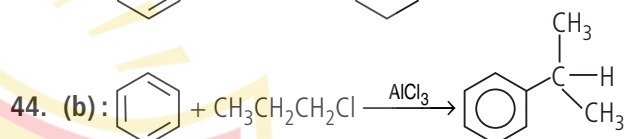
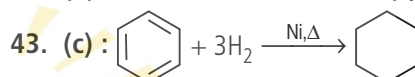
38. (c)

39. (b)

40. (d)

41. (b)

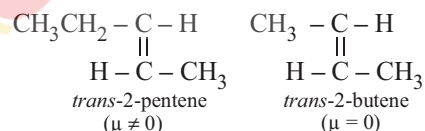
42. (c)



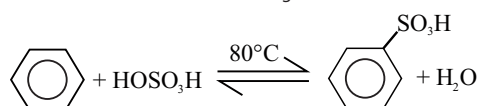
46. (a) : Greater is the molecular mass, greater is the magnitude of van der Waal's forces of attraction and hence higher the boiling point.

47. (b) : The acidic nature in acetylene is described on the basis of higher electronegativity of sp hybridized carbon atom which pulls the C-H bond pair more effectively to lose H^+ .

48. (b) : The vector sum of all polar bonds in *trans*-pent-2-ene is not zero but the vector sum is zero in *trans*-but-2-ene.



49. (a) : Sulphonation of benzene is an electrophilic substitution reaction in which SO_3 acts as the electrophile.



50. (d) : Sodium acetate on Kolbe's electrolysis gives ethane. It is formed at anode.

51. (a) : $\text{HC}\equiv\text{C}^-$ has 50% s -character and $\text{H}_2\text{C}=\text{CH}^-$ has 33% s -character. Stability of carbanions increases with an increase in the s -character at the carbanion. So, $\text{HC}\equiv\text{C}^-$ is more stable than $\text{H}_2\text{C}=\text{CH}^-$.

52. (c) : The two H-atom on first carbon and the two H-atoms on the third carbon atom are attached to sp^2 hybridised carbon atoms. The central carbon atom is sp -hybridized.

53. (a) : Wurtz reaction involves the formation of alkanes by heating alkyl halide with sodium in ether.

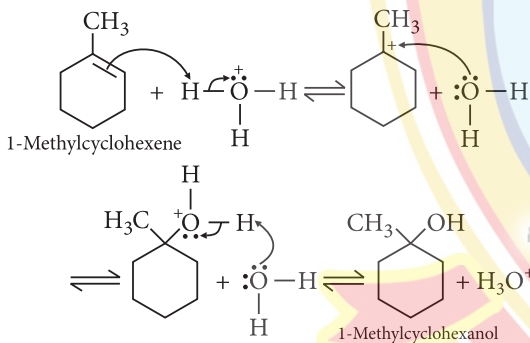
In this reaction two alkyl radicals join together to form an alkane. The net result in this reaction is the formation of symmetrical alkane ($R-R$) having an even number of carbon atoms.

54. (c) : Less reactivity of saturated hydrocarbons is due to the presence of single bonds between carbon atoms. Paraffins (alkanes) may have straight chain or branched chain isomers which have different parent names.

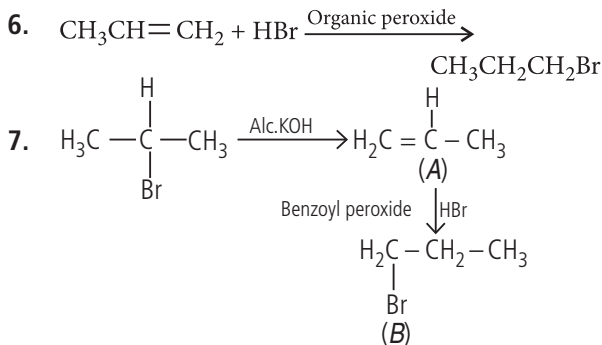
55. (b) : The Friedel Crafts reaction does not take place with nitrobenzene because the ring has been too greatly deactivated. Moreover, any coordination of $AlCl_3$ with unshared electrons of oxygen of NO_2 group would further deactivate the ring making $-NO_2$ electrophilic.

SUBJECTIVE TYPE QUESTIONS

- The product will be colourless.
- 1-Methylcyclohexanol will be formed because a 3° carbocation will be formed as an intermediate.

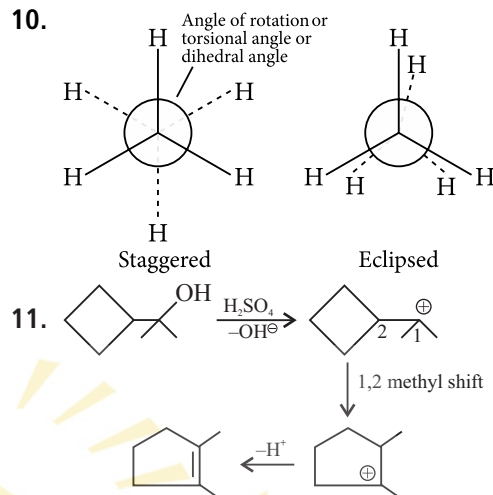


- Boiling point decreases with increase in branching due to decrease in surface area of the molecule.
- Wurtz reaction occurs between two alkyl halides to yield alkane. Methane has only one carbon atom, hence cannot be prepared by using Wurtz reaction.
- Due to sp -hybridisation of C-atom in acetylene, proton is strongly attracted by nucleus and cannot be abstracted easily therefore, it does not react with $NaOH$ or KOH .

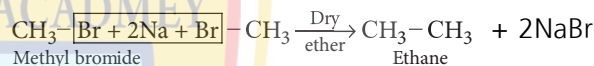


8. 2-Methylbutane, $(CH_3)_2CH-CH_2-CH_3$.

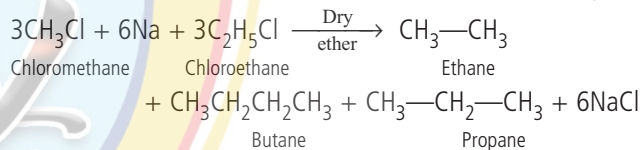
9. In Wurtz reaction, pure sodium is used which is highly violent towards water therefore, dry ether is used.



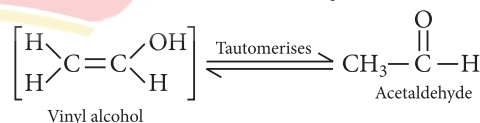
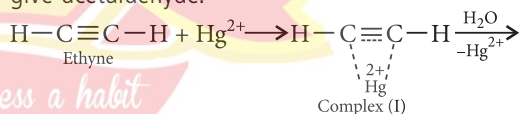
12. Two moles of methyl bromide react with sodium metal in presence of dry ether as solvent to give ethane. This reaction is known as Wurtz reaction.



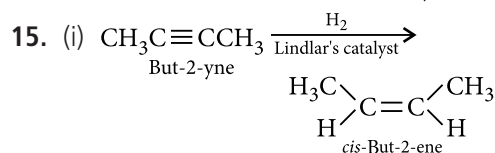
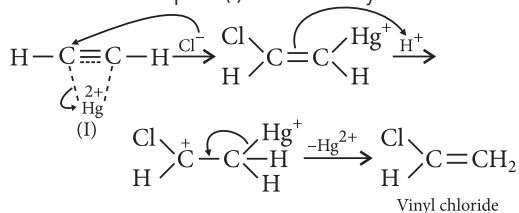
13. It is because mixture of alkanes will be formed e.g.,

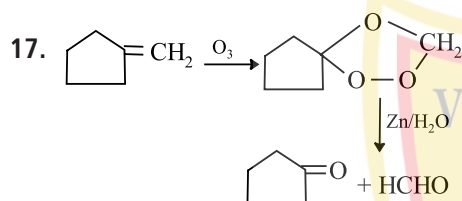
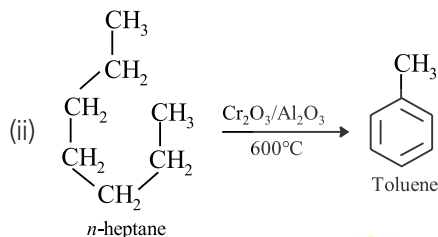
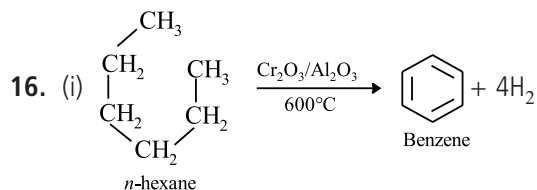
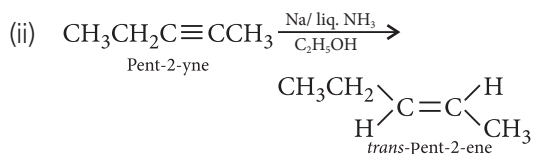


14. Mercuric ion forms a complex (I) with acetylene. Since, H_2O is more nucleophilic than SO_4^{2-} ion, it attacks the complex (I) to form first vinyl alcohol which further tautomerises to give acetaldehyde.

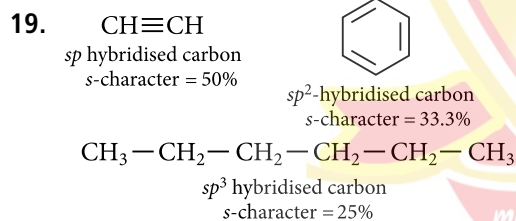


In case of dil. HCl , Cl^- ion is more nucleophilic than H_2O , it reacts with complex (I) to form vinyl chloride.



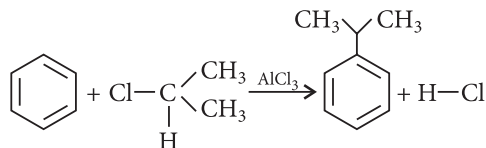


18. On heating with *alc.* KOH in inert solvent, the triple bond of 1-alkyne is shifted towards the centre to form an isomeric 2-alkyne. On heating with sodamide (NaNH_2 in liq. NH_3) the triple bond shifts towards end.



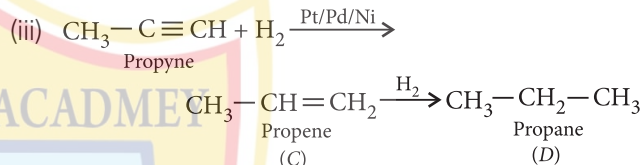
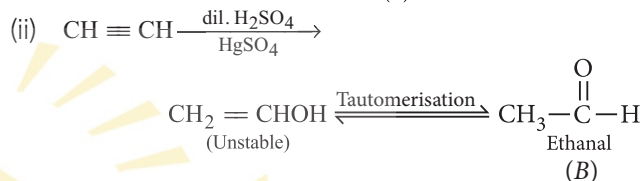
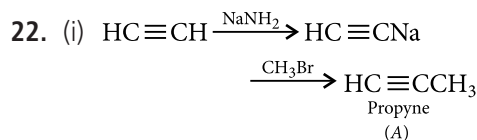
Since s -orbitals are closer to the nucleus, hence due to more s -character in ethyne (sp hybridised), the hybridised orbital is nearest to this carbon atom in comparison to sp^3 or sp^2 hybridised carbon. This leads to the movement of $\text{C}-\text{H}$ bond pair more towards sp hybridised carbon, leading to the development of partial positive charge on the hydrogen attached to sp hybridised carbon. Thus, such a hydrogen behaves as acidic hydrogen. Hence, order of acidic nature is, ethyne > benzene > *n*-hexane.

20. Friedel-Crafts alkylation is a Lewis acid-catalyzed electrophilic aromatic substitution reaction that allows the synthesis of alkylated products via the reaction of arenes with alkyl halides. With anhydrous aluminium chloride as a catalyst, the stable alkyl carbocation is generated which attacks the benzene ring. An example of this type of reaction is shown below :

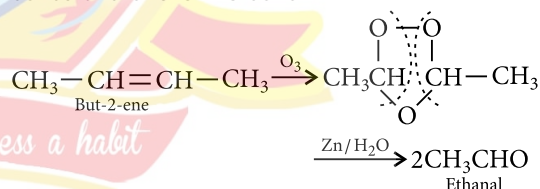


21. Alkyne X is C_5H_8 . Since it does not react with sodamide or ammoniacal cuprous chloride, the triple bond cannot be terminal.

$\therefore \text{X}$ is $\text{CH}_3\text{CH}_2\text{C}\equiv\text{CCH}_3$ (Pent-2-yne)

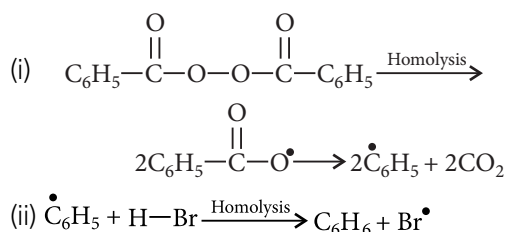


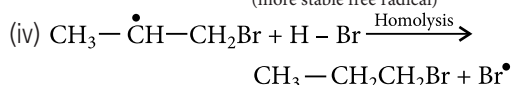
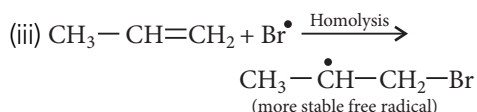
23. Alkene A contains $3\text{C}-\text{C}$, $8\text{C}-\text{H}$ and one $\text{C}=\text{C}$ bonds. An aldehyde containing one $-\text{CHO}$ group and having molar mass of 44 amu has to be CH_3CHO and since two moles of CH_3CHO are obtained by ozonolysis of alkene A , the alkene has to be joined by two $\text{CH}_3\text{CH}-$ groups by a double bond. It has to be $\text{CH}_3-\text{CH}=\text{CH}-\text{CH}_3$, i.e., but-2-ene. But-2-ene contains $3\text{C}-\text{C}$ σ bonds, $8\text{C}-\text{H}$ σ bonds and one $\text{C}=\text{C}$ bond.



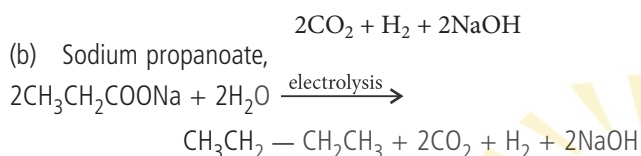
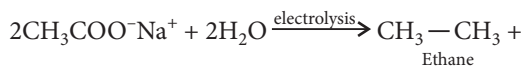
24. (a) Stability of carbocations decreases as $3^\circ > 2^\circ > 1^\circ$. Alkyl groups have +I effect. When an alkyl group is attached to positively charged carbon atom of a carbocation, it tends to release electrons towards that carbon and reduces the positive charge on the carbon. Thus, positive charge gets dispersed. This dispersal of the positive charge stabilises the carbocation.

(b) Mechanism : Peroxide effect proceeds via free radical mechanism as given below :





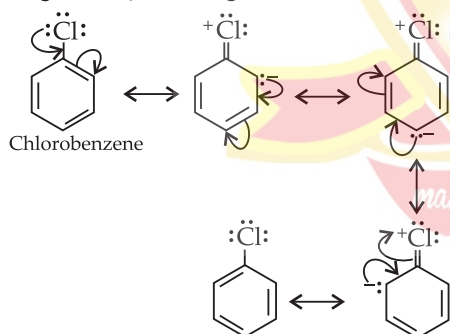
25. (a) Kolbe's reaction : In this reaction, an aqueous solution of sodium or potassium salt of carboxylic acid on electrolysis gives alkane having even number of carbon atoms at anode.



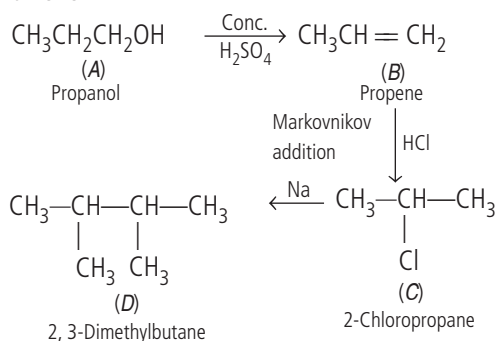
26. Due to higher dipole moment, the boiling point of *cis*-isomer is higher than the corresponding *trans*-isomer.

27. In case of aryl halides, halogens are little deactivating because of their strong $-I$ -effect. Therefore, overall electron density on the benzene ring decreases. In other words, halogens are deactivating due to $-I$ -effect. However, because of the $+R$ -effect, *i.e.*, participation of lone pairs of electrons on the halogen atom with the π -electrons of the benzene ring, the electron density increases more at *o*- and *p*-positions than at *m*-positions.

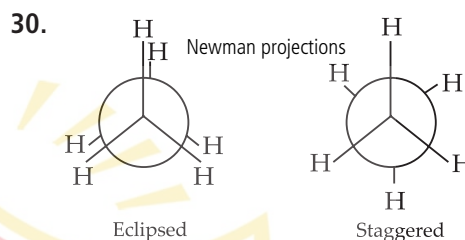
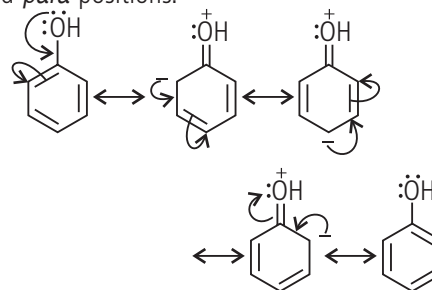
As a result, halogens are *o*, *p*-directing. The combined result of $+R$ -effect and $-I$ -effect of halogens is that, halogens are deactivating but *o*, *p*-directing.



28. Compound **A** is an alcohol which on reaction with H_2SO_4 gives alkene **B**.

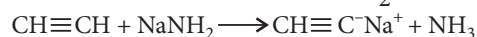
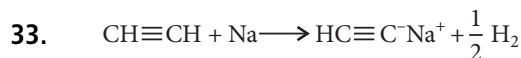
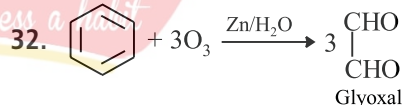
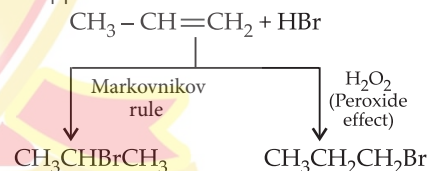


29. $-\text{OH}$ group attached to benzene ring release electrons and activate the benzene ring, direct the incoming groups to *ortho* and *para* positions.



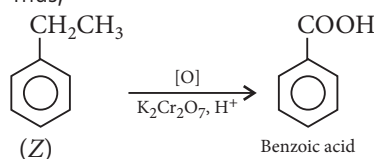
In staggered form of ethane, the electron clouds of carbon-hydrogen bonds are as far apart as possible. Thus, there are minimum repulsive forces, minimum energy and maximum stability of the molecule. On the other hand, when the staggered form changes into the eclipsed form, the electron clouds of the carbon-hydrogen bonds come closer to each other resulting in increase in electron cloud repulsions. Thus, the molecule has more energy and therefore, has lesser stability.

31. Peroxide effect : Addition of HBr in presence of peroxide gives products opposite to Markovnikov rule.

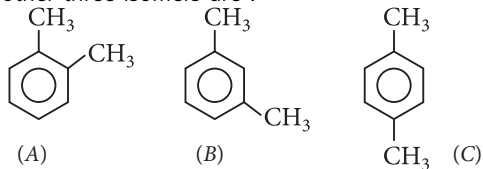


34. There will be no reaction between 2-butyne and Cu_2Cl_2 because it has no acidic hydrogen. In 1-butyne the terminal hydrogen is acidic ($\text{CH}_3\text{CH}_2-\text{C}\equiv\text{CH}$) so it will give a red ppt. with ammoniacal Cu_2Cl_2 .

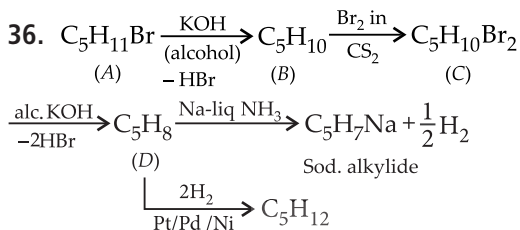
35. Since, it does not decolourise bromine water, it is arene. Thus,



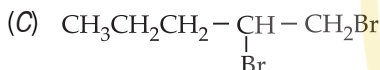
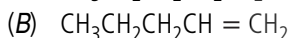
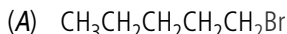
The other three isomers are :



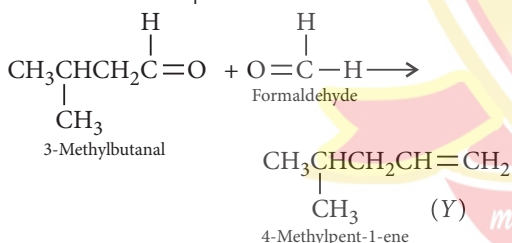
1, 2-Dimethylbenzene (o-Xylene) 1, 3-Dimethylbenzene (m-Xylene) 1, 4-Dimethylbenzene (p-Xylene)



Since, (D) an alkyne reacts with sodium in liq. NH_3 , to give alkylide, it shows that (D) is a terminal alkyne. It yields straight chain alkane on hydrogenation it shows it is a straight chain terminal alkyne. Hence,



37. Hydrocarbon 'Y' is alkene because it decolourises bromine water. From the products of ozonolysis, the structure of alkene can be predicted.

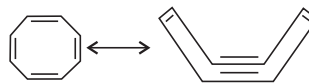


38. (a) $\text{C}_6\text{H}_5\text{CH}=\text{CH}_2$ it has 6π -electrons but not in the ring and one carbon atom has sp^3 -hybridisation, hence it is non-aromatic.

(b) In C_5H_6 , due to the presence of sp^3 -hybridised carbon (carbon 3) and only four π electrons, it does not contain planar delocalised cloud of $(4n+2)\pi$ electrons. Hence, it is non-aromatic compound.

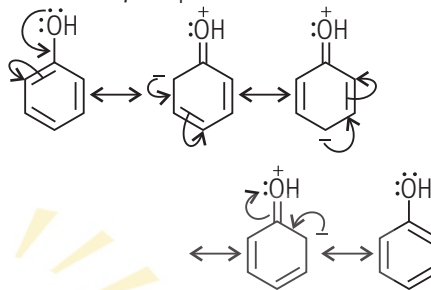
(c) Cyclooctatetraene (COT) is not aromatic because of its non-planar tub-shaped structure. Although according to electron-count it seems to be an anti-aromatic compound but,

infact it is non-aromatic compound.

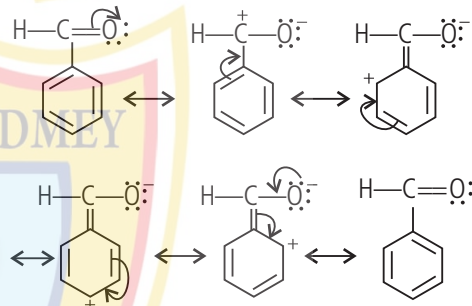


39. Resonating structures of phenol :

—OH group attached to benzene ring release electrons and activate the benzene ring, direct the incoming groups to *ortho*- and *para*-positions.

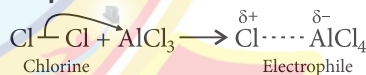


Resonating structures of benzaldehyde :

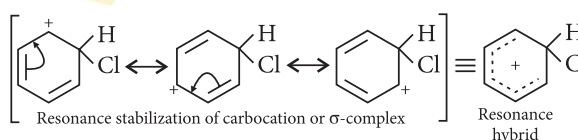
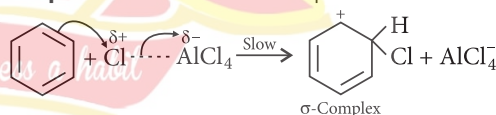


40. The halogenation of benzene proceeds by the following mechanism :

Step 1. Generation of an electrophile.

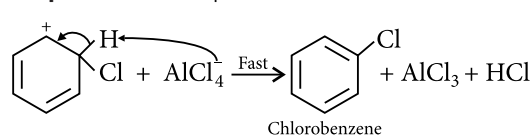


Step 2. Formation of σ -complex or carbocation intermediate.



This step is slow and hence is the rate-determining step of the reaction.

Step 3. Loss of a proton from the carbocation intermediate.



This step is fast and hence does not affect the rate of the reaction.

